
Learning to Separate: Set-Based Neural Reconstruction of Overlapping Electron Showers in LDMX

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Master's Thesis, 30 ECTS credits

Master of Science in Engineering: Engineering Physics

Abstract

The *Light Dark Matter eXperiment* (LDMX) searches for sub-GeV dark matter through missing momentum. Multiple electrons in one readout window can produce overlapping calorimeter showers and obscure the origin of individual hits. This thesis evaluates Transformer and GravNet for supervised hit-level reconstruction of simulated two- and three-electron pile-up events represented as unordered variable-size detector-object sets.

On held-out test samples, the Transformer achieved permutation-invariant pooled hit accuracies of 84.20 % for two-electron events and 74.84 % for three-electron events, or 89.76% and 83.18% respectively for the energy weighted variants. GravNet performed similarly, with a small observed Transformer advantage. The lower three-electron performance is consistent with greater shower overlap.

The *machine learning particle flow* (MLPF)-inspired *Transformer for Reconstruction and Attribution of Calorimeter Energy* (TRACE) retained nearly the same hit assignment performance as the baselines while also predicting electron slots, contributor sets and energy fractions. It distinguished two from three electrons with 99.983 % accuracy, while exact contributor-set accuracy was 72.41 %. Mixed deposits remained the main reconstruction challenge.

Inference-time trigger-pad ablations produced statistically resolved changes, but could not distinguish useful detector information from the distribution shift caused by removing a complete modality. The results establish set-based learning as a feasible offline approach to per-electron reconstruction in simulated LDMX pile-up. The reusable `ml_ldmx` framework supports controlled detector-modality studies and future reconstruction tasks.

18th August 2026

Acknowledgements

My sincerest gratitude goes to my supervisor Lene-Kristian Bryngemark, for their continuous guidance, discussions and support. I am also grateful to my co-supervisor, Ruth Pöttgen, who also supported me and the group of thesis students working within LDMX and provided valuable feedback throughout the project.

I would also like to thank my fellow thesis students in LDMX for the productive discussions during our workshops, their encouragement and our shared determination to bring our theses to completion.

I want to direct my thanks to my partner Tilde Bonnevier Wallstedt for her support and inspiration to write a thesis in particle physics. My friend Axel Helgstrand also deserves my thanks for the same reason and for introducing me to the LDMX research group at Lund University specifically.

Last, and also least, I would like to thank my two cats Shiro and Nemo, who assisted my work by performing random walk on my keyboard; Expanding the search space of possible solutions.

OTHER TITLE SUGGESTIONS:

Separating Overlapping Electron Showers in LDMX: A Set-Based Neural Reconstruction Study

From Hits to Showers: Set-Based Neural Reconstruction under Electron Pile-Up in LDMX

Resolving Electron Pile-Up in LDMX: Hit-Level Reconstruction with Transformers and GravNet

Resolving the Overlap: Hit-Level Electron Shower Reconstruction with Transformers and GravNet

Learning to Separate: Neural Hit Assignment for Overlapping Electron Showers in LDMX

Untangling Electron Showers: Neural Reconstruction of Multi-Electron Events in LDMX

Learning the Origin: Hit-Level Reconstruction of Electron Showers under LDMX Pile-Up

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1 Introduction

Dark matter (DM) is an invisible form of matter whose subatomic nature remains unknown, motivating searches for physics beyond the *Standard Model* (SM). The *Light Dark Matter eXperiment* (LDMX) searches for sub-GeV dark matter through a missing-momentum signature in electron fixed-target interactions. Invisible-particle production would appear as an imbalance between the incoming electron momentum and the reconstructed visible final state. Reliable reconstruction of beam-electron activity is therefore essential [1–3].

In experimental particle physics, *reconstruction* uses detector measurements to infer physically meaningful properties of the underlying particle event. At increased beam intensity, several electrons can enter the same detector readout window and produce overlapping showers in the *electromagnetic calorimeter* (ECal). Their detector activity can then become difficult to associate with individual electrons. Rejecting all multi-electron events would discard part of the delivered beam, while reliable reconstruction could make a larger fraction of recorded events useful and improve the practical physics reach of the experiment [3].

This thesis investigates modern supervised *machine learning* (ML) methods for separating showers in simulated $2e$ and $3e$ events. The principal task assigns each selected ECal hit to the electron responsible for the largest simulated deposited-energy contribution. Each event is represented as an unordered, variable-size collection of reconstructed ECal hits and *trigger-pad* (TPad)-track observations from the *Trigger Scintillator* (TS). This preserves the irregular detector structure and allows both subdetectors to be processed jointly without projecting the event onto a fixed image.

A GravNet *graph neural network* (GNN) and a Transformer encoder provide two fixed-multiplicity baseline architectures. Their reconstruction quality, model size and observed training behaviour are compared, and sensitivity to the TPad input is evaluated through ablation. An exploratory *machine learning particle flow* (MLPF)-inspired model, named the *Transformer for Reconstruction and Attribution of Calorimeter Energy* (TRACE), extends the task to a mixed $2e/3e$ sample. It predicts electron-slot validity, the set of electron contributors to each hit and their deposited-energy fractions. Electron count and dominant hit origin are then derived consistently from these outputs [4–6].

The study is an offline proof of concept based on simulated non-signal events. It does not perform a dark-matter search, establish detector-data performance or deploy an online trigger algorithm, because that is out of scope. A further contribution is the reusable `ml_ldmx` platform, which connects LDMX data preparation, supervised target construction, model training and reproducible evaluation and future studies, and is intended to support future extensions of the reconstruction methods developed here [7].

1.1 Aim and Study Strategy

The work was organised around the following objectives:

- **Investigate shower separation:** determine how reliably supervised models can assign ECal hits to their originating electrons in simulated $2e$ and $3e$ pile-up events.
- **Build a reusable research platform:** develop the `ml_ldmx` pipeline for extracting LDMX simulation data, constructing truth targets, training models and evaluating predictions.
- **Compare model architectures:** evaluate modern set-based neural networks choices GravNet and Transformer baselines using common inputs and targets, considering both reconstruction quality and practical computational properties.
- **Identify limiting conditions:** relate reconstruction performance to electron multiplicity, shower overlap, ECal depth and model confidence.
- **Assess heterogeneous detector information:** test whether reconstructed trigger-pad observations provide useful context beyond the ECal measurements.
- **Explore a broader reconstruction task:** extend the fixed electron-count hit assignment with the mixed-multiplicity TRACE model, inspired by machine-learning-based particle flow at CMS, which predicts electron-slot validity, contributing-electron sets and energy fractions. Electron count is derived from the valid slots.

The intended outcome is an offline proof-of-concept and a documented basis for continued reconstruction studies, rather than a production-grade trigger algorithm, a complete detector reconstruction chain or a dark-matter sensitivity study.

2 Background

2.1 Dark Matter Particle Physics

Dark matter (DM) is the name given to a proposed form of invisible matter that does not emit, absorb or scatter light. Astronomical and cosmological measurements reveal gravitational effects that, within general relativity, cannot be explained by the observed distribution of ordinary matter alone. DM is the leading explanation for these observations, but their cause has not been directly established. In particular, no DM particle or non-gravitational interaction with the particles of the *Standard Model* (SM) has been confirmed in an experiment. This gap between well-established astronomical observations and an unknown subatomic explanation makes DM one of the largest unsolved problems in modern physics [1, 8].

This subsection briefly traces the introduction of DM into particle physics before presenting thermal freeze-out, *light dark matter* (LDM) and the dark bremsstrahlung process that is the basis for the *Light Dark Matter eXperiment* (LDMX).

2.1.1 A Brief History of Dark Matter: from Astronomy to Particle Physics

At the frontier of observation, a new tool can change what is visible. When Galileo Galilei pointed his telescope at the sky in the 17th century, he resolved the faint glow of the Milky Way into stars and observed four satellites in orbit around Jupiter, all previously invisible to the naked eye. These were not discoveries of DM, but they illustrate two important ideas: nature may contain matter beyond the reach of ordinary perception, and a new instrument can move that reach [8].

DM as a concept encapsulates this idea of matter that exists, but cannot be perceived by ordinary means. The history of the physical phenomenon now called DM, however, begins with measurements that revealed more gravitating mass than astronomers could see. Astronomers could estimate the visible mass of a galaxy or galaxy cluster from its light and separately estimate the total mass required to explain how its stars or galaxies moved. Several studies found that the visible mass was not enough. These studies included the recently rediscovered work by Lund University astronomer Knut Lundmark, whose 1930 comparison of the luminosities and mass estimates of five galaxies preceded Fritz Zwicky’s more famous 1933 study of the Coma cluster. Zwicky borrowed the virial theorem from thermodynamics, modeling celestial bodies as particles, to show that the galaxies in Coma moved too rapidly for the cluster to be held together by its visible mass alone. Later measurements of galactic rotation curves showed a related discrepancy: stars far from galactic centres moved faster than expected from the visible matter [8–10]. For these astronomers, however, “dark matter” generally meant unseen astronomical material such as faint stars, cold gas or other non-luminous bodies, not necessarily a new elementary particle.

The transition from unseen astronomical matter to particle physics was not immediate nor automatic. It required both new evidence and a change in scientific culture. Half a century ago, cosmology was still regarded by many physicists as a “fringe science” with

limited predictive power and testability, and the two communities rarely contributed to each other's fields [8].

As cosmology became a recognized quantitative science, measurements of primordial element abundances and the cosmic microwave background showed that baryonic matter could not provide all the required mass. Within the standard cosmological model, baryonic matter, meaning ordinary matter composed primarily of protons and neutrons, accounts for only about 16% of the total matter in the Universe. The remaining 84% is attributed to DM [1]. Studies of structure formation further indicate that the dominant component must have moved slowly enough to behave as cold matter and attempted modified-gravity explanations have also been explored, but none has accounted for the full range of observations as successfully as the DM framework. These results helped transform DM from a problem of unseen astronomical objects into a problem that could require new particle physics. [1, 8].

The SM describes the known elementary particles and their strong, weak and electromagnetic interactions, while gravity is disconnected described separately by general relativity. No known SM particle simultaneously has the properties required to account for the observed cold DM. A particle interpretation therefore points beyond the SM, to possibly a DM sector. A viable DM particle candidate must have three key properties:

- 1) **Mass.** A DM particle candidate must have mass higher than zero, accounting for the invisible mass in the Universe, and it explains its gravitational influence.
- 2) **Stability.** Observations in the cosmic microwave background imply the existence of DM in the early Universe and it remains abundant today. It must therefore be stable, or at least have a lifetime long enough to survive until the present day.
- 3) **Almost no interaction with the SM.** Gravity is described separately from the SM, but it is the only established interaction between DM and ordinary matter. *If* DM has any additional interactions with SM particles, they must be sufficiently weak to have escaped detection in previous experiments. In particular, any electromagnetic interaction must be extremely weak.

2.1.2 Thermal Relics and the Sub-GeV Frontier

Possible DM candidates span an enormous range of masses. Figure 1 gives a schematic overview of several frequently discussed regions. In this thesis, LDM refers to particle dark matter below approximately the GeV mass scale. The range of greatest relevance to LDMX is roughly from MeV to GeV masses. [3, 11].

The *thermal freeze-out* hypothesis is one possible explanation for the amount of DM observed today. In this model one assumes that there exists an interaction that connects the SM and the dark sector. Given this assumption, in the hot and dense early Universe, particle collisions were frequent and energetic enough to create DM particles. DM particles could also annihilate back into SM particles. These reactions occurred rapidly enough to keep the two sectors in *thermal equilibrium*.

As the Universe expanded, it became cooler and less dense. Creating massive DM particles became increasingly difficult, while existing DM particles continued to annihilate. Eventually, the particles became so dilute that the annihilation rate fell below the

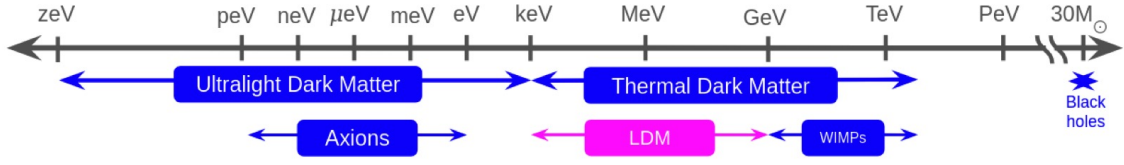


Figure 1. Representative mass ranges for several DM candidates and frameworks. The LDM region relevant to LDMX is highlighted in magenta, while WIMP benchmarks occupy a higher-mass part of the broader thermal DM region. The boundaries are illustrative and model-dependent. Reproduced from Figure 4 of Ref. [12], which was inspired by Figure 1 of Ref. [11].

expansion rate of the Universe. Annihilation then became rare, and the number of DM particles in a volume expanding with the Universe remained approximately constant. Their abundance had *frozen out*, and the surviving population is called a *thermal relic*.

The surviving thermal relic depends on the strength of the interaction. A stronger annihilation process remains effective for longer and leaves fewer DM particles, while a weaker process causes an earlier freeze-out and leaves more particles. For a specified particle model and mass, the observed DM abundance therefore defines a target interaction strength [1]. Thermal freeze-out does not define one universal DM mass range because the result also depends on the particle model, its interactions and the history of the early Universe.

The historically best-known thermal relic candidate is the *weakly interacting massive particle* (WIMP). A particle with a mass near the electroweak scale with a weak-strength interaction can freeze out with approximately the observed abundance, a coincidence known as the *WIMP miracle*. Searches have strongly constrained many WIMP models without finding a confirmed signal. Below the GeV scale, the nuclear recoil produced by a DM particle becomes difficult for traditional WIMP searches to detect, motivating alternative searches for LDM [1, 3, 13].

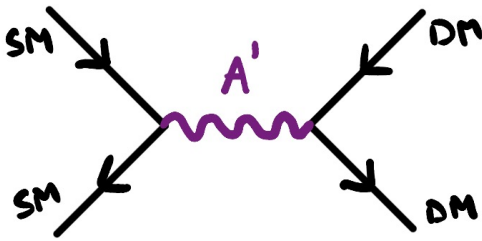


Figure 2. Feynman diagram of conversion of a pair of SM particles into a pair of DM particles, mediated by a dark photon A' .

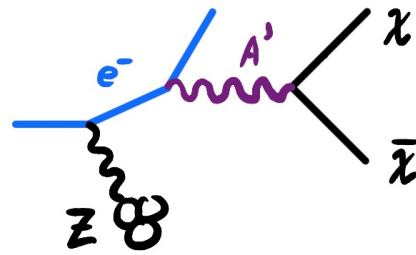


Figure 3. Feynman diagram-sketch of dark bremsstrahlung. The electron scatters and radiates a dark photon A' , which decays into an invisible DM particle-antiparticle pair $\chi\bar{\chi}$. The recoiling electron provides a visible signature by the law of conservation of momentum. Inspired by Ref. [14].

One way to connect LDM to the SM introduces a new dark-sector gauge-boson mediator called the *dark photon*, denoted by A' in Figure 2. It can be thought of as a

dark-sector relative of the ordinary photon. A small kinetic mixing with the ordinary photon allows the dark photon to interact weakly with electrically charged SM particles. In the invisible dark photon benchmark considered here, the dark photon decays into a pair of DM particles, denoted by $\chi\bar{\chi}$ [1, 3].

In an electron fixed-target experiment, the benchmark production reaction is *dark bremsstrahlung*, illustrated in Figure 3. It is a dark-sector version of ordinary bremsstrahlung. An incoming electron scatters in the field of a target nucleus and radiates a dark photon, which then decays into $\chi\bar{\chi}$. The DM particles leave no measurable detector signal. Their production can instead be inferred from the imbalance between the incoming electron momentum and the reconstructed visible final state. Unexplored thermal targets motivate the missing-momentum search for which LDMX is designed, that precisely plans to expand the region of search [2, 3].

Dark bremsstrahlung turns the original astronomical question into a tangible experimental one: can missing momentum reveal an invisible final state after ordinary sources of imbalance have been rejected? The next subsection introduces the detector built to answer that question.

2.2 The Light Dark Matter eXperiment

2.2.1 Search Strategy and Physics Goal

LDMX is an electron fixed-target *missing momentum* experiment designed to search for sub-GeV DM through the dark-bremsstrahlung process introduced above. The experiment is planned for End Station A at *SLAC National Accelerator Laboratory* (SLAC) outside of San Francisco, California and is developed by an international collaboration with institutions in the United States and Sweden. An 8 GeV electron beam will strike a thin tungsten target. Tungsten is used because its high atomic number gives a large production rate for low-mass dark photons and helps reduce nuclear backgrounds at the chosen target thickness [3].

In the dark bremsstrahlung benchmark, the beam electron radiates a dark photon, denoted A' . The mediator couples weakly to electrically charged SM particles and can decay into a pair of dark-sector particles, denoted by χ . It therefore provides a possible bridge between the SM and a dark sector. The DM particles do not produce a measurable detector signal. LDMX instead tests the model by searching for the recoil of the electron and the missing momentum left by their production [2, 3].

In the benchmark process, a candidate event has three main features. The electron leaves the target with less energy than it entered with, it receives a transverse (perpendicular to the beam direction) momentum kick, and no other visible final-state particle accounts for the lost momentum. The combination is important because ordinary SM interactions can mimic parts of this signature when a visible particle escapes detection. The trackers therefore measure the incoming and recoil-electron momenta, while the calorimeters test whether additional visible activity explains the apparent missing momentum [2, 3].

The planned beam structure provides approximately 20 million beam-designated buckets per second, with an average of one to two electrons in each bucket. This corresponds to roughly 20 to 40 million electrons reaching the target per second during operation. The exposure, measured in *electrons on target* (EoT), is not a universal event count required to reject a null hypothesis. Rather, increasing the exposure increases sensitivity to rarer production processes and weaker interactions. The attainable exposure also depends on running time, detector performance, backgrounds and radiation tolerance. LDMX is designed for a staged programme beginning around 4×10^{14} EoT and extending towards exposures of order 10^{16} EoT, with projected sensitivity up to three orders of magnitude beyond earlier searches [3].

Useful exposure depends not only on how many electrons are delivered, but also on which events can be triggered, stored and reconstructed. The ability to retain events containing several electrons can therefore improve the practical physics reach. This connection is developed further in Section 2.2.7.

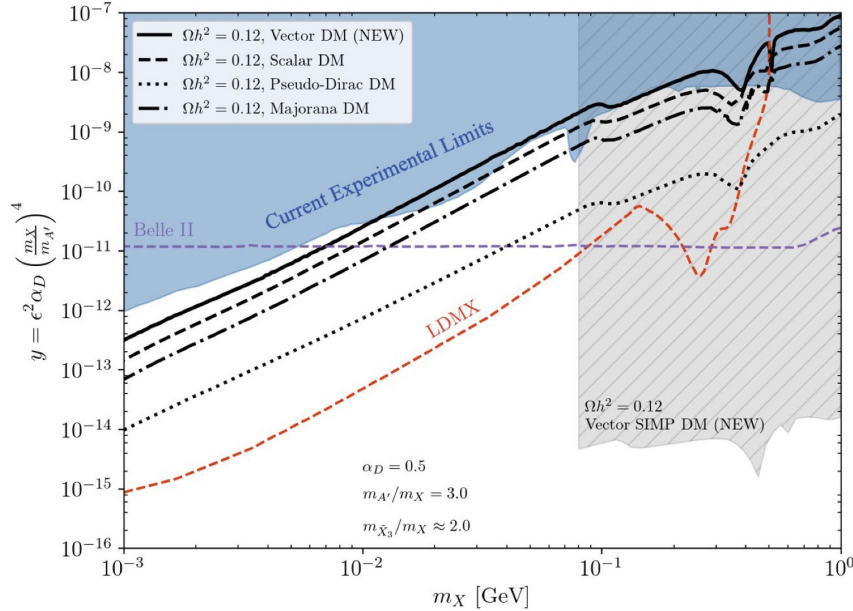


Figure 4. Representative comparison of thermal-relic targets with existing constraints and projected experimental sensitivity. The horizontal axis gives the dark-matter mass and the vertical axis gives the benchmark interaction quantity y . Blue shows approximate existing constraints, purple shows the Belle II projection, red shows the LDMX projection, and the black curves and grey region show several model-dependent DM thermal targets. Reproduced from Refs. [3, 15].

Figure 4 gives a representative view of the projected reach. The horizontal axis shows the dark-matter mass, while the vertical axis combines the assumed interaction strength and particle masses into the benchmark quantity y . The blue region represents parameter combinations already constrained by earlier searches. The red curves show the smaller interaction strengths that LDMX could test, thus expanding the frontier of search of DM, while the black curves and grey region show thermal-relic targets for several representative DM models. The figure is therefore a comparison within specified model

assumptions, not a model-independent exclusion of DM by mass alone [3, 15, 16].

2.2.2 Detector Layout

Figure 5 shows the physical scale and overall arrangement of the detector. The conceptual drawing in Figure 6 illustrates the path of the beam electron and the missing-momentum signature. The labelled cutaway in Figure 7 shows the detector geometry along the beam direction.

Two *Trigger Scintillator* (TS) modules are placed upstream of the silicon tagging tracker, and a third is placed directly before the target. The tagging tracker measures the incoming electron momentum. The electron then passes through the thin tungsten target inside a dipole magnet. Its magnetic field bends charged particles transversely according to their momentum, so a recoil electron that has lost energy is deflected more strongly than an electron retaining the full beam energy. A silicon recoil tracker downstream of the target measures the momentum and direction of the outgoing charged particles.

The baseline target is 0.1 radiation lengths thick and has a transverse size of $4\text{ cm} \times 10\text{ cm}$. Its small thickness limits multiple scattering while retaining a useful production rate. The nominal $20\text{ mm} \times 80\text{ mm}$ beam spot reduces peak detector occupancy and helps to separate the showers from different electrons [3].

The *electromagnetic calorimeter* (ECal) is placed behind the recoil tracker. It measures electrons, photons and their shower shapes. The *hadronic calorimeter* (HCal) surrounds the ECal at the sides and rear. Its main role in the dark-matter search is to identify penetrating hadrons and muons that could otherwise imitate missing momentum. Together, the trackers and calorimeters measure the recoil electron and *veto*, meaning reject, candidate events containing other visible particles. This combined information distinguishes an invisible final state from ordinary interactions in which visible particles were missed by one detector system [3].

2.2.3 Electromagnetic Calorimeter and Electron Showers

The main purpose of the ECal is to measure the energy and spatial development of electrons and photons, allowing the visible final state to be reconstructed and backgrounds to be rejected. Its high-granularity silicon-tungsten design is a compact "sibling detector" of the *High Granularity Calorimeter* (HGCAL) at the *Compact Muon Solenoid* (CMS) experiment, from which LDMX adopts detector-module and readout technology [3].

The ECal is a sampling calorimeter with a total depth of about 40 radiation lengths. A radiation length X_0 is the characteristic material scale for energy loss by a high-energy electron through bremsstrahlung. Tungsten layers initiate and develop electromagnetic showers, while interleaved silicon layers sample their energy deposits. The final design contains 32 silicon layers and the `ldmx-det-v15-8gev` geometry used in this thesis likewise produces reconstructed ECal hits at 32 layer positions. The performance simulations in the current (and newer) LDMX design report instead used an earlier geometry with 34 layers [3].

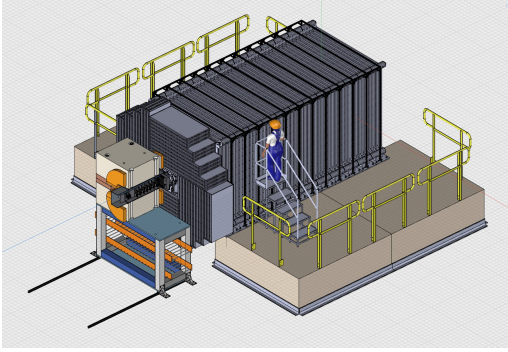


Figure 5. Solid-model overview of the LDMX detector installation. The compact tracking and target region is located at the left, in front of the calorimeter systems. Sourced from Figure 3.1 of the 2025 LDMX design report [3].

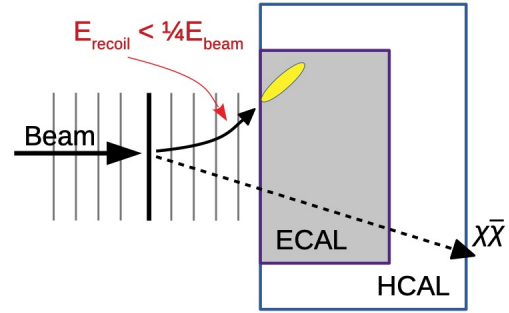


Figure 6. Conceptual illustration of the LDMX missing-momentum signature. After the beam electron passes through the tagging tracker and scatters in the thin tungsten target, a recoil electron carrying less than one quarter of the beam energy enters the ECal, while an invisible $\chi\bar{\chi}$ pair carries away momentum. The ECal and HCal test for additional visible activity. Sourced from Figure 8 of the 2018 LDMX design report [2].

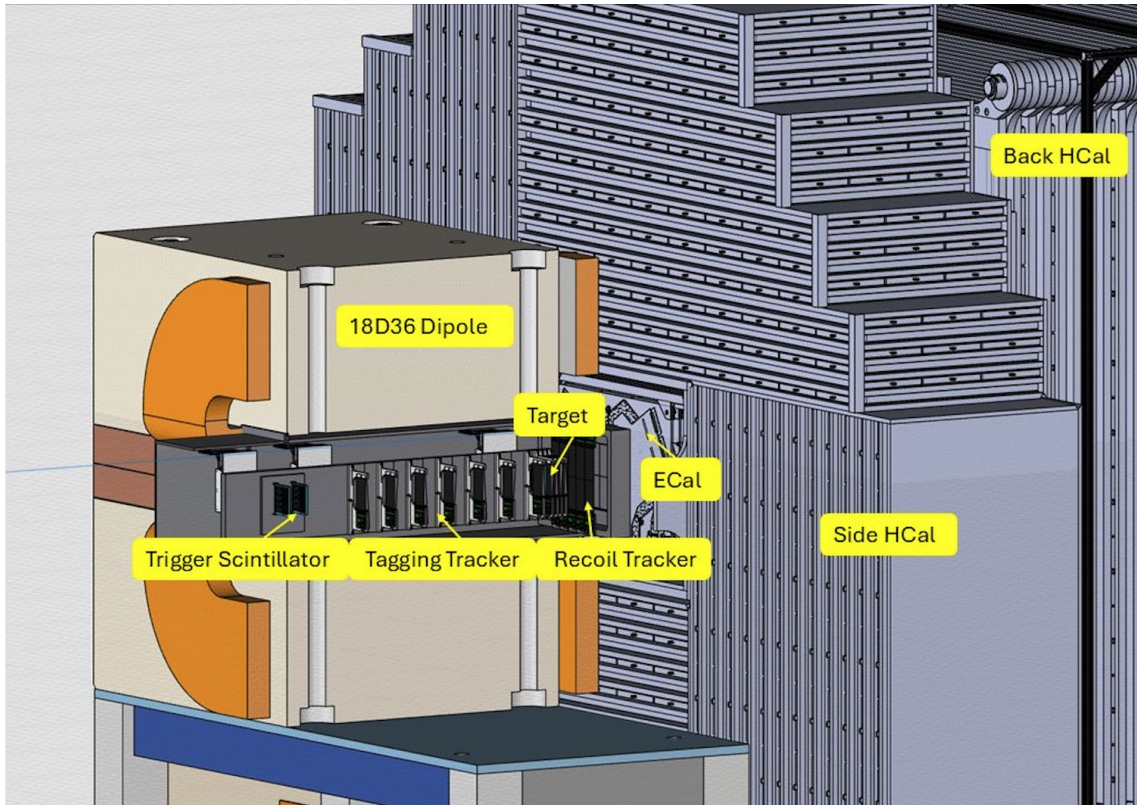


Figure 7. Cutaway view of the LDMX detector. From left to right, it shows the Trigger Scintillator, tagging tracker, target inside the dipole magnet, recoil tracker, ECal, and the side and back HCal. Sourced from Figure 3.2 of the 2025 LDMX design report [3].

An energetic electron radiates photons through *bremsstrahlung* as it passes through tungsten. These photons can produce electron–positron pairs, which radiate further photons. Repetition of these processes creates an *electron shower*. One incoming electron therefore produces energy deposits in many cells and layers rather than a single detector hit. The energy contributions in the 3D pattern captures information about the electron position, energy and shower development [17].

The transverse size of an electromagnetic shower is commonly described by the Molière radius R_M . On average, about 90% of the shower energy is contained within a cylinder of radius R_M [17]. For the layered LDMX ECal, the design report gives an effective Molière radius of approximately 2.5 cm. It also reports that 68% of the shower energy is contained within less than 1 cm during approximately the first 15 layers [3]. These values will provide useful reference scales for interpreting shower overlap.

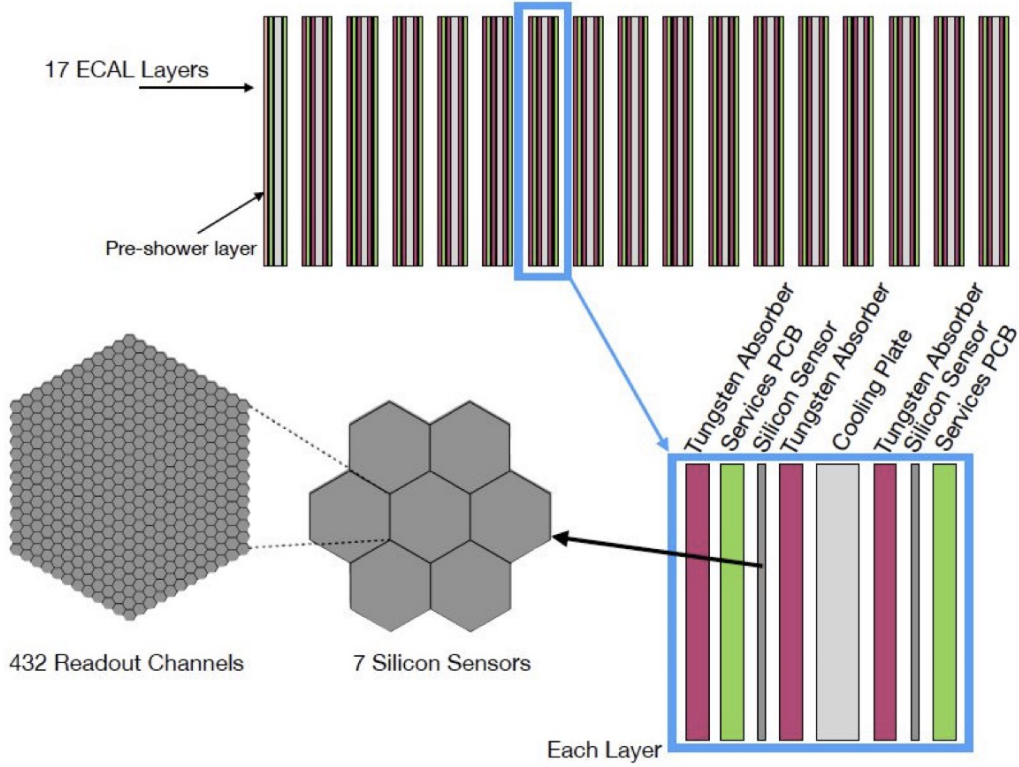


Figure 8. Schematic of the full-resolution LDMX ECal modules and the components of a bilayer. Sourced from the LDMX design report [3]

Each ECal sampling plane contains seven hexagonal modules, and each module contains 432 silicon readout cells. A local cell identifier connects every measured energy deposit with a fixed detector position. Figure 8 shows this full-resolution segmentation and a bilayer construction. For the real-time trigger, groups of nine neighbouring readout cells are summed into 48 coarser *trigger cells* per module. This grouping, which is not drawn in the figure, retains less spatial detail for the *online reconstruction* compared to the individual-cell readout available to *offline reconstruction* [3].

The fine segmentation helps separate nearby showers and identify small energy deposits from background particles, making it central to this thesis. When several beam electrons arrive in the same readout window, their showers may overlap in the same layers or cells. Reconstruction must then use the spatial and energy pattern of the full-resolution hits to determine which electron produced each part of the event and recover per-electron shower information from the combined response.

2.2.4 Trigger Scintillator and Trigger-Pad Tracks

The TS estimates how many beam electrons enter the target in the same readout window and where they pass through the beam region. It consists of three modules: two directly upstream of the tagging tracker and one directly upstream of the target. Each module contains two staggered layers of plastic scintillator bars. A traversing electron deposits energy in a bar and produces scintillation light that is measured as a photoelectron signal. Adjacent bar hits are joined into clusters, and clusters across the three modules are combined when their positions are compatible with an electron travelling along the expected beam direction [3]. Forming these track candidates is itself a rule-based reconstruction method: low-level bar signals are converted into estimates of electron position and multiplicity.

In this thesis, the reconstructed TS track candidates are referred to by their name in the LDMX codebase, `TriggerPadTracks`, or simply *trigger-pad* (TPad) in flowing text. The term describes objects derived from TS measurements, not a separate detector, and should not be confused as so. Each candidate provides a one-dimensional position estimate and a photoelectron count. Together, the candidates provide information about the positions, energy and the multiplicity of the incoming electrons. They are reconstructed detector objects rather than simulation-truth labels, and correspond one-to-one with the incident electrons only when the reconstruction makes correct prediction [3].

2.2.5 Triggering

The trigger decides in real time which detector responses should be fully read out and stored. Beam electrons arrive in short radio-frequency buckets separated by 26.9 ns, corresponding to a 37.14 MHz detector clock. The full detector response cannot be recorded at this rate, and almost all interactions are ordinary processes rather than candidate dark-matter production. The trigger therefore has to reject most buckets using only compact information available before full readout [3].

For the missing-energy decision, the TS supplies an estimated incoming electron count n , while the ECal supplies coarse trigger-cell energy sums. A schematic trigger-level missing-energy estimate is

$$E_{\text{miss}}^{\text{trig}} = nE_{\text{beam}} - E_{\text{ECal}}^{\text{trig}}, \quad (1)$$

where E_{beam} is the nominal energy per beam electron and $E_{\text{ECal}}^{\text{trig}}$ is the energy measured through the coarse ECal trigger readout. An overestimated n can imitate missing energy, while an underestimated n can reduce the estimated imbalance and cause a signal

candidate to be rejected. The trigger reduces the input to an average full-readout rate of approximately 10 kHz [3].

The distinction between online/offline reconstruction is set by this deadline. Online reconstruction occurs before storage and must satisfy strict limits on latency, memory and transferred information. It therefore uses reduced quantities such as ECal trigger-cell sums and Trigger Scintillator track candidates. Once an accepted event has been fully read out, offline reconstruction (which technically has no deadline) can use individual ECal cells and more computationally demanding algorithms to construct calibrated hits, tracks, showers and analysis quantities [3].

The methods developed in this thesis use simulated ECal hits at the full cell granularity and due to this they therefore belong to offline reconstruction research. Their results can show whether electron showers can be separated under multi-electron conditions, contributing to ascertaining what the ideal Poisson sampling might be. Deployment in the online trigger would be a separate task because it would require reduced trigger inputs and a dedicated latency and hardware study.

2.2.6 Pile-Up and Reconstruction

In this thesis, *pile-up* means primarily *in-time pile-up*. Two or more beam electrons occupy the same radio-frequency bucket, so their detector responses are recorded in one readout window. If the number of electrons N in a bucket follows a Poisson distribution with tunable mean μ , then

$$P(N = m) = \frac{\mu^m e^{-\mu}}{m!}. \quad (2)$$

For example, $\mu = 1$ gives $P(N = 1) = e^{-1} \simeq 36.8\%$ and $P(N \geq 2) = 1 - 2e^{-1} \simeq 26.4\%$, while the remaining 36.8% are empty. An exactly-one-electron selection would therefore retain 36.8% of all buckets, or 58.2% when conditioning on a bucket being non-empty. Lowering μ reduces the multi-electron fraction, but also reduces the number of EoT delivered per unit time. The engineering challenge of this trade-off is therefore to benefit from a higher beam intensity without discarding every bucket that contains several electrons [3].

Separate beam electrons can be treated as propagating independently to an excellent approximation because direct interactions between them are negligible. The propagation is partly deterministic (momentum) and partly stochastic (i.e. nondeterministically reversible). Let $Z^{(k)}$ denote the simulated deposits from electron k .

$$Z^{(k)} \perp\!\!\!\perp Z^{(\ell)} \quad \text{for } k \neq \ell. \quad (3)$$

This makes it possible to simulate their particle histories separately and *overlay* them before detector digitisation (detector response simulation) and reconstruction. The observed hit collection can be written as

$$\mathcal{H} = \mathcal{R}(Z^{(1)} \oplus \dots \oplus Z^{(m)}), \quad (4)$$

where $\oplus$ denotes overlay by detector channel and $\mathcal{R}$ includes digitisation, detector granularity, thresholds, noise and hit reconstruction. The particle histories may be inde-

pendent, but their observed responses need not remain separable. Deposits can share a cell or showers can overlap across several layers, so a simple geometric boundary does not in general reconstruct the original $Z^{(k)}$.

This creates an inverse reconstruction problem. From the combined detector objects $\mathcal{H}$, one would like to infer the incoming multiplicity and the contribution of each electron to the observed activity. At trigger level, an incorrect multiplicity changes the missing-energy estimate. Offline, overlap obscures ECal topology and may hide low-energy deposits relevant to background rejection. This thesis isolates one part of the inverse problem by testing whether full-resolution ECal hits can be assigned to their electron of origin. The experimental TRACE model also infers the restricted $2e$ -versus- $3e$ multiplicity and predicts contributions from more than one electron to the same hit.

2.2.7 The Practical Physics Reach

Pile-up is not only a reconstruction complication. If multi-electron events can be retained, it is also an opportunity to increase the useful electron throughput. The LDMX design report estimates that approximately 4×10^{14} EoT can be accumulated in twelve months at an average multiplicity of one electron per sample, while exposures of order 10^{16} EoT would take many years at the same effective rate. Rejecting all multi-electron buckets would discard a substantial part of the delivered beam and further extend the required running time [3].

The connection to signal acceptance has also been studied at trigger level. For a 0.1 GeV dark-photon benchmark and mean multiplicity $\mu = 1$, accepting events with up to four reconstructed TS tracks increased the total signal-trigger efficiency from 0.58 to 0.88 at approximately the same 1 kHz trigger budget. That result used the energy in the first 20 ECal layers and belongs to a particular trigger study, but it demonstrates why usable multi-electron events can improve the practical reach of the experiment, thus improving the *physics reach* as well [3].

Reliable pile-up reconstruction is therefore an enabling step rather than a dark-matter measurement by itself. If overlapping showers can be separated, more accepted events can retain interpretable per-electron information and contribute to the accumulated exposure. This thesis tests that premise for simulated wo-electron and three-electron events as direct extensions of the single-electron case and higher multiplicities are outside its scope. The study does not itself search for DM or optimise the online trigger, the latter would require its own study. Instead, it asks whether modern machine-learning methods can recover per-electron shower structure from the combined full-resolution detector response.

2.3 Machine Learning for Set-Based Detector Reconstruction

2.3.1 Learning a Reconstruction Mapping

The range of *machine learning* (ML) applications in *high-energy physics* (HEP) has expanded rapidly [18, 19]. Popularity alone is not a scientific motivation for using ML,

its relevance here is rather that simulation supplies paired detector observations and particle-level truth provides a natural application for classification tasks to be explored, while overlapping showers create associations that are difficult to express through a set of geometric rules designed "by hand".

The same idea in Equation 4 can be stated more generally for an intuitive illustration for the reader. Let Y be the generated particles in an event, let $\mathcal{S}$ be the detector simulation, and let X be the detector signals produced by that simulation. Following the functional description used for *machine learning particle flow* (MLPF) [5], detector simulation maps Y to X , while reconstruction tries invert and to map X back to Y :

$$\mathcal{S}(Y) = X, \quad \mathcal{R}_\theta(X) \simeq \mathcal{S}^{-1}(X) = Y, \quad (5)$$

Here, $\mathcal{R}_\theta$ is the learned reconstruction model, where θ denotes the learnable parameters. The inverse notation is conceptual rather than literal. The forward map is stochastic and loses information through finite detector granularity, thresholds and noise, so an exact and unique $\mathcal{S}^{-1}$ generally does not exist.

Rule-based reconstruction encodes detector knowledge through explicit operations, while learned reconstruction estimates a parameterised mapping from examples. The latter can combine many correlated inputs without every decision rule being specified manually, but it is not automatically optimal and depends on the simulation, target definition and training distribution, so the approaches are complementary. The models in this thesis receive ECal hits and trigger-pad tracks that have already passed through conventional detector reconstruction, and learn an additional association and prediction stage on top of those objects.

2.3.2 Machine-Learned Particle-Flow as a Reference Case

Particle-flow reconstruction combines complementary tracking and calorimeter measurements to reconstruct individual final-state particles. MLPF, developed for the CMS experiment, replaces parts of this association and particle-level prediction procedure with a learned model, making it a relevant example of machine learning used inside reconstruction rather than only as a final event classifier. The original study used a *graph neural network* (GNN) architecture under high-pile-up conditions [5]. The latest iteration from the start of this year transitions the architecture to a Transformer integrated into the CMS reconstruction software and evaluates it on both simulation and collision data [6]. The transition also illustrates the growing importance of accelerator¹-friendly computation for high-dimensional, variable-size event data.

ML and rule-based reconstruction are complementary rather than exclusive. MLPF receives calorimeter clusters, *detector objects*, that already include outputs from earlier reconstruction stages, then learns associations and particle-level predictions from those objects. The high occupancy and heterogeneous inputs of CMS particle flow are substantially broader than the problem studied here. The present thesis studies a narrower

¹"Accelerator" in this context refers to specialized hardware such as *graphics processing units* (GPUs) or even ML specialized architectures such as the relatively new *tensor processing unit* (TPU) that accelerate the speed of computation for applicable processes [20]

LDMX-specific problem to explore: finding how overlapping electron showers can be separated at hit level, but adopts the same general view of an event as a variable-size collection of heterogeneous detector objects.

2.3.3 Detector Events as Unordered Sets

Under pile-up, several particles can contribute to nearby or even identical detector cells. Reconstruction must therefore use the pattern formed by many detector objects, rather than classify each hit from its own features alone. For one event, these objects can be written as a variable-size set

$$\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}, \quad (6)$$

where $\mathbf{x}_i$ contains the measured features of one ECal hit or trigger-pad track. The storage index i has no physical meaning.

This representation avoids projecting the three-dimensional ECal and the different trigger-pad objects onto one regular image. A *convolutional neural network* (CNN) is naturally suited to a fixed two-dimensional grid and local translation symmetry. By contrast, LDMX events contain a variable number of objects from heterogeneous detector systems, and the ECal itself has a hexagonal cell layout. Graph neural networks and Transformer encoders can instead retain every object as a node or token and learn how the objects should exchange information [21, 22].

For the per-object models used here, reordering the input should only reorder the corresponding outputs. If $\mathbf{P}$ is a permutation matrix and $\mathbf{F}$ denotes the network, this requirement is

$$\mathbf{F}(\mathbf{P}\mathbf{X}) = \mathbf{P}\mathbf{F}(\mathbf{X}). \quad (7)$$

Both GravNet and a Transformer encoder without sequence-position encodings have this permutation-equivariant property. Their principal difference is how they choose the other objects from which each node/token receives information.

2.3.4 Graph Neural Networks and GravNet

A graph represents detector objects as *nodes* and connections between them as *edges* [23]. A GNN updates every node using information from its neighbours. The same learned operation is applied to all nodes, and the neighbour information is combined in a way that does not depend on the order of the neighbours. Stacking several layers allows information to propagate through a larger part of the event [24]. Graph-based models have therefore become a natural approach to particle tracking and reconstruction tasks in HEP [21].

GravNet was selected as a baseline because it was developed specifically for sparse, irregular particle-detector data and first evaluated on the clustering of calorimeter showers [4]. Related dynamic-graph ideas have also been used successfully in modern high-energy-physics models. The 2021 MLPF model used nearest-neighbour graph construction and graph convolutions for particle-flow reconstruction, while ParticleNet used a dynamic graph for jet tagging [5, 25]. These models do not use the same graph layer, but together they motivate learned graphs as an established approach for unordered

detector and particle data. Using GravNet alongside a Transformer therefore compares a detector-oriented graph architecture with a modern attention-based architecture on the same shower-separation task.

GravNet differs from a GNN with a fixed graph because it learns which nodes should be neighbours. In each layer, the network places every node in an internal coordinate space and connects it to its k nearest neighbours, a tunable integer hyperparameter. This learned space is not a physical detector coordinate. It is a representation developed during training so that nodes useful to one another can become neighbours [4].

Each node then receives information from these neighbours. Nearby neighbours in the learned space have greater influence, while mean and maximum summary operations capture typical and strong neighbour features. These summaries are combined with the node's existing representation. A new graph is constructed in every layer, so the network can learn a different notion of similarity as the information is processed [4]. Figure 9 illustrates this sequence. The implementation used in this work is provided by PyTorch Geometric [26].

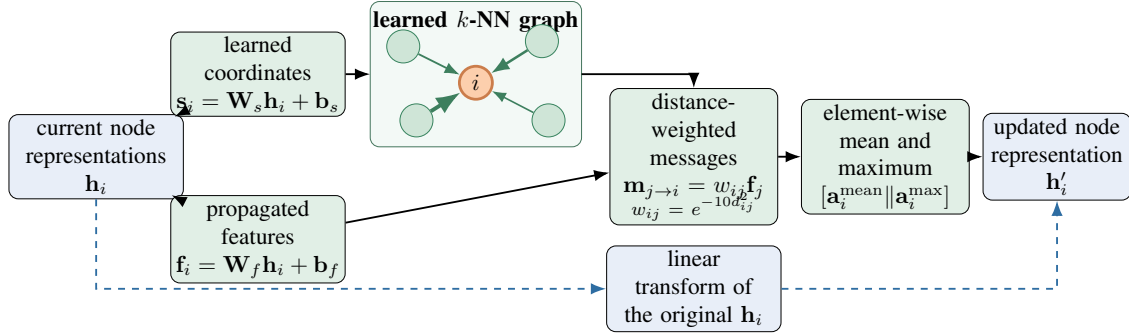


Figure 9. Data flow through one GravNetConv operation. Each node is projected into a learned coordinate space and connected to its k nearest neighbours within the event. Neighbour features are weighted by learned-space distance, aggregated by their element-wise mean and maximum, and combined with a transformed copy of the original node. Original diagram based on the GravNet operation introduced in Ref. [4].

The models in this thesis use either ECal hits alone or ECal hits and trigger-pad tracks as nodes. After several GravNet layers, an output network makes a prediction for each ECal hit. Because each node communicates with only k neighbours per layer, the graph remains sparse, although a nearest-neighbour search must be performed. The complete model architecture is specified in Section 3.5.

2.3.5 Transformer Encoders and Self-Attention

A Transformer represents its inputs as *tokens* [27]. Attention-based models can also operate on unordered sets rather than only on ordered sequences [22]. In this thesis, a token is an ECal hit or a trigger-pad track. The same input network first converts the measured features of every token into an internal representation. The use of a Transformer for the newer MLPF reconstruction model provides a recent application of this architecture to heterogeneous detector objects in HEP [6].

Self-attention then allows each token to collect information from all valid tokens in the event [27]. The network produces a *query*, *key* and *value* for each token. A query is

compared with the keys to determine which other tokens are relevant, while the values contain the information to be combined. The resulting attention weights are normalised, so each updated token becomes a weighted mixture of information from the event [27].

Multi-head attention repeats this process several times, allowing different heads to represent different relations between detector objects. After attention, a feed-forward network processes every token independently. Residual connections preserve earlier information, while layer normalisation is used within each encoder block [27]. In the pre-layer-normalised design used here, normalisation is applied before the attention and feed-forward operations. Several such encoder blocks are stacked, alternating information exchange across the event with processing of each token. The exact block used in this work is specified in Section 3.5 and shown in Figure 10.

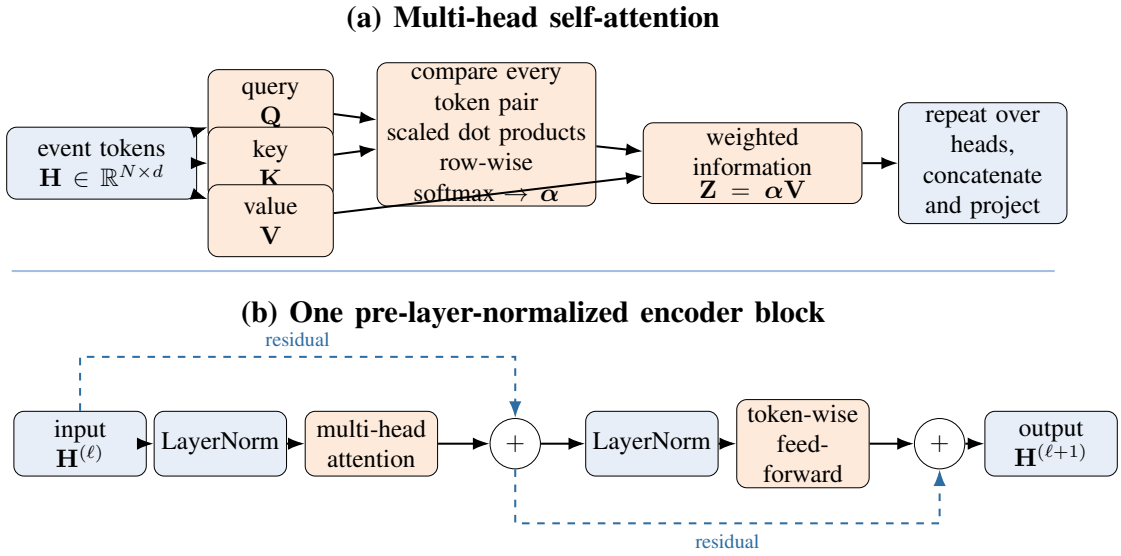


Figure 10. Transformer operations used in this thesis. Panel (a) shows how one attention head compares every valid token pair and forms a weighted mixture of their value vectors; several heads repeat the operation with independent projections. Panel (b) shows the pre-layer-normalized encoder block used by the reported models. Diagram conventions are adapted from NNTikZ [28]; the operations follow Ref. [27].

Only the Transformer *encoder* is used. The decoder from the original architecture is unnecessary because the task is not to generate a sequence. Encoding of sequence order is also omitted because the storage order of detector objects has no physical meaning. Physical position is instead provided directly among the detector features. This treatment follows the set-based use of attention, for which the input order should not determine the result [22]. Unlike GravNet, self-attention compares every pair of valid tokens. This gives global event context in one layer, but its memory and computation grow quadratically with the number of tokens [27, 29].

2.3.6 GravNet–Transformer Analogy

The Transformer is a special case of a GNN [29]. Here, GNN is understood in the generic message-passing sense. A generic GNN can be moulded into the mathematical

equivalent of a Transformer by enforcing fully connected graphs and implementing self-attention to determine the importance of every connection between nodes. Each node then exchanges information with every node in the event, including itself, just as each Transformer token attends to every valid token, deconstructing the separated analogy of nodes and tokens. The formal equivalence is derived in Ref. [29] and its qualitative consequence is that a Transformer can be viewed as a GNN whose graph is complete and whose edge weights are learned through attention.

Both architectures therefore update one representation per detector object by exchanging information between objects. The Transformer uses the complete graph described above, whereas GravNet learns a sparse k -nearest-neighbour graph and weights its edges through distance in the learned coordinate space. This mathematical correspondence does not make their concrete implementations or computational costs identical however.

Table I
OPERATIONAL COMPARISON OF THE INTERACTION LAYERS USED IN THIS THESIS. HERE N IS THE NUMBER OF DETECTOR OBJECTS AND k IS THE NUMBER OF GRAVNET NEIGHBOURS.

Property	GravNet	Transformer encoder
Input object	Node	Token
Objects contacted in one layer	k nearest neighbours in a learned space	Every valid token in the event
Interaction weight	Gaussian function of learned-space distance	Softmax of query-key similarity
Information reach	Local in one layer; expands when layers are stacked	Global within one layer
Pairwise interactions	$O(Nk)$ after graph construction	$O(N^2)$ token pairs per head
Output	One updated representation per input node	One updated representation per input token

From a computational perspective, Transformers are described as GNNs that are “currently winning the *hardware lottery*”² [29]. The current industry shift towards accelerator computing has produced highly optimized GPU and TPU support for the dense matrix multiplications used by Transformers. In practice, all token interactions can be evaluated through large parallel tensor operations.

The development of MLPF is consistent with this hardware-lottery interpretation, although the CMS study does not establish hardware alignment as the sole reason for its architectural change. The newer implementation explicitly replaces the earlier GNN layers with Transformer layers accelerated by FLASHATTENTION and reports roughly 5,000 input elements in a pile-up event [6]. FLASHATTENTION evaluates the same exact attention in tiles without materialising the full attention matrix, reducing memory traffic and runtime while retaining the all-to-all interaction pattern. By comparison, the LDMX samples studied here contain only a few hundred ECal hits and a small number of trigger-pad-track tokens per event. Full attention is therefore considerably more tractable for the present input sets.

²The term *hardware lottery* describes an architecture benefiting because its operations match the available software and hardware ecosystem, rather than this alone demonstrating that it is universally superior [30].

GravNet evaluates fewer pairwise messages after graph construction, but it must construct a nearest-neighbour graph and perform indexed gather–scatter operations for sparse message passing. These operations are less directly aligned with the dense-computing path emphasized by current accelerators and software libraries [26]. Although full self attention is likely not mathematically cheaper, the number of token pairs still grows as N^2 , compared with Nk edges after GravNet graph construction, the Transformer could still compute faster in real time. Which model is faster for this study still depends on event size, batch size, implementation and hardware and is therefore measured empirically and discussed in the Results and Discussion chapter. The complete architectures, including their chosen widths, depths, heads and neighbour counts, are specified in Section 3.5.

3 Methodology

This chapter describes the complete experimental method: simulated event production, construction of supervised targets and model-ready tensors, the three neural-network architectures, their training procedures and the metrics used for evaluation. In simpler terms it could be said that this chapter will answer the question for the reader “what was built, what was tested and how was it done?”.

This chapter also presents and describes the intended reusable `ml_ldmx` framework [7], with the aim of separating data conversion and target construction from any individual model. Simulation, large-scale conversion and training were performed on the COSMOS Linux *high-performance computing* (HPC) cluster at Lund University with *Central Processing Unit* (CPU) and GPU resources available [31].

The chapter follows the processing chain from simulated events to supervised targets, model inputs, neural-network training and evaluation. Technical details that are more in-depth in nature necessary to reproduce this study are collected in Appendix 7.

3.1 Workflow and Reconstruction Tasks

3.1.1 Simulation and Data Pipeline Overview

Event production begins in the open-source *LDMX-Software* (`ldmx-sw`) framework [32], which uses *GEometry ANd Tracking 4* (Geant4) to simulate particle transport and detector response [33]. The physics single-electron $1e$ events are independently simulated and afterwards *overlaid* to form $2e$ and $3e$ readout windows. Afterwards the detector response is simulated (“digitisation”) and reconstruction algorithms are then performed on the combined detector deposits generating objects but also calculating energy readout. The selected reconstructed quantities and simulation truth are extracted, within the `ml_ldmx` framework, from the resulting ROOT files and converted to variable-size PyTorch tensors and stored in ML model-ready “shards”. Figure 11 summarises this workflow from simulation to training in a high-level overview.

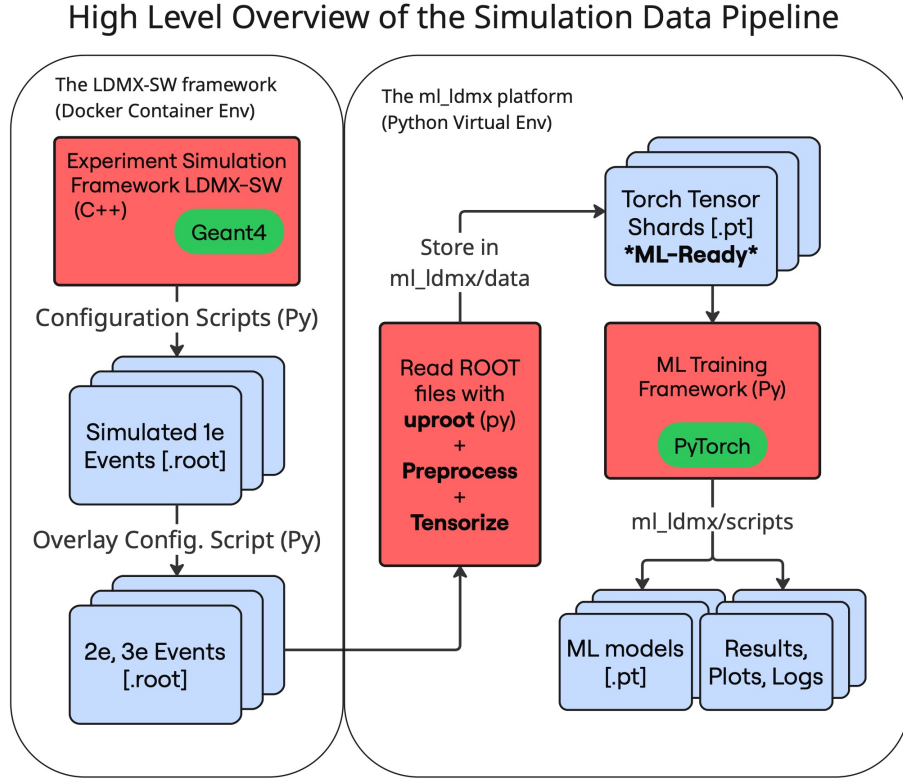


Figure 11. High-level data flow pipeline from LDMX simulation through target construction and tensorization to model training and evaluation.

3.1.2 Reconstruction Formulation

For one event, the observed ECal response is represented as the variable-size set

$$\mathcal{H} = \{(\mathbf{r}_i, E_i)\}_{i=1}^{N_{\text{ECal}}}, \quad \mathbf{r}_i = (x_i, y_i, z_i), \quad (8)$$

where $\mathcal{H}$ is the hit collection extracted from the ECal subdetector, E_i is the reconstructed energy of hit i and $\mathbf{r}_i = (x_i, y_i, z_i)$ is the coordinates in 3D space. The standard event representation also includes the variable-size set of reconstructed TS track observations

$$\mathcal{T} = \{(c_j, n_j^{\text{pe}})\}_{j=1}^{N_{\text{TS}}}, \quad (9)$$

where $\mathcal{T}$ is the collection extracted from the TS subdetector, c_j is the reconstructed object of a one-dimensional track centroid and n_j^{pe} is the corresponding reconstructed number of photoelectrons. The complete event input set is therefore represented by the pair $\mathcal{X} = (\mathcal{H}, \mathcal{T})$, using an ECal-only model view corresponds to the input set $\mathcal{X} = \mathcal{H}$.

The principal supervised task is a *node classification* task [21] and each selected ECal hit is assigned to the electron contributing the largest simulated deposited energy in that cell. As a simplified starting point, this fixed-multiplicity task is studied with GravNet and Transformer baselines. Because a reconstruction method that is useful in practice should not require a known electron count in advance, this motivated the more

experimental *Transformer for Reconstruction and Attribution of Calorimeter Energy* (TRACE) model, which additionally predicts the valid electron slots in a mixed $2e/3e$ sample, the set of contributors and their energy fractions, drawing inspiration from particle-flow reconstruction.

3.2 Simulation

The samples used in this study were produced with `ldmx-sw`. Its versioned Docker³ container release captures the framework and its software dependencies, including Geant4. The relevant configuration is summarized in Table II; the exact simulation, overlay and job-launch scripts are archived with the thesis code [7].

Table II
SOFTWARE AND EXECUTION CONFIGURATION USED FOR EVENT PRODUCTION.

Item	Configuration
Simulation framework	<code>ldmx-sw</code> container <code>ldmx/pro:v4.7.3</code>
Detector geometry	<code>ldmx-det-v15-8gev</code> with the minimal scoring planes enabled
Beam configuration	One 8 GeV electron generated upstream of the tagging tracker
Execution environment	COSMOS Linux HPC cluster; jobs submitted with Slurm

3.2.1 Single Electron ($1e$) Event Simulations

The single-electron source sample was generated with the configuration in Table II, using the `single_8gev_e_upstream_tagger()` beam generator. No interaction biasing or event-level filter was applied.

The resulting events are referred to as *inclusive*: they were not selected for a particular interaction process. Instead, an electron was delivered from the upstream beam position and its subsequent interactions were determined by the transport simulation. The sample is therefore dominated by electrons without a hard target interaction, while SM interactions remain present at their simulated rates [3]. No dark-sector signal process was injected. For clarity in this report, the electrons drawn from this inclusive sample are called *non-signal electrons*. In the broader LDMX context they may also be called *pile-up electrons* when used as additional “uninteresting” interactions in an overlay. An existing source pool of about 10^9 such events was available for this study.

3.2.2 Multi-Electron ($2e$ and $3e$) Event Simulation

Multi-electron, or *pile-up*, events were constructed by overlaying independently simulated single-electron interactions. The `ldmx-sw` overlay combines their simulated physics and hit contributions by detector channel before the digitisation and reconstruction. The particles are transported independently (since $Z^{(k)} \perp\!\!\!\perp Z^{(\ell)}$), but simulated detector response and reconstruction are repeated after overlay because overlapping detector deposits must be processed jointly.

³A Docker container is a portable software environment that packages an application together with the dependencies needed to run it reproducibly across different systems [34].

The configuration was based on the `ldmx-sw` in-time pile-up validation workflow [35]. Poisson sampling (of electron multiplicity) was disabled: every output contained one main electron and either one or two overlaid electrons. Disjoint input source files were assigned so that no simulated electron was reused in the generated multi-electron sample, a principle of *unique* simulated electrons in every input event were held. The production logic details is found in Appendix 7.1.

Figure 12 provides an example view of this construction. The $1e$ event contains one electron shower, whereas the $2e$ and $3e$ events contain several truth-labelled hit groups in the same readout. Their spatial overlap is the hit-classification problem addressed in this thesis.

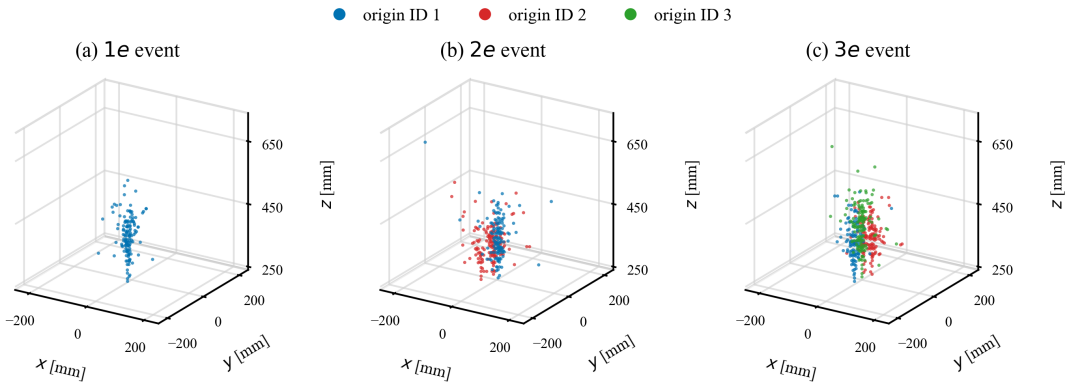


Figure 12. Three-dimensional ECal event displays of randomly selected $1e$, $2e$ and $3e$ events. Each marker represents a selected ECal hit and is coloured by the electron origin with the largest summed deposited-energy contribution to that hit. Noise hits are omitted and only simulation truth is shown; the event displays also represent a would-be perfectly predicted electron shower separation.

The target of 10^7 events for each pile-up multiplicity was selected to provide a large independent pool for model development and the study of infrequent reconstruction failures. The resulting dataset contains 10^7 events of each multiplicity. This sample size is not a direct test of the experiment-level background rejection required by LDMX: those requirements are process-specific and can reach twelve orders of magnitude for backgrounds associated with few-GeV bremsstrahlung [36]. The overlay release included an origin-identifier fix implemented by the author so that source-electron labels remain available after overlay [37]. Table III summarises the generated samples.

Table III
SIZES AND CONSTRUCTION OF THE INCLUSIVE (NON-SIGNAL) EVENT SAMPLES.

Sample	Event construction	Events per file	Total events	Release
$1e$	Inclusive event 8 GeV beam	10^4	$\sim 10^9$	<code>ldmx-sw v4.6.2</code>
$2e$	1 main + 1 overlay electron	10^4	10^7	<code>ldmx-sw v4.7.3</code>
$3e$	1 main + 2 overlay electrons	5×10^3	10^7	<code>ldmx-sw v4.7.3</code>

3.3 Data Extraction and Supervised Targets

This section defines the reconstructed detector quantities supplied to the models and the simulation-only quantities used to construct targets. The simulation and reconstruction output is stored in files with the `.root` extension. ROOT is a C++ data-analysis framework widely used in high-energy physics. Within each LDMX file, a `TTree` stores every event as one entry and organises its quantities into compressed, column-oriented branches. This provides space-efficient storage of the full event record while allowing selected branches to be read efficiently [38–40].

ROOT reading, tensor and target construction, and model training are deliberately kept as separate stages. This is a conscious design choice: the ROOT files retain the broader simulation record, whereas conversion extracts only the quantities required by this study into a reusable model-ready representation. The conversion is therefore performed once rather than repeated for every training run. Besides reducing repeated I/O and preprocessing work, this keeps the target definition independent of the model architecture and allows new models to use the same processed events.

Figure 13 shows the boundary between detector inputs and simulation truth. Exact ROOT fields, reading logic and intermediate data structures are given in Appendix 7.2, together with the complete conversion diagram in Figure 37.

3.3.1 Selected Detector Data and Simulation Truth

Only quantities available from reconstructed detector data are given to the model. The trigger-pad tracks provide context from another subdetector but do not have their own hit-origin targets. Only ECal hits have these targets. The networks still produces a vector of class scores, called *logits*, for every input row. Before the loss and metrics are calculated, the `ecal_mask` selects only the ECal rows. Trigger-pad rows can therefore influence the ECal predictions by context, but their own output scores are ignored.

Detector-cell identifiers are used to match *reconstructed hit quantities* (RecHit) and *simulated hit quantities* (SimHit) in the ECal field. The deposited-energy and origin arrays describe the energy in each stored contribution and the overlay origin to which it belongs. Separate track identifiers describe the simulated tracks. The noise flag is also used only when targets and dataset views are prepared. Truth is therefore used to calculate the supervised loss and evaluation metrics, not as a model input.

Simulation truth is used to construct targets and evaluation quantities, but is never supplied as a model feature, as that would be cheating because these quantities are impossible to know in real life. Detector-cell identifiers match reconstructed and simulated ECal hits, after which deposited-energy and origin arrays determine the supervised targets. Trigger-pad tracks provide event context but are not matched to the individual origin electron labels since that information was not implemented in the `ldmx-sw` simulations, thus not extractable. Event multiplicity is retained as source metadata and, where required, as an evaluation target. The exact input collections and fields are listed in Table XI in Appendix 7.2.

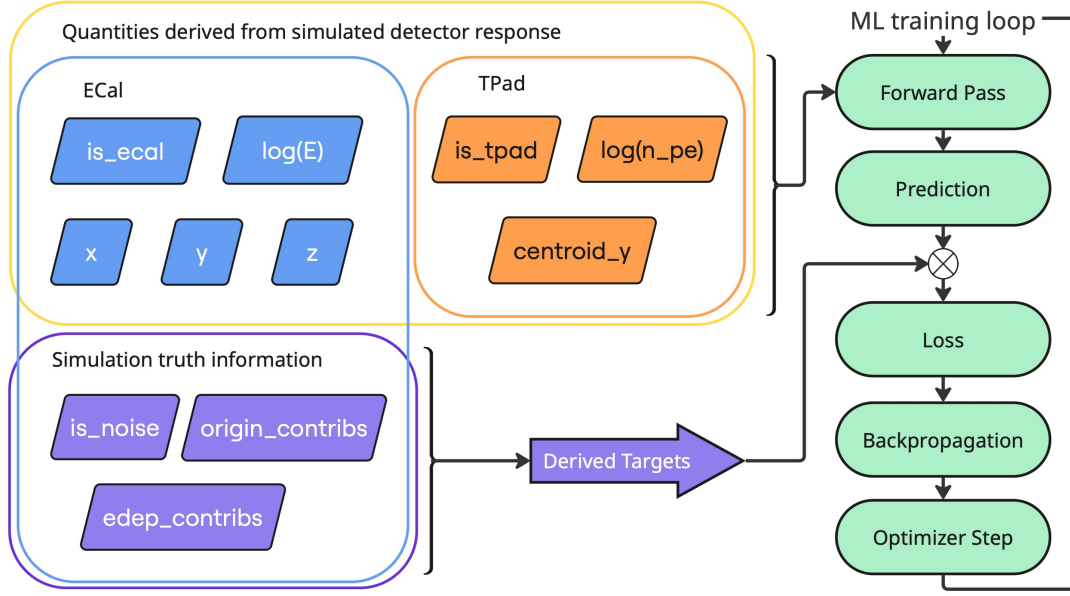


Figure 13. Use of detector-derived and simulation-only information in the supervised pipeline. Detector quantities form the model input, whereas simulation truth is used only to construct training targets and evaluation metrics.

3.3.2 Reconstructed-to-Simulated Hit Matching

Reconstructed and simulated ECal collections can use different row orders. They are therefore matched by detector-cell identifier while preserving the reconstructed-hit order used by the model. A non-noise reconstructed hit must have a corresponding simulated contribution in order to receive an origin target; otherwise conversion stops rather than inventing a label. The chunk-wise reading and intermediate record used for this operation are described in Appendix 7.2.

3.3.3 Target Definition and Baseline Simplifications

The principal task assigns each selected ECal hit to the electron with the largest deposited-energy contribution. To isolate this hit-assignment problem, the baseline models make three simplifying assumptions:

- 1) **Known electron count.** Each fixed-multiplicity baseline is trained and evaluated on either $2e$ or $3e$ events. The number of electron classes is therefore indirectly known before reconstruction and this isolates the hit-assignment task. A perhaps more desirable full-event reconstruction model (especially for online reconstruction) would also be assigned the task of predicting the number of electrons present. This lead to the experimental mixed-multiplicity model that relaxes this assumption and also infers the count.
- 2) **One hard label per hit.** An ECal cell may contain energy from several overlaid showers. For the baseline loss, the hit is assigned to only one origin label in

$\{1, 2, 3\}$, called the *dominant origin*. It is the origin with the largest total deposited energy after contributions from the same origin have been summed.

- 3) **Noise-free baseline input.** Simulated noise hits are kept in the reusable production shards but removed before data are given to a baseline model. The experimental model can instead retain them through a separate background class 0.

3.3.4 Dominant-Origin and Energy-Fraction Targets

One reconstructed hit can contain several simulated energy contributions, including multiple contributions from the same electron. Contributions with the same overlay origin are therefore added first. Let D_{ij} denote the total simulated energy deposited in hit i by electron j .

The dominant-origin target is simply the electron that contributed the most energy:

$$y_i^{\text{origin}} = \text{electron } j \text{ with the largest } D_{ij}. \quad (10)$$

This single class label is used by the baseline classifiers and by the dominant-origin evaluation of TRACE.

The energy-fraction target retains the information discarded by the hard label. For electron j , it is

$$f_{ij} = \frac{D_{ij}}{D_i^{\text{total}}}, \quad (11)$$

where D_i^{total} is the total simulated energy deposited in the hit. The fractions sum to one when the configured origins account for all of that energy. The contributor-set target contains every electron with a stored contribution to the hit. Thus, a hit can have one dominant-origin label while still being represented as shared by several electrons.

An earlier implementation selected the origin associated with the largest individual simulated contribution rather than first combining contributions from the same electron. The summed rule was adopted because the target should depend on the total energy from each physical origin, as that has a more meaningful physical interpretation. This correction produced no measurable change in the reported model performance however.

3.3.5 Noise Handling and Electron Ordering

The production shards keep RecHits that carry the simulation noise flag and mark them with a Boolean noise target. Here, “noise” denotes a category defined by simulation truth, not an estimate of unmodelled noise in detector data. A noise hit is assigned to background class 0, and all of its electron fractions are set to zero. Noise hits are removed before data are given to a baseline model, whereas TRACE can retain them as background. Retained noise hits remain part of the event context, but they are excluded when the set and ordering of electron origins are constructed.

The origin identifiers created during overlay merely distinguish the source electrons; their numerical values have no consistent physical meaning between events. Without a common convention, the same output class could refer to the upper shower in one event and the lower shower in another. The electrons are therefore assigned canonical labels using their transverse y -positions.

For each electron, the ordinary, unweighted mean y -coordinate is calculated from the non-noise ECal hits for which that electron is the dominant origin. The electrons are then sorted from the lowest to the highest mean y -coordinate and assigned consecutive classes: the lowest becomes class 1, the next class 2, and, in a $3e$ event, the highest class 3. The energy-fraction columns are reordered in the same way.

This changes the class labels only. It does not sort the tensor rows, weight positions by energy, use trigger-pad centroids or match trigger-pad tracks to electron origins. The saved shards contain the original physical-origin labels. Canonical- y labels are constructed when the data are loaded. The original origin identifiers and fraction-column order are kept so the mapping can be traced.

3.4 Tensorization and Model-Ready Data

Each temporary event record is converted into numerical arrays, masks, targets and metadata suitable for model training. Events keep their variable size and the stored order within each detector collection.

3.4.1 Combined Event Tensor

Let N_E and N_T be the total number of retained ECal hits and trigger-pad tracks in an event respectively. The ECal rows are placed first and the trigger-pad rows follow:

$$\mathbf{X} \in \mathbb{R}^{(N_E+N_T) \times 8}, \quad \mathbf{x}_n = \begin{cases} (1, 0, x_i, y_i, z_i, \tilde{E}_i, 0, 0), & n = i, \\ (0, 1, 0, 0, 0, 0, c_j, \tilde{n}_j^{\text{pe}}), & n = N_E + j. \end{cases} \quad (12)$$

The first two columns are the row-type flags `is_ecal` and `is_tpad`. Columns 3–6 contain `ecal_x`, `ecal_y`, `ecal_z` and `ecal_energy`; the final two contain `tpad_centroid` and `tpad_pe`. Fields that do not apply to a row are filled with zeros. The type columns and corresponding `ecal_mask` and `tpad_mask` identify the row type. Hit-level targets and losses apply only to the N_E ECal rows, while trigger-pad rows provide context.

An ECal-only model view selects $(x_i, y_i, z_i, \tilde{E}_i)$ from the same stored event rather than using a second processed dataset, so the combined event tensor can be utilized by all trained models. The collection order is preserved in storage, but the row index has no physical meaning for the models. ECal hits and TPad tracks are treated as unordered sets with variable size.

3.4.2 Preprocessing of Energy Input Features

The production shards used in this study apply a log transform of the energy-based features that are present in the input

$$\tilde{E} = \log(1 + \max(E, 0)), \quad \tilde{n}^{\text{pe}} = \log(1 + \max(n^{\text{pe}}, 0)), \quad (13)$$

using the `log1p` function in `PyTorch` [41]. Both input distributions are right-skewed, as is typical for intensity-valued energy features [42]. These transforms reduce their

numerical range while preserving the order of the values, making them numerically more suitable for ML. Deposited energies used only to construct truth targets remain on their original scale. The input distributions before and after transformation are shown in Figure 38 in Appendix 7.3.

3.4.3 Model-Ready Dataset

Converted events are stored in versioned `PyTorch` shards containing the model inputs, detector masks, targets, raw values needed for evaluation and source metadata. Unlike the complete `ROOT` event record, these shards are a task-specific training representation and do not duplicate unused simulation content. Training can therefore read the converted tensors directly without reopening the source files or reconstructing targets, while the retained metadata preserves traceability to the source event. The stored schema, provenance manifests, indexing and caching strategy are documented in Appendix 7.3.

3.4.4 Data Split and Training-Time Preparations

The event indices are shuffled with a fixed seed and divided into training, validation and test sets. The proportions are 80%, 10% and 10% respectively. Only the index lists are split, so the shards do not need to be rewritten. For a baseline model, noise rows are removed and the physical-origin labels are mapped to canonical- y classes when each event is loaded.

After the split, the six continuous columns 3–6 in Equation (12) are standardized using means and standard deviations calculated from the training set only. The two binary detector-type columns are not standardized. The same training statistics are applied to validation and test data and saved with the model run. Scaling is fitted on the common heterogeneous representation before a model-specific view is selected; implementation details are given in Appendix 7.3.

3.5 Model Architectures

3.5.1 Capacity-Selection Strategy

Model capacity was selected with a performance-first strategy. The principal objective was to establish whether the proposed inputs and interaction mechanisms could achieve useful reconstruction performance, rather than to identify the smallest possible network. Widths and depths were therefore chosen conservatively from development runs so that insufficient capacity was unlikely to be the dominant limitation. The scale of contemporary Transformer-based MLPF implementations provided a qualitative reference for a reconstruction model with heterogeneous inputs [6]. The present LDMX task nevertheless has *much* smaller input sets and also a narrower output scope than full CMS particle-flow reconstruction.

This study did not either include a broad hyperparameter or architecture sweep, nor did it attempt pruning, quantisation, knowledge distillation or another model compression procedure. Thus, the reported parameter counts should not be interpreted as

minimum capacities required by the reconstruction tasks. Establishing the accuracy–latency–memory trade-off is left to future work.

The architectures reported below are the configurations used for the million-event training campaigns found in the results. Separate $2e$ and $3e$ models were trained within each baseline family, but their architectures and hyperparameters are identical except for the width of the final classifier: $C = 2$ for the $2e$ task and $C = 3$ for the $3e$ task. Each baseline model used one million fixed-multiplicity events, whereas the contributor-set slot model used one million events balanced between the $2e$ and $3e$ sources not informing the model the multiplicity count beforehand.

3.5.2 GravNet Baseline

The GravNet baseline embeds every eight-feature ECal or trigger-pad node in a 128-dimensional latent space. Four residual `GravNetConv` blocks then learn event-local neighbourhoods and propagate information between nearby nodes in the learned space [4]. A linear per-node head produces the origin logits, of which only the ECal rows are supervised. The graph-assignment vector described in Section 3.6.1 prevents message passing between events. The selected baseline architectures are summarised side by side in Figure 14.

3.5.3 Transformer Baseline

The Transformer baseline maps the same eight input features to a 128-dimensional representation and applies three pre-layer-normalised encoder blocks. Within each event, full self-attention lets every valid token interact with every other valid token and the padding mask prevents padded rows from contributing, and the leading batch dimension is processed independently. A linear token head produces the ECal hit-origin logits. This model adds no sequence positional encoding and the detector coordinates are already included in the input features, and the row order carries no meaning. This set-to-set construction follows the interaction pattern used in recent MLPF models while retaining and restricting it to a single LDMX-specific output task [6]. The selected baseline architectures are summarised side by side in Figure 14.

3.5.4 TRACE: Multi-Task Contributor Reconstruction

The proposed model uses one shared Transformer backbone for the mixed $2e/3e$ dataset. A learned detector-type embedding distinguishes ECal and trigger-pad tokens. Two ECal-supervised heads predict the subset of electron slots contributing to each hit and their energy fractions. In parallel, a mask-aware event summary predicts which of the three electron slots are valid. Coherent post-processing combines these outputs to derive the electron count, dominant hit origin and pure-versus-mixed classification and these are not learned by separate heads. The three training losses: contributor-set, fraction and slot-validity, were assigned equal weight.

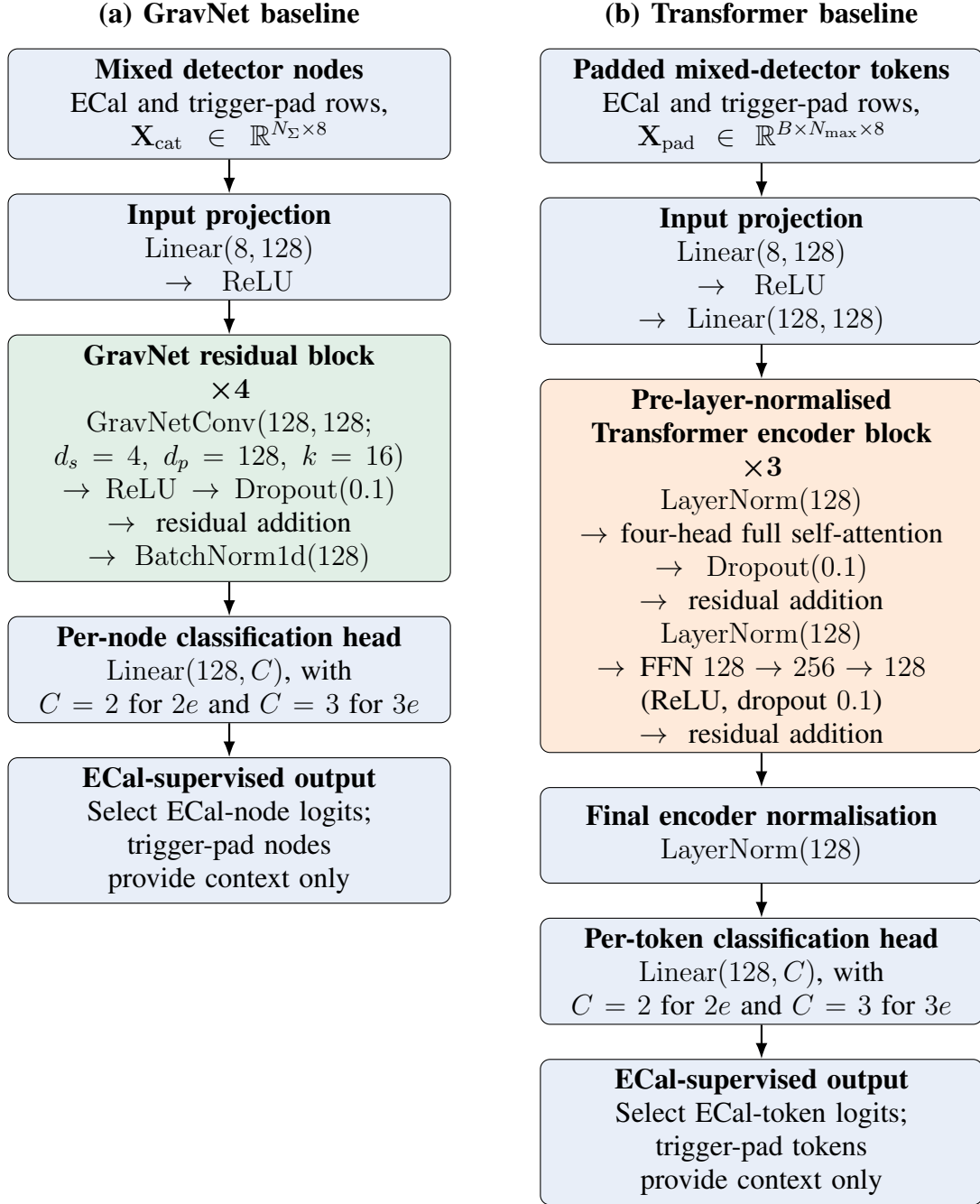


Figure 14. Architectures of the selected (a) GravNet and (b) pre-layer-normalised Transformer baselines. For GravNet, the learned-space dimension is $d_s = 4$, the propagated-feature dimension is $d_p = 128$, and each learned neighbourhood contains $k = 16$ nodes. The Transformer has model width 128, feed-forward width 256, and four attention heads of dimension 32.

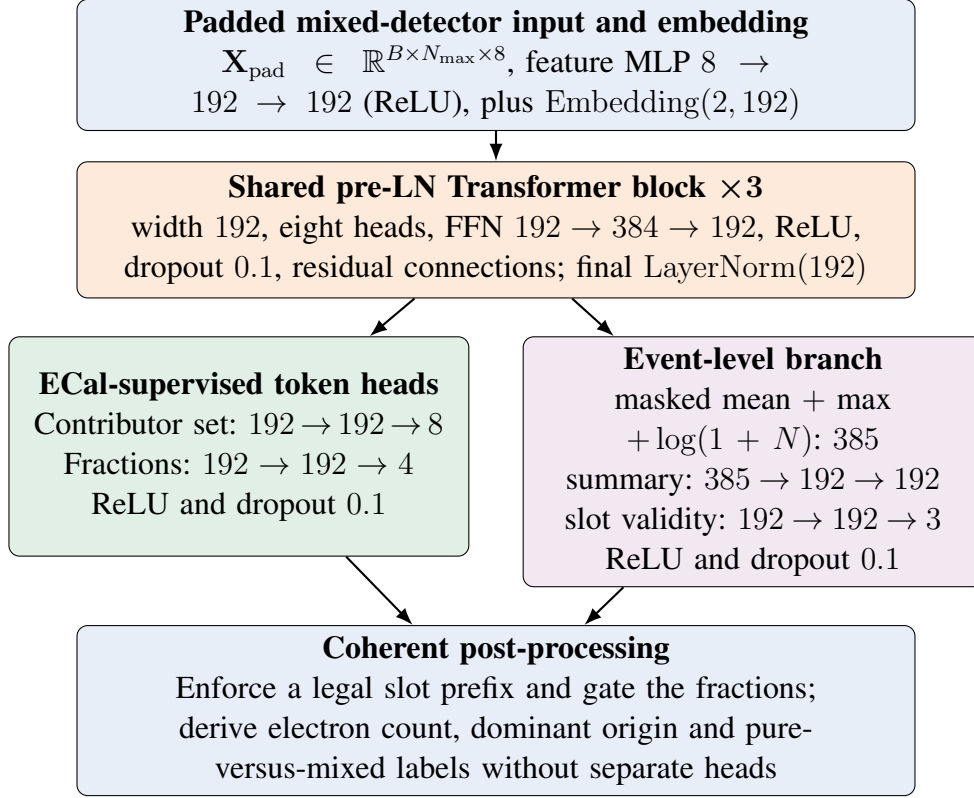


Figure 15. Architecture of the selected MLPF-inspired contributor-set slot model. The eight contributor-set classes encode all subsets of three electron slots; the four fraction components represent background and the three slots.

3.6 Training Procedure

3.6.1 Mini-Batch Construction for Variable-Sized Events

Each mini-batch contains at most B events, where the batch size counts events rather than detector objects. For event b , the model-specific input is a variable-size matrix $\mathbf{X}_b \in \mathbb{R}^{n_b \times F}$. Transformer inputs are padded only to the longest event in the current batch and accompanied by validity and ECal masks. For the Transformer, the leading batch dimension indexes independent event sequences and is not part of the attention contraction and the GravNet inputs are concatenated, with an event-index vector that prevents neighbourhood construction across events. In both cases, trigger-pad rows provide context while the ECal mask selects the supervised outputs. The complete collation and update sequence is shown in Figure 39 in Appendix 7.4.

3.6.2 Training Objectives

For the baseline classifiers, the batch loss shown in Figure 39 is categorical cross-entropy over the retained ECal hits. The contributor-set model instead gives every event

equal weight, though it could be tuned. For event b , its three loss terms are

$$\mathcal{L}_{\text{sup}}^{(b)} = \frac{1}{N_b} \sum_{i \in \mathcal{E}_b} w_{s_{bi}} \text{CE}(\widehat{\mathbf{s}}_{bi}, s_{bi}), \quad (14)$$

$$\mathcal{L}_{\text{frac}}^{(b)} = -\frac{1}{N_b} \sum_{i \in \mathcal{E}_b} \sum_{j=0}^3 f_{bij} \log \widehat{f}_{bij}, \quad (15)$$

$$\mathcal{L}_{\text{slot}}^{(b)} = \frac{1}{3} \sum_{j=1}^3 \text{BCE}(\widehat{v}_{bj}, v_{bj}). \quad (16)$$

Here, $s_{bi} \in \{0, \dots, 7\}$ encodes the subset of three electron slots contributing to hit i , $w_{s_{bi}}$ is a fixed class weight derived from the training split, $f_{bi0}, \dots, f_{bi3}$ are the background and electron energy fractions, and v_{bj} is the validity target for slot j . The optimized mini-batch objective is

$$\mathcal{L}_{\text{batch}} = \frac{1}{B} \sum_{b=1}^B \left(\lambda_{\text{sup}} \mathcal{L}_{\text{sup}}^{(b)} + \lambda_{\text{frac}} \mathcal{L}_{\text{frac}}^{(b)} + \lambda_{\text{slot}} \mathcal{L}_{\text{slot}}^{(b)} \right), \quad (17)$$

with all three coefficients set to one in the reported model. Electron count, dominant origin and pure-versus-mixed labels are derived during post-processing and therefore have no separate training losses.

3.6.3 Optimizer and Regularization

All reported models were trained with AdamW, using an initial learning rate of 3×10^{-4} and a weight-decay coefficient of 10^{-4} . AdamW uses the adaptive first- and second-moment updates of Adam but decouples weight decay from the loss-gradient update. The regularisation is therefore applied directly to the parameters instead of being rescaled by the optimizer’s adaptive gradient statistics [43]. It was selected as a robust general-purpose optimizer for both the Transformer and GravNet campaigns. This was a pragmatic common choice, not the outcome of a dedicated optimizer comparison.

The reported configurations use dropout with probability 0.1 and clip the total gradient L^2 norm to at most 1.0 before each optimizer step. Weight decay and dropout regularise the model, while gradient clipping limits occasional large updates. The baseline runs retain the initial learning rate. For the multi-task slot model, validation-loss plateau scheduling is enabled with a patience of one epoch and a reduction factor of 0.5.

3.6.4 Early Stopping and Checkpoint Selection

The validation loss is evaluated after every training epoch and controls early stopping. The reported campaigns permit at most 15 epochs, require at least five, and stop after three consecutive epochs without a validation-loss improvement of at least 10^{-4} . The test split does not influence this decision.

Two related checkpoint criteria are retained deliberately. A resumable latest checkpoint and an epoch checkpoint are written after each epoch. Separately, the best checkpoint is replaced whenever the validation loss reaches a new strict minimum. The

latter improvement may be smaller than the 10^{-4} threshold used to reset early-stopping patience. Thus, the best available parameters are preserved even when several improvements are too small to justify extending training. Interrupted runs can resume with the optimizer, learning-rate scheduler, data split and early-stopping history restored.

3.7 Evaluation Methodology

Model selection and exploratory failure analysis are performed on the validation split. The test split remains held out for the final comparison of the selected models. Metrics are calculated both after pooling all retained ECal hits and separately for each event. The event-wise quantities are important in this problem because a single aggregate accuracy can conceal a broad distribution containing both easily separated and strongly overlapping showers.

The following math was quickly simplified and rewritten, in line with LK feedback

3.7.1 Hit-Assignment Accuracy

The baseline task predicts one electron label for every retained ECal hit. For a single event, ordinary hit accuracy is simply

$$A_{\text{hit}} = \frac{N_{\text{correct}}}{N_{\text{hits}}}, \quad (18)$$

where N_{correct} is the number of hits whose predicted label matches the target label. Accuracy is calculated separately for every event to show the distribution from easy to difficult events. A pooled accuracy is also reported after combining the hits from all events; this gives events with more hits greater weight.

The electron labels require one additional consideration. Their canonical ordering is useful during training, but the numerical names of the predicted shower groups are not themselves physical. A model can separate the hits into the correct showers while exchanging two complete labels. To distinguish such an exchange from incorrect grouping, every possible *global* label mapping is tested for each event: two mappings for a $2e$ event and six for a $3e$ event. The mapping giving the highest ordinary hit accuracy is retained. Equivalently,

$$A_{\text{perm}} = \max_{\pi} A_{\text{hit}}(\pi), \quad (19)$$

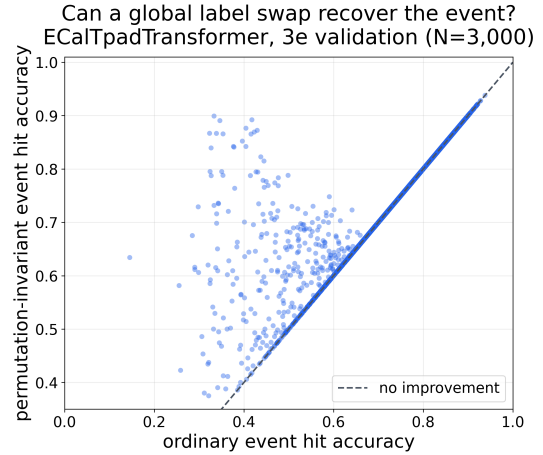


Figure 16. Canonical-label and permutation-aligned accuracy for 3,000 $3e$ validation events. Alignment changes 297 scores (9.9%) and increases the mean from 0.722 to 0.737.

where π denotes one global relabelling of the predicted groups. The same mapping must be used for every hit in the event, so this cannot correct individual classification errors. Unless stated otherwise, the reported hit-assignment accuracies use this permutation-invariant alignment. Training curves remain unaligned because training is performed against the canonical targets.

Figure 16 shows the practical effect of the alignment. Most event scores are unchanged, while a minority improve because one global exchange better matches the truth labels.

A second accuracy gives more importance to hits carrying more reconstructed energy. After applying the same event-level label mapping as above, it is

$$A_E = \frac{\sum_{\text{correct hits}} E_h}{\sum_{\text{all hits}} E_h}, \quad (20)$$

where E_h is the untransformed reconstructed ECal energy. Thus, $A_E = 1$ means that all reconstructed energy is carried by correctly assigned hits. Comparing A_E with the unweighted accuracy indicates whether errors preferentially occur on low- or high-energy hits. Pooled confusion matrices are used separately to show which origin groups are confused.

3.7.2 Prediction Confidence

The model converts its output scores into normalised class scores (logits) with a softmax function. The largest score is called the hit confidence. Normalised entropy for hit h is

$$H_h = -\frac{1}{\log K} \sum_{j=1}^K p_{hj} \log p_{hj}, \quad (21)$$

where p_{hj} is the softmax score for electron class j and K is the number of classes. Dividing by $\log K$, the largest possible entropy, places H_h between zero and one. It is zero when one class receives all weight and one when all classes are equally likely. Mean confidence and entropy are calculated for each event and used as supporting diagnostics of uncertainty and difficult events; neither changes the predicted hit label.

3.7.3 Projected Shower Geometry

To relate performance to physical overlap, every ECal hit is projected onto the transverse x - y plane. For readability, the event index is omitted below. Hit h , with transverse position $\mathbf{r}_h$, contributes the following energy to truth shower j :

$$w_{hj} = E_h f_{hj}, \quad (22)$$

where E_h is the untransformed reconstructed hit energy and f_{hj} is the simulated fraction deposited by electron j . Consequently, a mixed hit contributes to several truth showers in proportion to their deposited energy.

The centroid μ_j is the energy-weighted centre of shower j , and s_j is its root-mean-square transverse width:

$$\mu_j = \frac{\sum_h w_{hj} \mathbf{r}_h}{\sum_h w_{hj}}, \quad (23)$$

$$s_j = \sqrt{\frac{\sum_h w_{hj} \|\mathbf{r}_h - \mu_j\|_2^2}{\sum_h w_{hj}}}, \quad (24)$$

For two showers i and j , their centroid distance d_{ij} is divided by their combined width to obtain the dimensionless separation S_{ij} :

$$d_{ij} = \|\mu_i - \mu_j\|_2, \quad (25)$$

$$S_{ij} = \frac{d_{ij}}{\sqrt{s_i^2 + s_j^2}}. \quad (26)$$

For an event with several electrons, the smallest pairwise value is used:

$$S_{\min} = \min_{i < j} S_{ij}. \quad (27)$$

A small $S_{\min}$ means that at least two showers are close compared with their widths and are therefore strongly overlapping. Geometry is evaluated either over all ECal layers or over an explicitly stated early-layer subset. Because the weights use simulated energy fractions, this is a truth-assisted difficulty measure rather than an observable available on detector data.

3.7.4 Difficulty Profiles and Multi-Task Metrics

Accuracy profiles use approximately equal-population bins in shower separation or model confidence. Means and 95 % confidence intervals are estimated by bootstrap resampling events within each bin. Events are also ranked by permutation-aligned hit accuracy to select lower-tail, median and high-performing examples.

For the multi-task model, the predicted energy fractions are evaluated with the mean absolute error (MAE):

$$\text{MAE}_f = \text{mean}_{h,j} \left| \hat{f}_{hj} - f_{hj} \right|, \quad (28)$$

where f_{hj} and $\hat{f}_{hj}$ are the true and predicted fractions for hit h and output slot j . The mean covers every retained ECal hit and the four fraction components: background and the three possible electron slots. An MAE of zero is perfect; an MAE of 0.10 means that a predicted fraction differs from its target by 0.10, or ten percentage points, on average. The metric is reported both before and after invalid electron slots are masked during post-processing.

The remaining multi-task metrics have direct counting interpretations. Exact contributor-set accuracy requires the complete predicted set of contributing electrons to match the truth set. Dominant-origin accuracy considers only the electron with the largest predicted fraction. Mixed-hit precision is the fraction of predicted mixed hits that are truly mixed,

while recall is the fraction of truly mixed hits that the model identifies. Electron-count accuracy is the fraction of events with the correct number of valid electron slots.

Selected events are inspected with Plotly-based interactive HTML displays that preserve the detector z -coordinate. A menu switches among truth, prediction, correctness, confidence, entropy and energy, while hover labels expose hit-level values. These displays support qualitative checks and failure-mode discovery; an example is shown in Figure 40 in the Appendix. 7.5.

4 Results and Discussion

This chapter first characterises the simulated overlay events and presents the geometric shower-overlap problem at hand. It then evaluates Transformer and GravNet baselines with fixed electron multiplicity before turning to the additional counting, contributor-set and energy-fraction tasks of the experimental model inspired by particle-flow reconstruction. Results and their immediate interpretation are presented together.

4.1 Event Characteristics and the Shower-Overlap Problem

Before comparing reconstruction models, this subsection establishes the detector, longitudinal and transverse scales of the problem that they must solve. Unless stated otherwise, the descriptive figures use deterministic cache prefixes containing 1,000,000 $2e$ events and 1,000,000 $3e$ events.

4.1.1 Detector Occupancy and Longitudinal Shower Development

Figure 17 shows the pooled transverse distribution of reconstructed ECal hits, in detector coordinates. The hits are pooled together along the beam direction flattened into a two-dimensional image. Hits are registered over the entire field, but the highest occupancy and energy density occur near the beam region (which can be reminded is $8\text{cm} \times 2\text{cm}$) located at the center. The width of the highest density seems to be wider than 2cm which is explained by the vertical magnetic field inferred by the dipole magnet that distorts the beam in the horizontal direction due to Lenz's law. The energy density is also comparatively more gravitated to the center compared to the transverse distribution of hits. The energy is weighted to the center because the first hits are more energetic and located near the beam region. As the shower develops through the layers it spreads in the transverse direction but this also corresponds to loss of energy. The discrete pattern of populated bins follows the hexagonal cell geometry and should not be interpreted as a continuous image. On average, the selected $2e$ events contain 247.2 reconstructed hits and 16.01 GeV of reconstructed energy, compared with 354.6 hits and 24.02 GeV for $3e$.

Simulation-flagged noise hits are rare in the production sample. Across the one-million-event mixed-multiplicity sample used for the final contributor model, 27,603 of 300,828,260 retained ECal hits carry the noise flag. This corresponds to 0.0092 %

of the hits, or 0.0276 flagged noise hits per event on average (approximately one per 36 events). The latter is a hit-level rate and should not be interpreted as the fraction of events containing noise. The baseline models and the descriptive shower figures remove these hits, whereas the contributor model retains them as a background class for prediction.

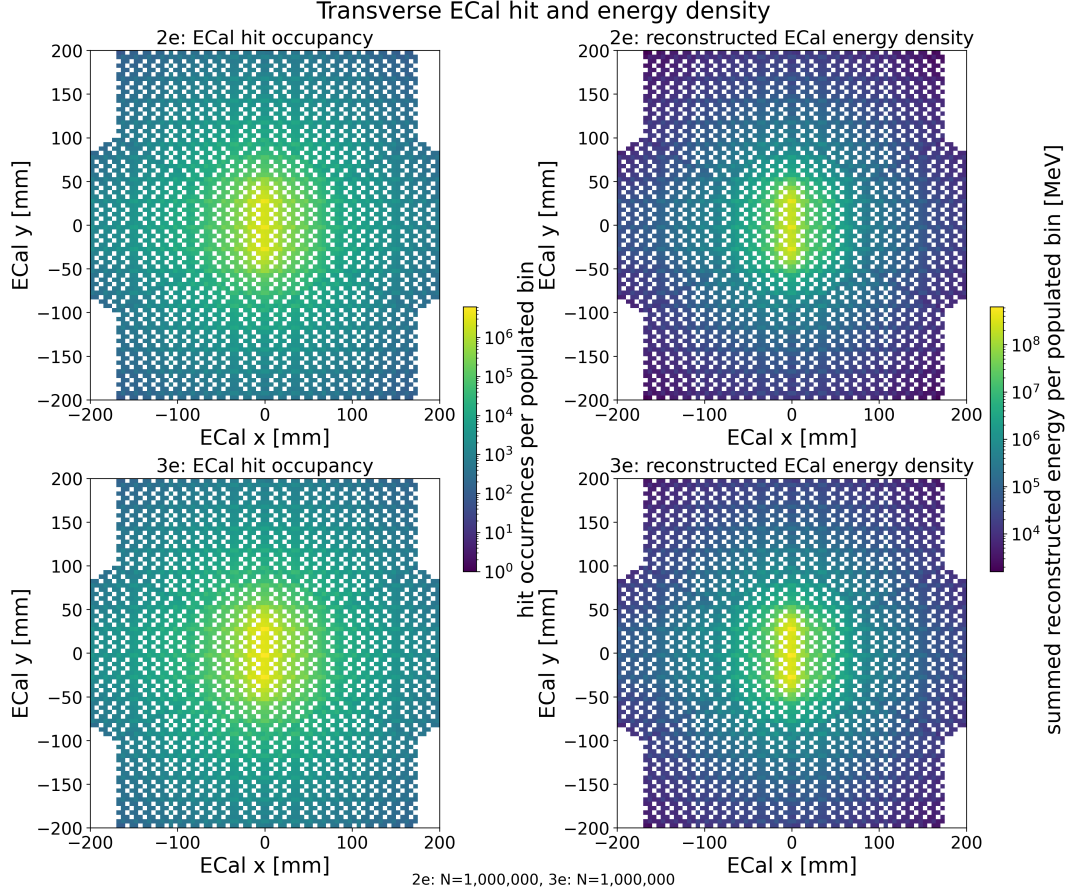


Figure 17. Transverse ECal hit occupancy (left) and summed reconstructed energy density (right) for 1,000,000 $2e$ events (top) and 1,000,000 $3e$ events (bottom). The histograms use the detector x - y coordinates and logarithmic colour scales. White regions are unpopulated spatial bins in this coordinate system, but in reality the hexagonal cell pattern of the ECal fill the entire transverse region.

The same showers are resolved in depth in Figure 18. For both multiplicities, the mean hit count is largest at ECal layer 9, whereas the mean reconstructed energy reaches its maximum at layer 8. These positions correspond to $z \simeq 362$ mm and $z \simeq 343$ mm, respectively. Both quantities subsequently decrease through a long tail in the calorimeter. The $3e$ profiles have larger amplitudes but nearly the same longitudinal shape as the $2e$ profiles. Adding an electron therefore primarily increases the number and energy of simultaneous deposits rather than shifting the characteristic shower maximum. The sizeable event-to-event *root mean square* (RMS) values near the peak also demonstrate that the mean profile is not representative of every individual event.

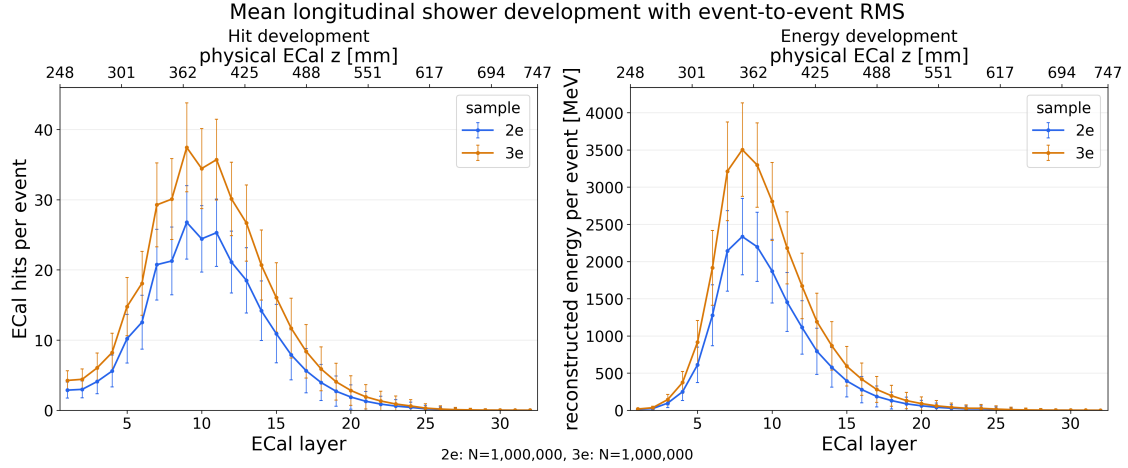


Figure 18. Mean longitudinal ECal hit multiplicity (left) and reconstructed energy (right) per event. Error bars show the event-to-event population RMS at each layer. The lower horizontal axes give ECal layer number and the upper axes give the corresponding physical z coordinate. Each profile contains 1,000,000 events.

4.1.2 Truth-Level Transverse Shower Geometry

The truth-assisted geometry (that we can know only because the data is simulated) is defined in Equations (22)–(27). It weights each hit by reconstructed energy and the simulated contribution fraction, so a mixed hit can contribute to several shower geometries. These quantities are unavailable in measured detector data and are used only to characterise the sample and interpret model performance.

Figure 19 combines the closest pair of resulting shower centroids in each event with the corresponding truth-shower width distributions. The left column resolves separation along the two transverse coordinates and then normalises the two-dimensional centroid distance by the combined radial shower width. The right column shows the individual shower widths in x , y and radius. The effective Molière radius $R_M = 25$ mm provides a reference scale. For $2e$ events, the median minimum separation is $0.253R_M$ (6.3 mm) in x and $0.946R_M$ (23.7 mm) in y . The corresponding $3e$ medians are $0.093R_M$ (2.3 mm) and $0.334R_M$ (8.3 mm). The difference between the x and y distributions is a property of the generated sample rather than a universal detector-resolution limit. Moreover, a $3e$ event contains three distinct combinations of electron pairs instead of one. Taking the closest pair therefore shifts the minimum-separation distribution towards smaller values even if the distribution for a single pair were unchanged.

The width panels separate the scale of an individual shower from the distance between showers. The median transverse widths are independent of event multiplicity: for both samples, $\sigma_x \simeq 0.67R_M$, $\sigma_y \simeq 0.65R_M$ and $\sigma_r \simeq 0.96R_M$. In contrast, the median minimum width-normalised separation from Equation (26) decreases from 0.724 for $2e$ to 0.353 for $3e$. The individual showers therefore have similar transverse dimensions in the two samples, but the showers occupy a substantially more overlapping regime when a third electron is present. This is the main physical explanation for expecting the $3e$ hit-assignment task to be more difficult, as more overlap implies more non-reversible ambiguity that complicates inference, yielding worse predictions.

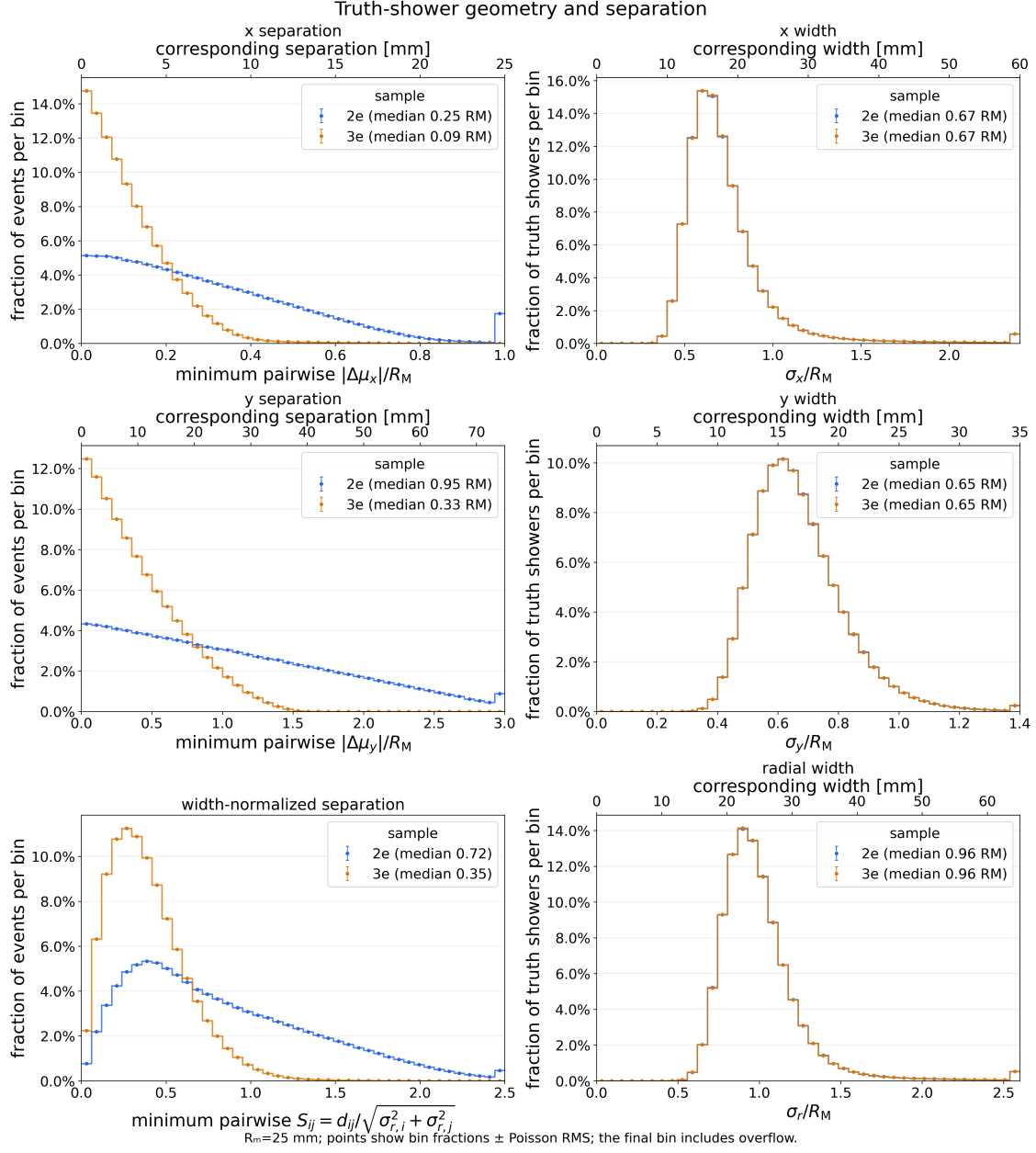


Figure 19. Truth-assisted transverse shower geometry for the 2e and 3e samples. The left column shows the minimum pairwise centroid separation in x and y , followed by the dimensionless minimum width-normalised separation from Equation (26). The right column shows the energy-fraction-weighted truth-shower widths in x , y and radius. Molière-scaled quantities include secondary physical-distance axes; the width-normalised separation is dimensionless. Points and vertical bars show bin fractions and Poisson RMS counting uncertainties, and final bins include overflow. Each sample contains 1,000,000 events.

4.1.3 Trigger-Pad Multiplicity as Event-Level Context

The TPad tracks provide a small set of online reconstructed upstream position and photoelectron measurements in addition to the offline reconstructed ECal hits. Their token multiplicity is already an informative electron-count proxy, but it is not a complete one. Figure 20 shows that the observed multiplicity equals the generated count in 85.7% of $2e$ events and 74.0% of $3e$ events. Most of the remaining $2e$ events contain one token (13.9%), while 23.7% of the $3e$ events contain only two. A further 2.2% of the $3e$ sample contains only one token, whereas overcounting is negligible for both multiplicities, which is an intended result of the design and implementation of the TPad. Across the balanced $2e/3e$ sample, direct equality gives 79.9% electron-count accuracy. The multiplicity therefore carries useful global information, but missing, merged or otherwise unresolved tracks prevent it from determining the electron count by itself. This motivates testing whether a model can combine the TPad observations with the ECal shower pattern, at least in offline reconstruction it is highly expected that a reconstruction model can improve the electron count.

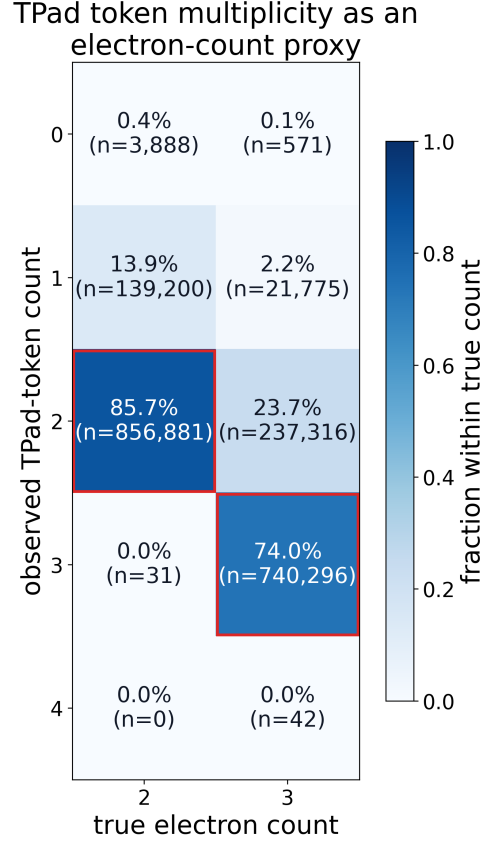


Figure 20. Confusion matrix with observed TPad-track token multiplicity vs true electron multiplicity. Each column contains 1,000,000 events. Red outlines show agreement between the token and generated counts.

4.1.4 Implications of the Event Characteristics

The event statistics establish the physical context for interpreting the model results. Individual shower widths are similar in the $2e$ and $3e$ samples, while the closest-pair separation is substantially smaller for $3e$ events. The greater opportunity for pairwise overlap therefore makes the $3e$ assignment problem intrinsically more crowded. The TPad multiplicity supplies complementary but incomplete event-level context. These observations motivate the performance analyses below, but do not by themselves determine how much of the event structure the models can recover.

4.2 Baseline Models and Establishing a Baseline for Shower Separation

4.2.1 Baseline Architecture and Performance Comparison

The headline baseline comparison uses 1,000,000 selected events for each electron multiplicity: 800,000 training, 100,000 validation and 100,000 test events. Both architectures receive the same log-transformed ECal and TPad-track features and use a hidden width of 128. The Transformer contains three pre-layer-normalised self-attention layers with four attention heads and a final layer normalisation. The GravNet model contains four neighbourhood aggregation blocks, each using $k = 16$ neighbours, with batch normalisation between blocks. Table IV summarises the resulting model sizes and observed training times.

Table IV
ARCHITECTURE AND PRACTICAL PROPERTIES OF THE MATCHED 1,000,000-EVENT BASELINE RUNS. PARAMETER STORAGE ASSUMES 32-BIT FLOATING-POINT VALUES. TRAINING TIME IS THE MEAN OPTIMISATION TIME PER EPOCH AND EXCLUDES VALIDATION. TRAINED ON NVIDIA A100 GPUS.

Model	Core configuration	Parameters	Weights (MB)	Checkpoint (MB)	Batch size	Train time/epoch $2e/3e$ (min)
Transformer	3 layers, 4 heads, $d = 128$	0.416 M	1.66	9.92	64	38.3/40.1
GravNet	4 blocks, $k = 16$, $d = 128$	0.268 M	1.07	8.15	32	46.8/49.7

As described in Section 3.5.1, these models were selected as performance-oriented reference configurations rather than through an exhaustive capacity search. The comparison below therefore evaluates the implemented architectures, not necessarily an optimal accuracy-parameter or accuracy-latency frontier. Simply put, these hyperparameter choices were the best found for the scope of this thesis, and one extra block was given to the GravNet model since it has a lower parameter count per layer compared to the transformer. In particular, the use of contemporary MLPF models as a qualitative scale reference does not imply matched capacity: the CMS particle-flow problem has a substantially larger size of input, different inputs, outputs and reconstruction scope. No pruning or other sophisticated compression study was performed.

The GravNet baseline has approximately 36 % fewer trainable parameters and correspondingly smaller model weights and checkpoints. In these runs, however, this did not translate into shorter epochs: Transformer optimisation took approximately 38–40 minutes per epoch, compared with 47–50 minutes for GravNet both training on NVIDIA A100 GPUs. The learned neighbour search and aggregation are more expensive than the parameter count alone suggests. This timing comparison remains descriptive rather than controlled because the Transformer used a batch size of 64 and GravNet used 32, though separate testing not presented here has not indicated that increasing the batch size of the GravNet to 64 would change this fact.

Table V compares the checkpoints selected by minimum validation loss on the held-out test sets. All assignment metrics use one optimal event-wide permutation of the predicted electron labels. Both architectures reach similar performance, while the $3e$ task is consistently more difficult than the $2e$ task. Energy weighting increases every

accuracy, showing that the misassigned hits carry less reconstructed energy on average than the correctly assigned hits.

Table V
PERMUTATION-ALIGNED PERFORMANCE OF THE MATCHED BASELINE MODELS ON THEIR 100,000-EVENT TEST SETS. ENERGY-WEIGHTED ACCURACY IS POOLED OVER THE RECONSTRUCTED HIT ENERGY. ALL VALUES ARE PERCENTAGES.

Multiplicity	Model	Pooled hit	Mean event	Median event	Energy-weighted
2e	Transformer	84.20	83.73	85.90	89.76
2e	GravNet	83.79	83.32	85.77	89.26
3e	Transformer	74.84	74.40	75.82	83.18
3e	GravNet	74.01	73.55	75.14	82.13

Because both models were evaluated on the same test events, their event-wise results can be compared directly. Table VI reports paired bootstrap intervals for this comparison. The Transformer has an advantage of 0.417 percentage points in mean 2e event accuracy and 0.850 percentage points for 3e. The corresponding energy-weighted differences are 0.505 and 1.048 percentage points. All four intervals exclude zero. The large test samples therefore resolve a consistent Transformer advantage for these trained models, but its magnitude remains about one percentage point or less. Moreover, the confidence intervals describe finite-event uncertainty for fixed trained models and it is important to underline that they do not include variation between random initialisations or training samples. The results support similar practical performance with a small advantage to the Transformer in this campaign, but this is not an exhaustive study and cannot with certainty be interpreted as an universal ranking of the two architectures.

Table VI
PAIRED TRANSFORMER-MINUS-GRAVNET DIFFERENCES IN PERMUTATION-ALIGNED METRICS ON THE COMMON 100,000-EVENT TEST SAMPLES. DIFFERENCES AND PERCENTILE-BOOTSTRAP 95 % CONFIDENCE INTERVALS ARE IN PERCENTAGE POINTS. THE BOOTSTRAP RESAMPLES COMPLETE EVENTS.

Multiplicity	Event metric	Difference	95 % CI
2e	Mean event hit accuracy	+0.417	[+0.403, +0.431]
2e	Mean event energy-weighted accuracy	+0.505	[+0.484, +0.525]
3e	Mean event hit accuracy	+0.850	[+0.834, +0.866]
3e	Mean event energy-weighted accuracy	+1.048	[+1.023, +1.072]

Since the two baselines were observed to behave very similarly in the coming study, and the Transformer gives the higher headline accuracy, the following detailed diagnostics use it as the representative fixed-multiplicity model instead of duplicating every plot for GravNet. They use the best Transformer checkpoints and the held-out 100,000-event test sets from the same million-event campaigns.

4.2.2 Representative Training Behaviour

Figure 21 shows the training history of the 2e pre-layer-normalised Transformer trained on 1,000,000 events. Of these, 800,000 were used for training and 100,000 for validation, in line with the 80:10:10 split. Each displayed point follows one complete epoch and

therefore does not include the untrained model. The same qualitative convergence pattern was observed for the other fixed-multiplicity baselines.

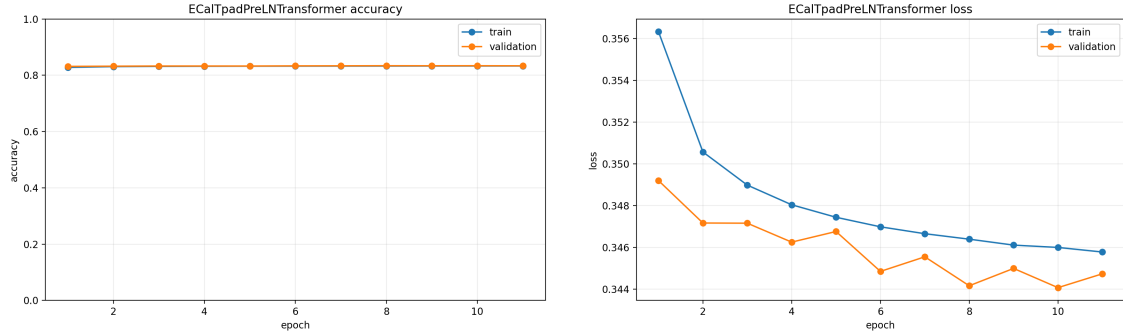


Figure 21. Training history for the 1,000,000-event $2e$ pre-layer-normalised Transformer. The left panel shows pooled native-label hit accuracy and the right panel shows cross-entropy loss for the 800,000-event training and 100,000-event validation sets. Validation-loss early stopping ended training after epoch 11 and the best validation loss was extracted from epoch 10.

The native-label accuracy is already 0.828 for training and 0.832 for validation after the first epoch. By epoch 11, the training accuracy has increased to 0.833, while the highest validation accuracy is 0.834 at epoch 8. The accuracy curve is clearly almost flat on the displayed scale. The loss gives a more sensitive view of the remaining optimisation: the training loss decreases from 0.3563 to 0.3458, a change of only 0.0106, and the validation loss decreases from 0.3492 to its minimum of 0.3441 at epoch 10. The small subsequent change in accuracy shows that most class decisions made by the final model were already established after the first recorded epoch and later epochs mainly adjusted the predicted class probabilities.

Early stopping monitored validation loss, with a minimum of five epochs, a patience of three epochs and a minimum improvement of 10^{-4} . The run stopped after epoch 11 rather than reaching the maximum of 15 epochs, and the epoch-10 parameters were retained as the best checkpoint. The validation loss is slightly lower than the training loss throughout. This is consistent with dropout being active while training metrics are accumulated, but disabled for validation, which is evaluated once after each epoch with fixed parameters. Smaller 10,000-event runs showed similar qualitative behaviour and were therefore useful for development, while all headline comparisons use the million-event campaigns to make sure the models have proper exposure to the input data.

4.2.3 ECal Hit Assignment Performance

The event-wise distributions in Figure 22 show that reconstruction quality varies substantially between events. The median (permutation-aligned) event hit accuracy is 0.859 for $2e$ and 0.758 for $3e$ events. The corresponding event means are 0.837 and 0.744, while the pooled hit accuracies are 0.842 and 0.748 (Figure 23). The additional electron therefore clearly makes the assignment task more difficult, but the broad distributions also show that electron multiplicity is not the sole factor for whether an event is reconstructed well or not.

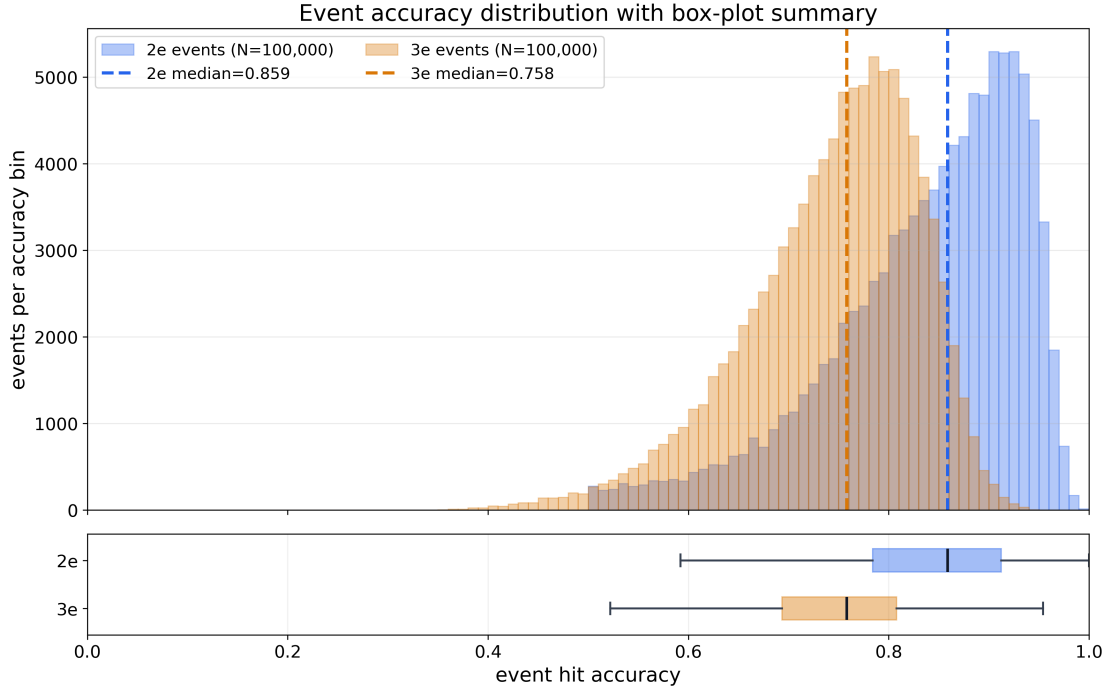


Figure 22. Distribution of permutation-aligned event hit accuracy for the 100,000 $2e$ and 100,000 $3e$ test events. Every event contributes equally to the histogram and box-plot summary. Dashed lines indicate the sample medians.

The confusion matrices in figure 23 show *pooled* hit accuracy, which means we are inspecting the model performance for predicting individual hits rather than whole events, and the accuracy is still reported as permutation invariant where the global swap that yielded the highest accuracy classifying the electron into groups is considered. The predicted group is still reported in the original canonical ordering and from the $3e$ events it is apparent that the performance is weaker in predicting the origin electron with the label 2. This is not a coincidence as this electron shower is the one with a centroid in between the other two in the y -direction. This explains why the accuracy is lower for the hits that belong to this shower, it is placed in between two showers and it should thus have more overlap and hits that are in between the showers have higher ambiguity. The confusion also show that for incorrect label predictions, the origin electron label is most likely to be confused with the nearest electron shower. The confusion matrices implies that closeness of the showers makes them harder to separate, this will be studied in subsequent parts.

The pooled hit accuracy across all layers of the ECal does not tell the whole story, as it is evident that there is a decrease in the last twelve ECal layers, visible for both multiplicities. The pooled accuracy is 0.705 for $2e$ and 0.565 for $3e$, compared with 0.844 and 0.751 in layers 1–20, and these are very close to the accuracy across all layers, mainly because most hits occur in this grouping. Layers 21–32 contain only 1.45 % and 1.51 % of the respective test-set hits, so their lower accuracies does not have the same statistical weight as the all-layer result, but there are enough to state that the models perform worse in the layers furthest in the ECal. This is likely due to the fact that the

hit assignment becomes more difficult further along the beam direction where electron showers have developed and widened increasing their overlap and increasing ambiguity. Layers 1–20 were chosen since they are also the depth used by the trigger-level ECal readout, although that system uses coarser trigger cells rather than the full-resolution hits evaluated here in this offline reconstruction analysis. These results lead to the next subsection where we investigate performance along shower depth further.

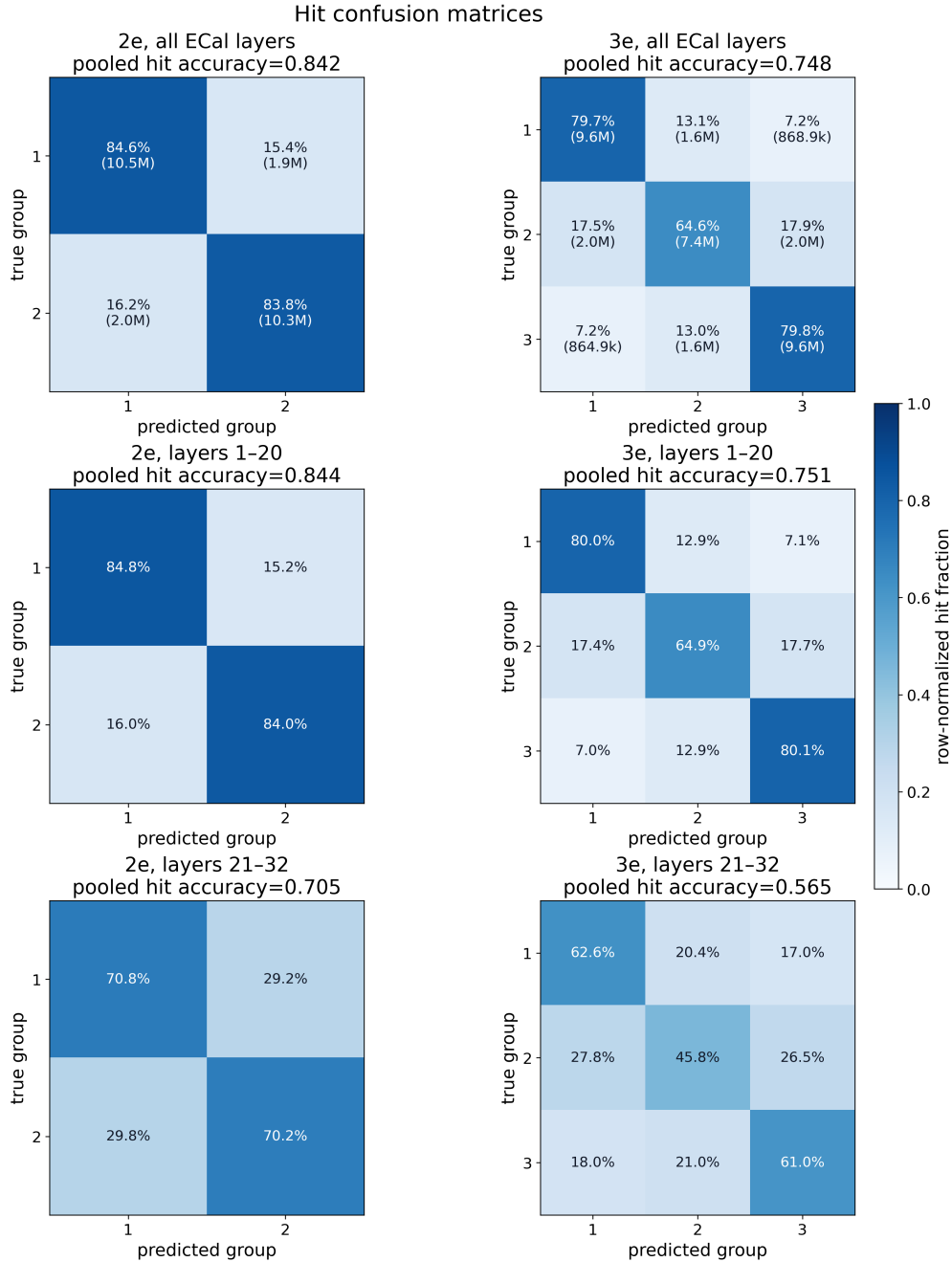


Figure 23. Row-normalised hit confusion matrices after one event-wide label alignment. Rows are true groups and columns are predicted groups. The top row uses all ECal layers, the middle row uses layers 1–20 and the bottom row uses layers 21–32. Percentages are conditional on the true group, and counts are pooled over test-set hits.

4.2.4 Performance Across the Electron Shower Depth

Figures 24 and 25 resolve hit-assignment performance by ECal layer and therefore along the depth of the electron showers. The mean hit accuracy decreases with ECal depth for both event multiplicities, with the $3e$ sample remaining below the $2e$ sample. The late-layer fluctuations and confidence intervals should be interpreted with some caution because fewer event-layer observations remain there. From the figure one can see that at this depth the model is almost always completely accurate 100% within a layer or completely inaccurate 0% within a layer where the rectangular bins are filled at the top or bottom of the figure. This pattern arises also due to the reason that there are few hits at this depth and most often there are only single hits causing this pattern where there are empty rectangular bins in between. For the $2e$ plot there is a horizontal line of rectangular bins at 50% and for the $3e$ plot there are horizontal lines of rectangular bins at 33% and 67% caused by the same combinatorics.

Over all layers, the pooled and *energy-weighted* accuracies are 0.898 for $2e$ and 0.832 for $3e$, both higher than their corresponding unweighted hit accuracies. Within the test samples, incorrectly assigned hits therefore carry less reconstructed energy on average than correctly assigned hits. The depth dependence nevertheless remains, showing that energy weighting does not remove the loss of separation quality in the later shower region. The rectangular bin pattern explained earlier is partly lost here since deposited energy are float values, smearing the accuracy values across the vertical direction so most bins are filled.

4.2.5 Dependence on Shower Separation

The following plots relate event accuracy to the truth-assisted shower geometry defined in Section 3.7. Figure 26 uses the smallest pairwise distance between the energy-fraction-weighted truth-shower centroids. Dividing by the reference Molière radius $R_M = 25$ mm supplies some interpretable detector scale, but the line at one Molière radius is not a hard resolution boundary. For these plots presented, the $2e$ plots have only empty rectangular bins below 50% since the global swap of the permutation invariant accuracy guaranties that a model guess cannot be made worse than this (imagine a situation where all hits were assigned the same label), and the $3e$ plots have for the same reason only empty rectangular bins below 33%.

The event mean rises as the minimum centroid distance increases. The trend is present for both multiplicities and is consistent with shower overlap being a major source of assignment ambiguity. The sizeable accuracy spread at a fixed distance also shows that centroid separation is not a complete description of event difficulty.

Figure 27 instead divides the minimum centroid distance by the combined energy-fraction-weighted RMS widths of the two showers, attempting to also capture the shower spread as a factor that factors prediction difficulty. This produces a dimensionless overlap measure in which values below unity indicate that the centroid distance is smaller than the combined transverse shower-width scale.

The profiles again show increasing accuracy with separation. The broad vertical distributions remain important: even after normalising by shower width, events with

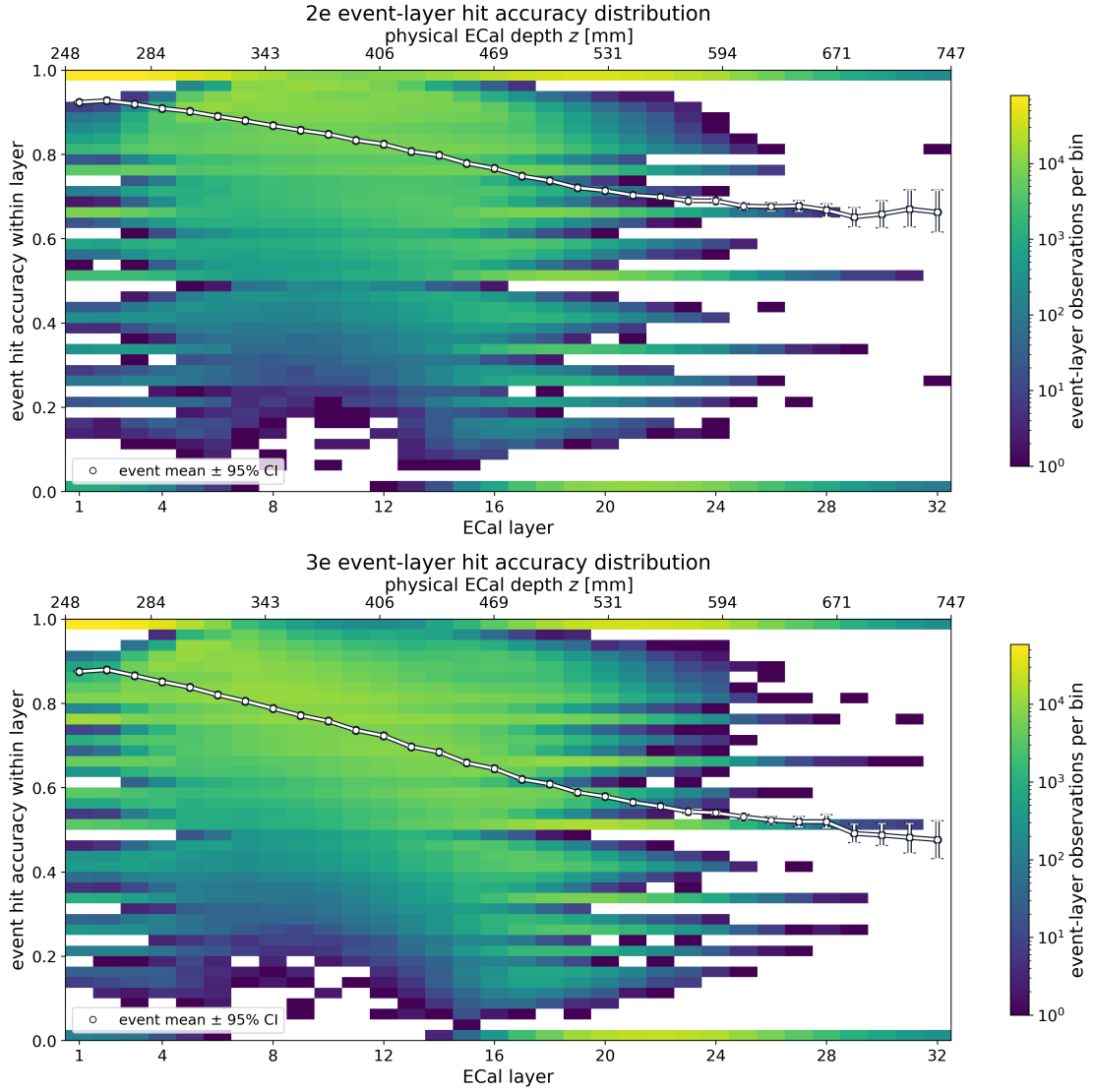


Figure 24. Distribution of permutation-aligned event-layer hit accuracy for 2e events (top) and 3e events (bottom). Colour shows the number of event-layer observations in each rectangular bin on a logarithmic scale. The colour scale starts at one observation. White points show event-balanced means with 95% confidence intervals. The lower horizontal axes give the ECal layer number, while the upper axes give the corresponding physical ECal depth z in millimetres.

similar geometric separation can have substantially different assignment accuracy. The observed relation is therefore treated still as a correlation with overlap, not as a fitted functional law. Generally it also seems that the 3e events have a broader vertical distribution compared to the 2e events.

4.2.6 Confidence, Entropy and Selective Performance

Next up metrics that are rather unique for ML-based models, compared to rule-based ones, shall be investigated and it will be explored how these could potentially be exploited, if one desires to artificially boost accuracy or if one is OK with omitting some

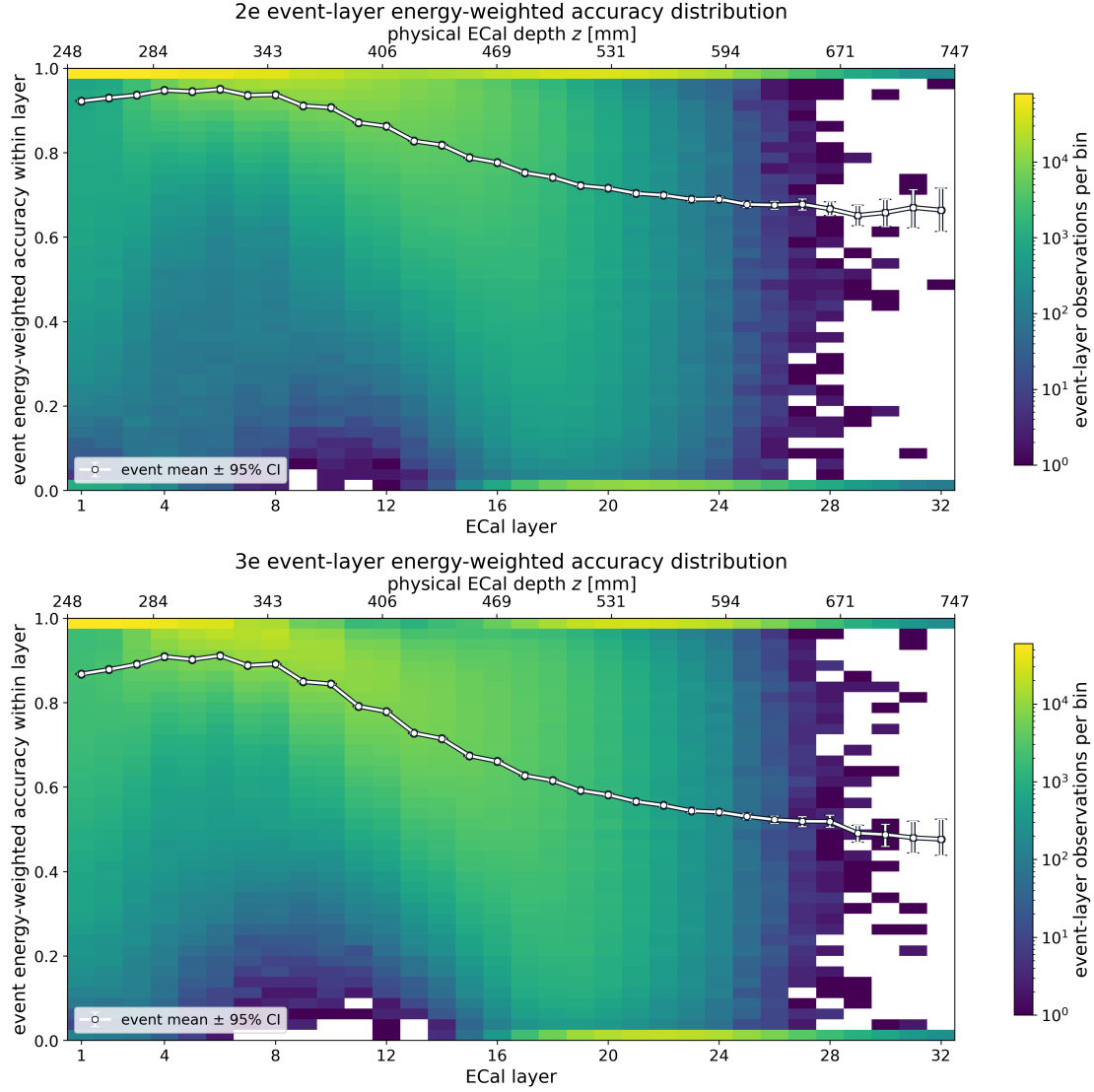


Figure 25. Distribution of raw-energy-weighted permutation-aligned accuracy at event-layer for $2e$ events (top) and $3e$ events (bottom), using the same event-wide label alignment as Figure 24. The colour scale starts at one observation. White points show event-balanced means with 95 % confidence intervals. The lower horizontal axes give the ECal layer number, while the upper axes give the corresponding physical ECal depth z in millimetres.

hits. Confidence and entropy are based on the logit outputs of the baseline models. The logit outputs are bounded between $p \in [0, 1]$ and they provide a metric that could be interpreted as a probability, though these two does not have a perfect correspondence and they should not be confused with each other. *Confidence* in Figure 28 is defined as the mean max-softmax confidence, i.e the largest logit value in the vector that holds the logits corresponding to each electron label, and then calculating the mean over the whole event. The plot shows that high mean hit confidence is associated with high event accuracy, while Figure 29 shows the complementary negative relation between normalised prediction entropy and accuracy. Entropy is perhaps the more interesting metric of "confidence" as it captures the spread of logits of the different

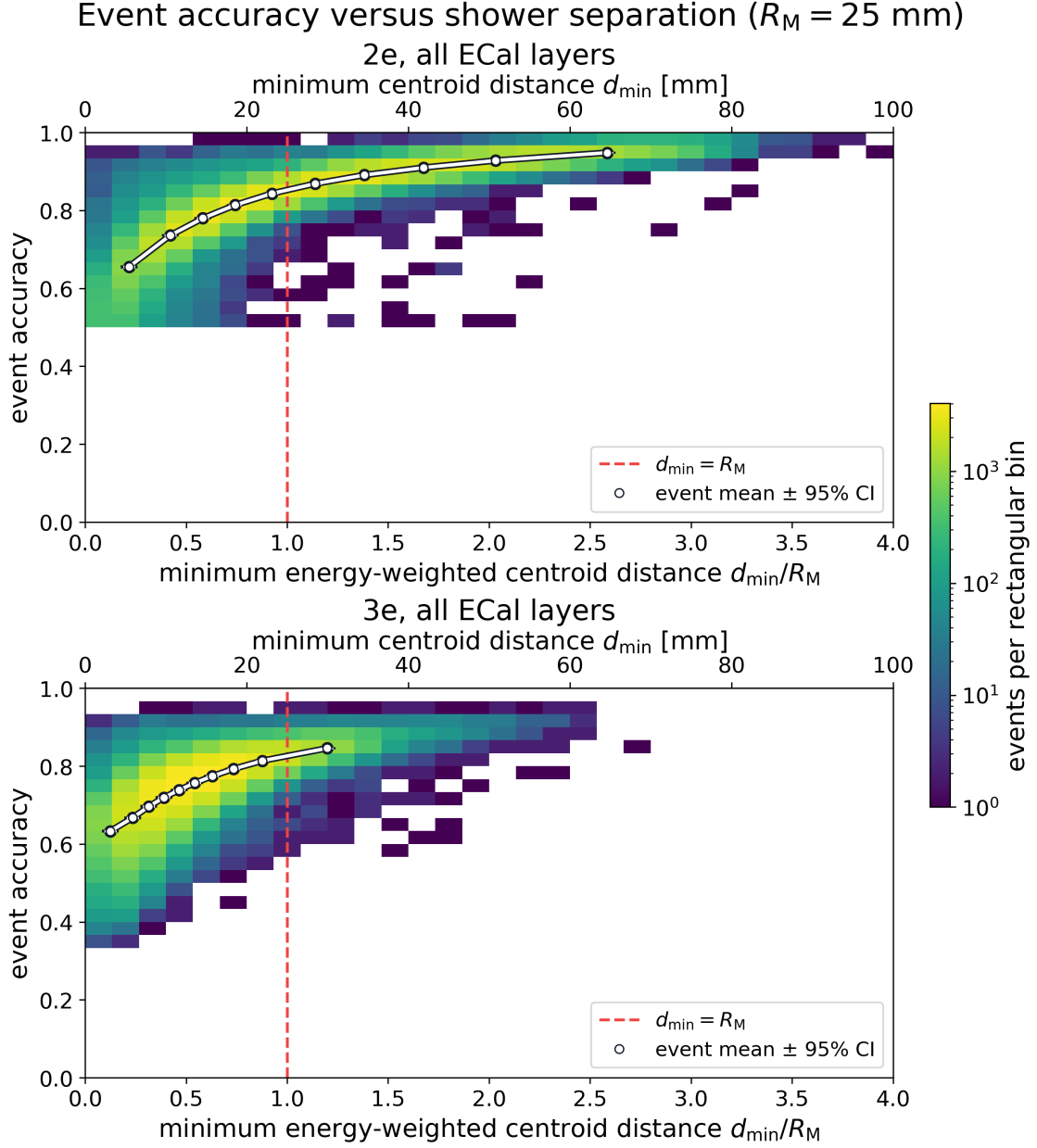


Figure 26. Permutation-aligned event hit accuracy as a function of the minimum energy-fraction-weighted truth-centroid separation in units of $R_M = 25$ mm, for 2e events (top) and 3e events (bottom). White points show equal-population bin means with 95 % confidence intervals. The red line marks one Molière radius as an interpretive scale. The lower horizontal axes show $d_{\min}/R_M$, and the upper axes show the corresponding centroid distance $d_{\min}$ in millimetres.

Event accuracy versus energy-weighted width-normalized separation

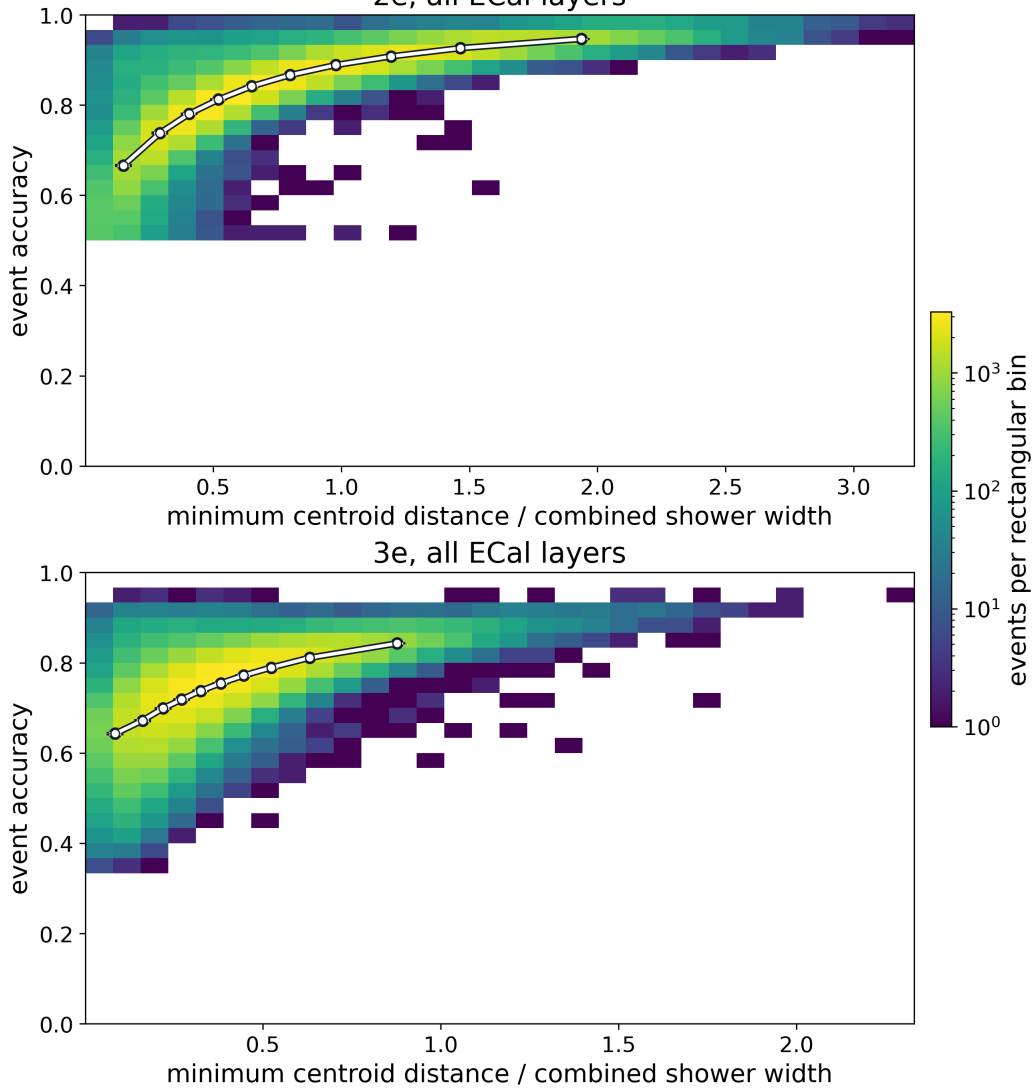


Figure 27. Permutation-aligned event hit accuracy as a function of minimum energy-fraction-weighted truth-centroid distance divided by combined shower width, using all ECal layers. Results for $2e$ events are shown above those for $3e$ events. White points show equal-population bin means with 95 % confidence intervals.

labels. Both summaries can therefore rank difficult events, although neither relation by itself establishes probability calibration. One also has to be somewhat careful with these metrics as they compare accuracy against a model derived metric, a metric only indirectly based on the actual properties of the event. This shown correlation is perhaps not surprising, if this pattern was not apparent the models would not be able to make successful predictions these go hand in hand, but these metrics could be interesting to explore deeper in how they could be exploited if a practical application arises.

Figure 30 makes the corresponding selection trade-off explicit. Retaining only high-confidence hits increases their accuracy, but reduces hit coverage. The retained energy fraction decreases more slowly than the retained hit fraction, indicating that many of

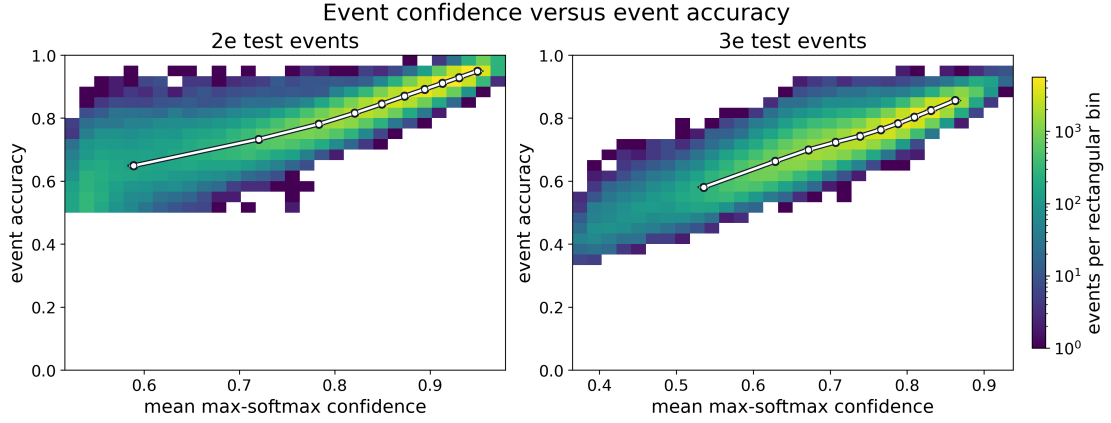


Figure 28. Mean hit confidence against permutation-aligned event hit accuracy for $2e$ events (left) and $3e$ events (right). Colour shows event density on a logarithmic scale beginning at one event. White points show equal-population bin means with 95 % confidence intervals.

the first hits removed carry less reconstructed energy. Selecting complete events by confidence gives a smaller but consistent increase in mean event accuracy as coverage is reduced.

The layer profiles in Figure 31 show that mean confidence and mean accuracy follow each other closely through most of the ECal. Their separation and uncertainty grow in the sparsely populated final layers, where the per-layer estimate is least stable. Agreement between the two layer means does not by itself establish hit-level probability calibration.

4.2.7 Trigger-Pad Ablation

A paired inference-time ablation was performed with the best checkpoint of the final $3e$ baseline Transformer, specifically choosing the $3e$ events since these are the hardest to separate showers and where potentially useful information outside of the ecal can help the most. The intact model and the same model with every TPad-track token removed were evaluated on the identical 100,000 held-out test events. Table VII reports permutation-aligned results, consistently with the other baseline hit-assignment metrics.

Table VII

PAIRED INFERENCE-TIME TPad ABLATION FOR THE FINAL $3e$ PRE-LAYER-NORMALISED TRANSFORMER ON ITS 100,000-EVENT TEST SET. ASSIGNMENT METRICS ARE PERMUTATION ALIGNED. DIFFERENCES ARE ECal-PLUS-TPAD MINUS THE SAME MODEL WITH ALL TPad TOKENS REMOVED, IN PERCENTAGE POINTS. INTERVALS ARE PAIRED 95 % PERCENTILE INTERVALS FROM 2,000 EVENT-BOOTSTRAP RESAMPLES.

Metric	ECal+TPad (%)	TPad removed (%)	Difference (percentage points)
Pooled hit accuracy	74.844	74.726	+0.119 [0.113, 0.124]
Mean event accuracy	74.396	74.272	+0.125 [0.119, 0.130]
Energy-weighted accuracy	83.274	83.032	+0.243 [0.231, 0.255]

Accuracy with the intact input exceeds the ablated evaluation by 0.119 percentage points for pooled hit accuracy, 0.125 points for mean event accuracy and 0.243 points for energy-weighted accuracy. All three paired bootstrap intervals exclude zero, showing

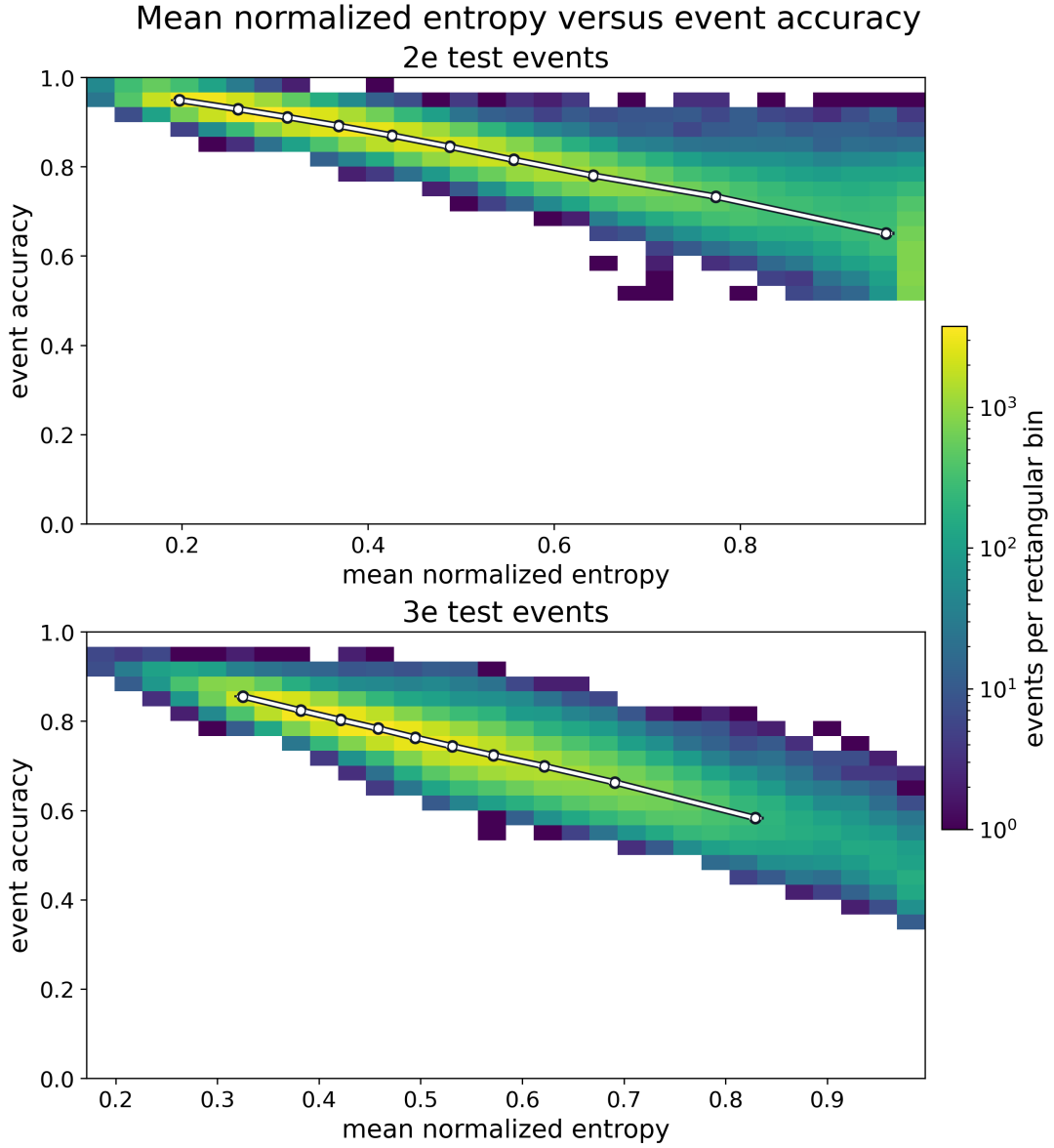


Figure 29. Mean normalized prediction entropy against permutation-aligned event hit accuracy for $2e$ events (top) and $3e$ events (bottom). White points show equal-population bin means with 95 % confidence intervals.

that the measured differences are statistically resolved for this fixed checkpoint and test sample. Their magnitude is nevertheless small: the largest difference is 0.243 percentage points, and the median event-level accuracy change is zero.

The ablation therefore demonstrates sensitivity to removing the TPad tokens, but it cannot establish that useful detector information contained in those tokens caused the difference. Removing an entire modality only at inference time changes the model’s input distribution and attention context. A controlled comparison between separately trained ECal-only and ECal-plus-TPad models, preferably across several random seeds, would be required to isolate the added value of TPad information. The confidence intervals quantify uncertainty for this specific checkpoint and they do not include

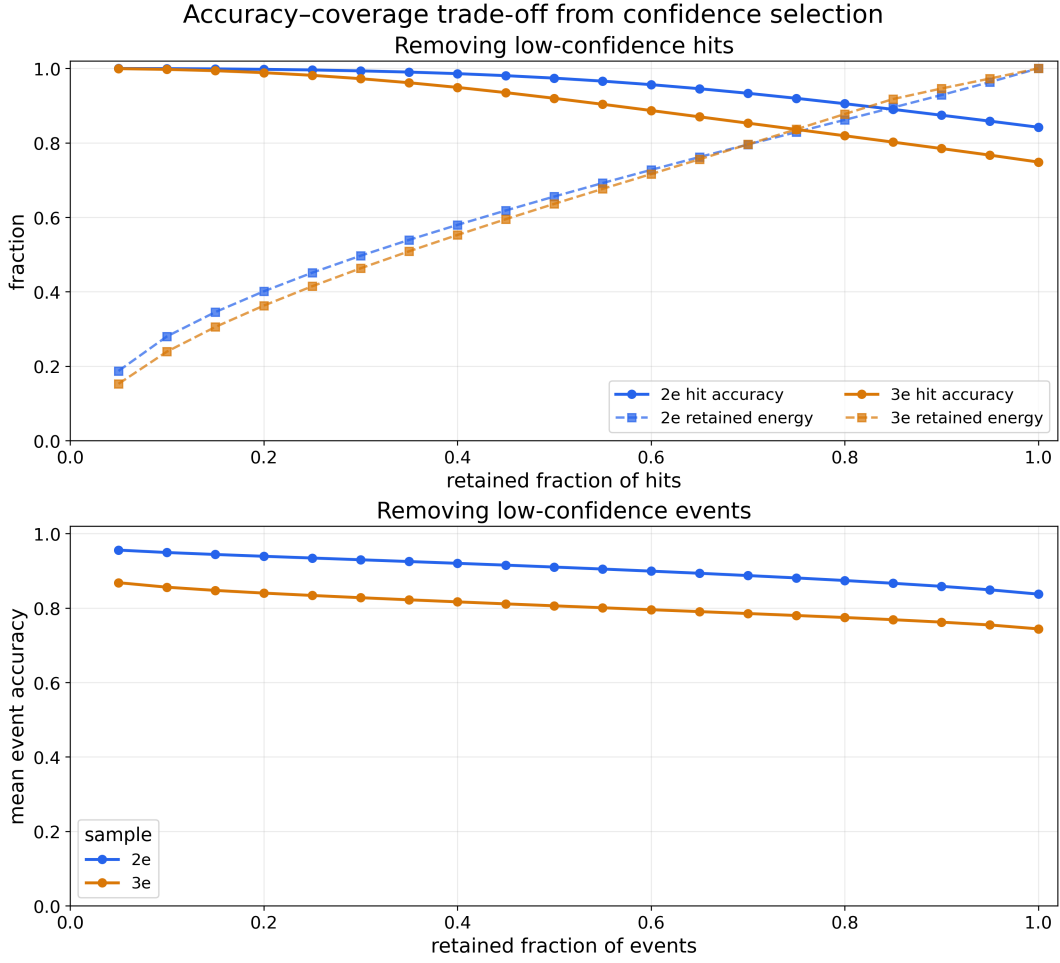


Figure 30. Accuracy-coverage trade-off obtained by ordering predictions by confidence. The upper panel removes low-confidence hits and shows both the accuracy and retained raw-energy fraction. The lower panel removes complete low-confidence events and shows the mean accuracy of the retained events.

variation from retraining with different random seeds or samples. The result is also specific to a $3e$ task whose electron count is supplied in advance.

4.3 TRACE Multi-Task Reconstruction

4.3.1 Multi-Task Reconstruction Performance

TRACE extends the fixed-multiplicity assignment problem to a particle-flow-like multi-task setting. It was trained on 500,000 $2e$ and 500,000 $3e$ events. The combined sample was divided into 800,000 training, 100,000 validation and 100,000 test events. With canonical ordering, $2e$ has roughly an equal distribution of ECal hits with labels 1 and $3e$ events has also roughly an equal distribution of hits labels 1, 2 and 3. This means that the dataset has a lower number of ECal hits with the label 3 and it means that all the hits with that specific label comes from a $3e$ event, while the hits with label 1 and 2 is mixed from both $2e$ and $3e$ events. Results use the epoch-15 checkpoint

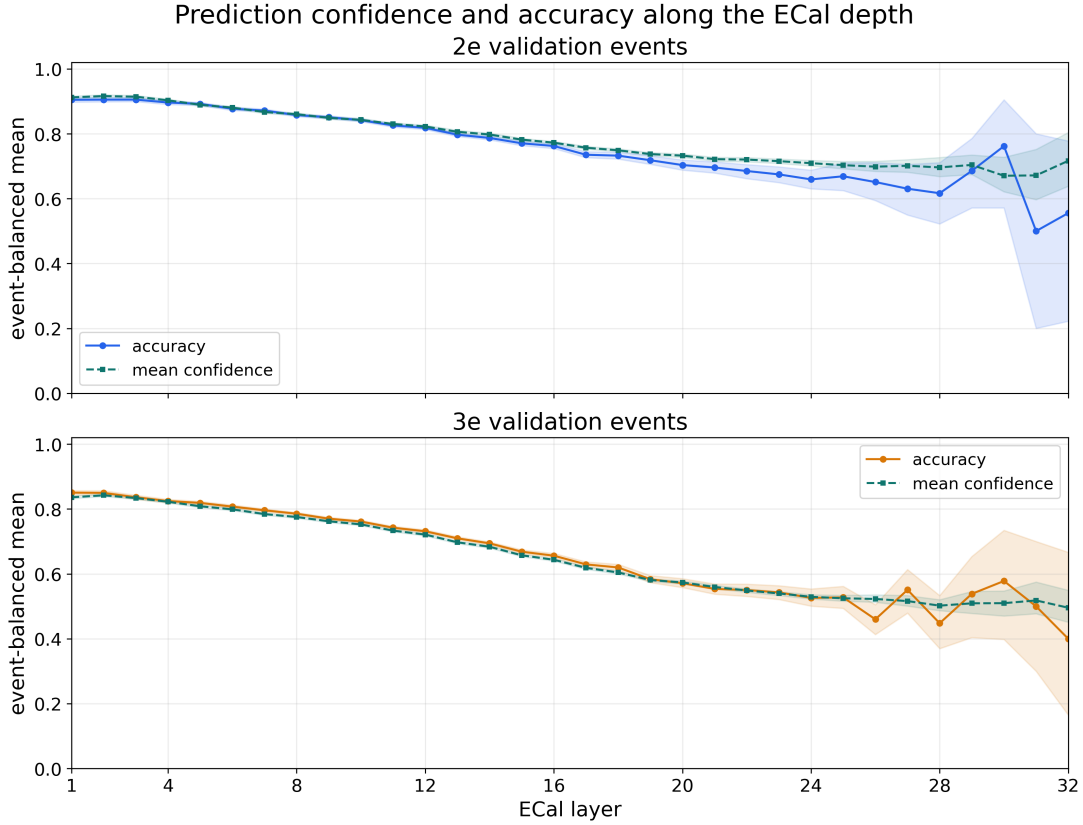


Figure 31. Event-balanced mean prediction confidence and permutation-aligned accuracy across the ECal depth for $2e$ events (top) and $3e$ events (bottom). Shaded bands show 95 % bootstrap confidence intervals.

selected by minimum validation loss. The backbone has hidden width 192, three pre-layer-normalised Transformer layers and eight attention heads (increased from 4 in the baseline model), corresponding to approximately 1.16 million trainable parameters. The test set contains 30,079,515 ECal hits.

The model outputs logits in pre-determined vector sizes, and loss is based on these float values and the reconstructed metrics are based on a post-processing assessment based on these, thus yielding a "slot based design" inspired by the MLPF model. The model predicts three electron-slot validity scores for each event and a contributor-set distribution and energy fractions for each hit. The electron count is derived from the valid slots (note that it is not produced by a separate count head as it was with the baseline model). Similarly, the dominant-origin label is derived from the predicted fractions. This post-processing keeps the reported quantities consistent with the model's underlying set-valued prediction. Table VIII summarises the native multi-task outputs.

Figure 32 resolves the exact contributor-set task. Pure one-electron contributions dominate the data and are generally reconstructed more reliably than mixed sets, but among them the hits with label 2 has the lowest accuracy due to the same effect discussed in 4.2.3 and about $1/3$ of these hits should come from a $3e$ event. The overall exact set accuracy is 72.41 %, so this is substantially more demanding than electron counting. A contributor-set error can still occur when the predicted dominant origin and the event

Table VIII
PERFORMANCE OF THE EPOCH-15 CONTRIBUTOR-SET SLOT MODEL ON ITS 100,000-EVENT TEST SET. COUNT, CONTRIBUTOR-SET AND DOMINANT-ORIGIN RESULTS ARE ACCURACIES; FRACTION ERRORS ARE MEAN ABSOLUTE ERRORS.

Metric	Value
Derived electron-count accuracy, all events	99.983 %
Derived electron-count accuracy, true $2e$	99.994 %
Derived electron-count accuracy, true $3e$	99.972 %
Exact per-hit contributor-set accuracy	72.41 %
Per-hit dominant-origin accuracy	77.04 %
Mixed-hit precision	48.08 %
Mixed-hit recall	64.41 %
Mixed-hit F_1 score	0.551
Fraction MAE, raw prediction	0.141
Fraction MAE, validity-masked prediction	0.115

electron count remain correct. The model performs the worst in predicting mixed hits with labels $1 + 3$, which has a geometric interpretation as these showers are in mean the furthest away, according to the canonical ordering.

Figure 33 shows two useful reductions of the same predictions. The native dominant-origin accuracy, including the noise class and without an event-wise label permutation, is 77.04 %. Mixed-hit detection reaches 64.41 % recall and 48.08 % precision, giving an F_1 score of 0.551. The model therefore identifies a majority of truth-mixed hits but also labels a sizeable number of pure hits as mixed. This task should not be summarised by ordinary accuracy alone because pure hits are much more common.

Applying the predicted slot-validity mask before normalising the energy fractions reduces their mean absolute error from 0.141 to 0.115. This improvement illustrates one purpose of coupling the tasks: the event-level validity prediction constrains which electron fractions are physically available during post-processing. Nevertheless, the contributor-set and mixed-hit results show that resolving overlap at hit level remains the main weakness of this preliminary model.

4.3.2 Comparison with Fixed-Multiplicity Hit Assignment

For a direct comparison with the principal baseline task, the contributor model’s dominant-origin predictions were separated by true event multiplicity, truth-noise hits were excluded, and the three slot labels were aligned once per event. Table IX shows that the resulting accuracies are close to those of the fixed-multiplicity Transformer. For $2e$, the pooled hit accuracies differ by 0.07 percentage points; for $3e$, they differ by 0.45 percentage points. The energy-weighted differences are similarly small. The comparison is descriptive rather than paired because the mixed-multiplicity model and the baselines use different held-out event selections.

Figure 34 shows that this agreement is not limited to a pooled average: the event-accuracy distributions retain the same broad shape and the same reduction from $2e$ to $3e$ seen for the baseline Transformer. The median aligned event accuracy is 0.858 for $2e$ and 0.754 for $3e$, compared with 0.859 and 0.758 for the fixed-multiplicity models.

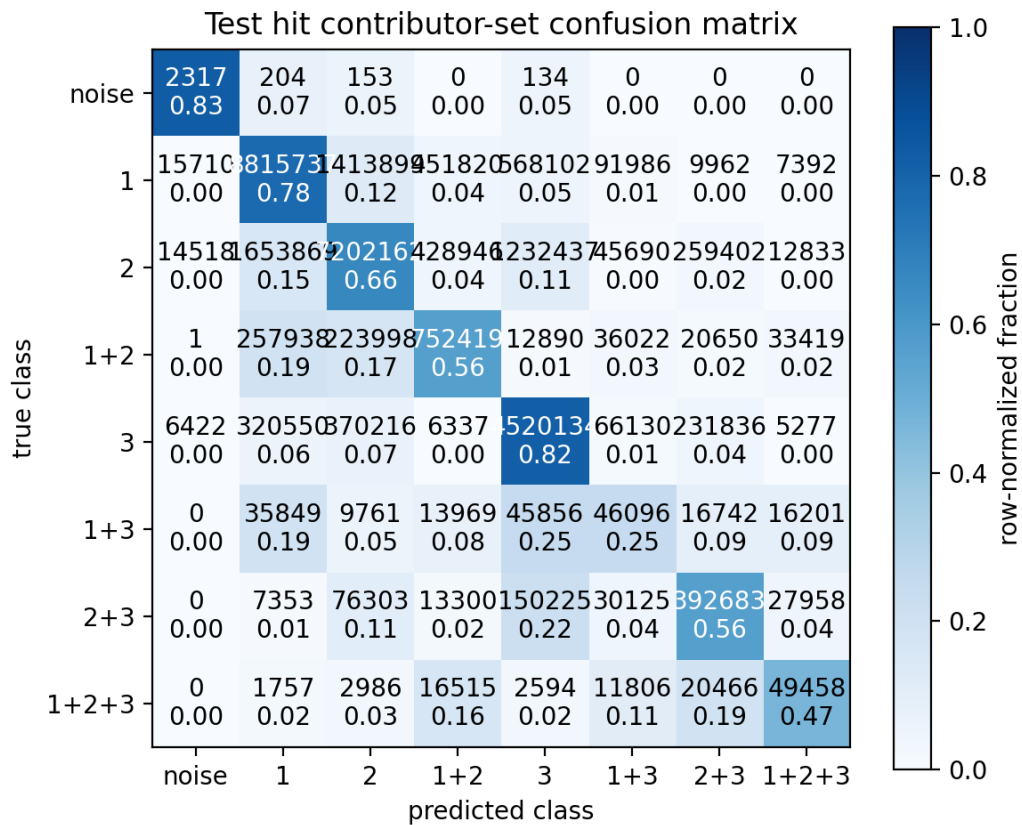


Figure 32. Row-normalised test-set confusion matrix for the per-hit contributor set predicted by TRACE. Each class denotes the exact subset of electron origins contributing simulated deposited energy to a hit. Cell annotations show row percentages.

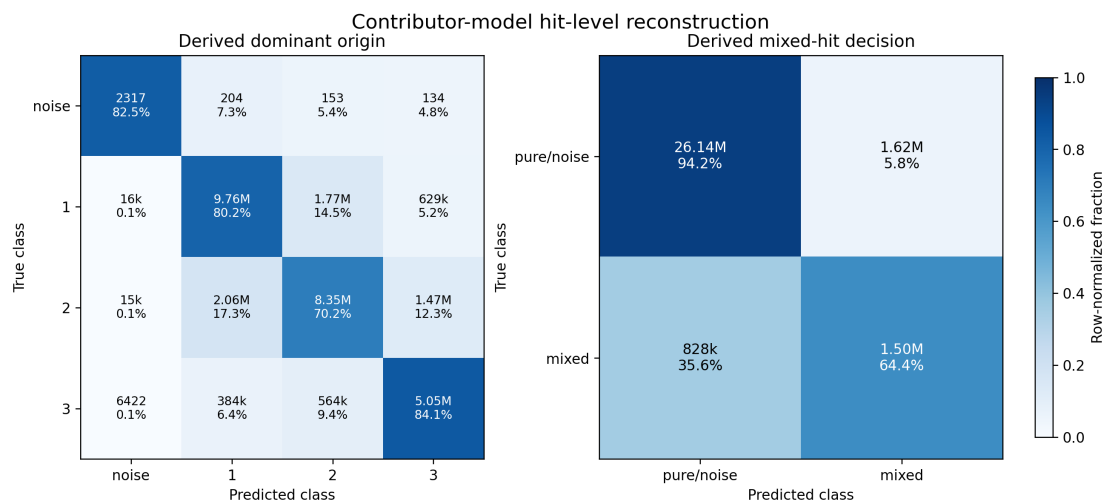


Figure 33. Test-set confusion matrices derived from the TRACE outputs. Left: dominant electron origin inferred from the predicted energy fractions. Right: binary pure-versus-mixed classification derived from the predicted contributor set.

Table IX

PERMUTATION-ALIGNED DOMINANT-ORIGIN ASSIGNMENT ON HELD-OUT TEST EVENTS. THE BASELINE TRANSFORMER IS EVALUATED ON 100,000 EVENTS OF EACH MULTIPLICITY. THE SLOT MODEL TRANSFORMER IS EVALUATED ON THE CORRESPONDING 49,934 $2e$ AND 50,066 $3e$ PORTIONS OF ITS MIXED-MULTIPLICITY TEST SET. TRUTH-NOISE HITS ARE EXCLUDED FROM THE CONTRIBUTOR-MODEL COMPARISON. VALUES ARE PERCENTAGES.

Multiplicity	Model	Pooled hit	Mean event	Median event	Energy-weighted
$2e$	Baseline Transformer	84.20	83.73	85.90	89.76
$2e$	Slot Model Transformer	84.13	83.67	85.78	89.73
$3e$	Baseline Transformer	74.84	74.40	75.82	83.18
$3e$	Slot Model Transformer	74.39	73.93	75.43	82.93

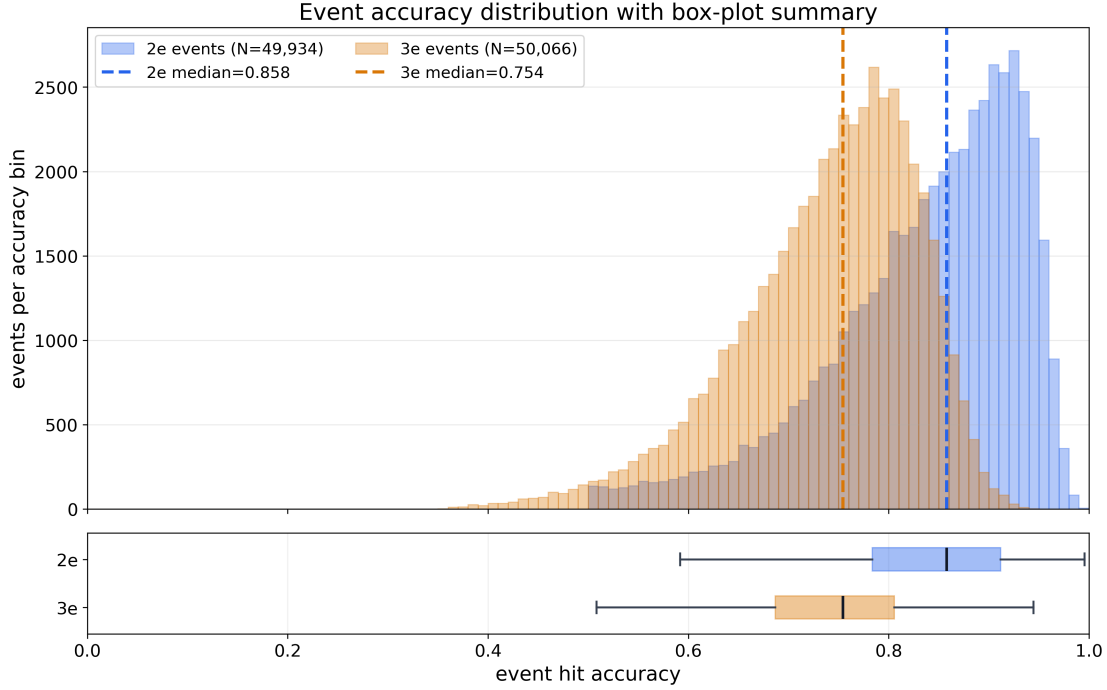


Figure 34. Permutation-aligned dominant-origin event accuracy for the TRACE model on the 49,934 $2e$ events and 50,066 $3e$ events in its test set. Truth-noise hits are excluded. Every event contributes equally to the histogram and box-plot summary.

The dependence on width-normalised truth-shower separation in Figure 35 also follows the baseline pattern: mean accuracy rises as the showers become more distinct, but a broad spread remains at fixed separation. Thus, the added validity, contributor-set and fraction objectives do not remove the overlap dependence, yet they preserve approximately the same dominant-origin performance while supporting the additional reconstruction outputs.

4.3.3 Electron Counting and the Contribution of Trigger-Pad Information

Electron counting is the strongest result of the contributor-set model. The derived count is correct for 99.983% of the 100,000 test events. Figure 36, errors occur in both directions but are rare. This result should be interpreted within the restricted scope of

Event accuracy versus energy-weighted width-normalized separation

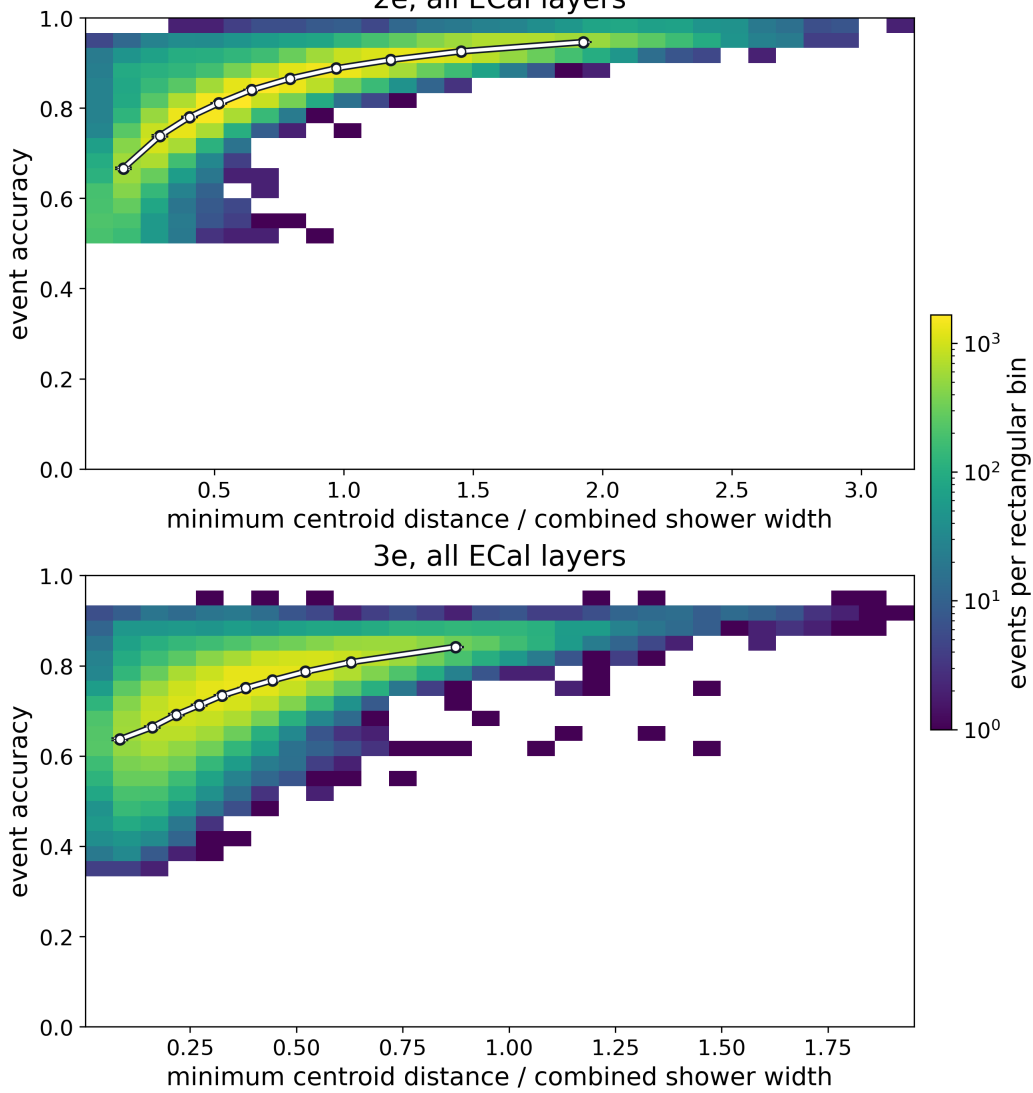


Figure 35. Permutation-aligned TRACE dominant-origin accuracy as a function of the minimum truth-centroid distance divided by combined shower width, for $2e$ events (top) and $3e$ events (bottom). White points show equal-population bin means with 95 % confidence intervals. Truth-noise hits are excluded.

the experiment. The model is evaluated only on $2e$ and $3e$ events and it should be recognized that this is a simpler task. The model essentially solves a binary multiplicity distinction through its three slot-validity predictions and it has not been trained to count an unrestricted range of incident electrons, as compared to the TPad for example.

The raw TPad multiplicity in Figure 20 provides detector context rather than a competing trained classifier. Its disagreement with the true electron count is strongly asymmetric and almost every mismatch is an undercount, while there are some spurious overcounts. This is consistent with the conservative TPad track construction, for which an incident electron may fail to produce a separate reconstructed candidate when signals are missing or merged. Direct token-count equality therefore reaches 79.9 % over the

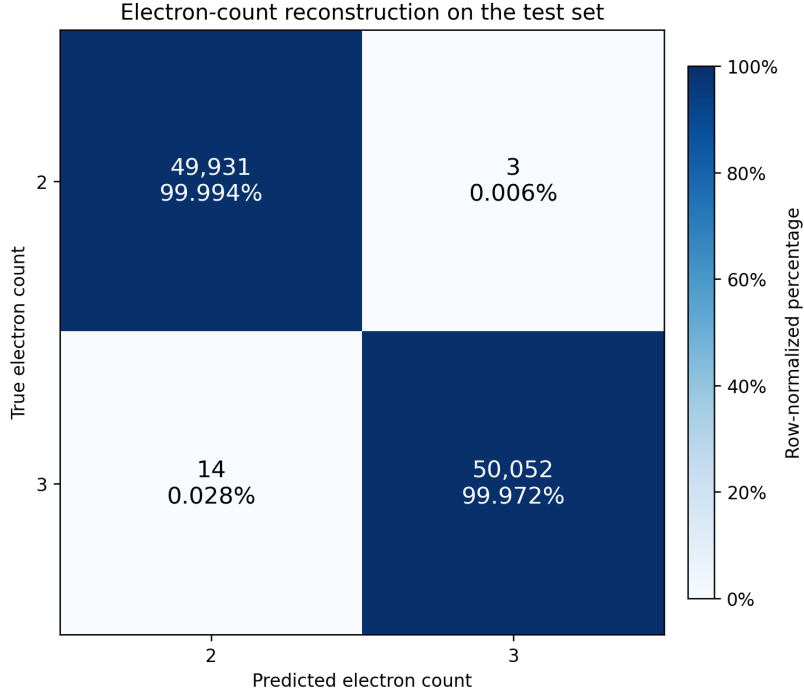


Figure 36. Electron count derived from TRACE’s predicted slot-validity mask on the 100,000-event test set. The model receives both ECal and TPad inputs. Rows are normalized within the true electron count; cell annotations show the event count and row percentage.

balanced $2e/3e$ sample, but it is not equivalent to the neural task. The neural model may combine the ECal shower pattern with TPad positions, photoelectron measurements and multiplicity, and it only has to distinguish $2e$ from $3e$. Its much higher accuracy should therefore not be read as an apples-to-apples improvement over a stand-alone TPad counting algorithm, but it does show promise to explore the electron counting assignment further in future work.

The contribution of TPad information is instead tested by removing its tokens from the same trained network at inference time. Table X reports this paired companion study using the earlier 100,000-event development model and its 10,000-event test set.

Table X
COMPANION ELECTRON-COUNT ABLATION USING THE EARLIER 100,000-EVENT DEVELOPMENT MODEL AND ITS 10,000-EVENT TEST SET. VALUES ARE PERCENTAGES; DIFFERENCES HAVE EVENT-BOOTSTRAP 95 % CONFIDENCE INTERVALS. “TPAD REMOVED” IS AN INFERENCE-TIME ABLATION.

Method or comparison	All	True $2e$	True $3e$
Neural model, ECal+TPad	99.79	99.98	99.61
Same model, TPad removed	98.15	100.00	96.36
ECal+TPad minus TPad removed	+1.64	−0.02	+3.25
95 % CI	[1.39, 1.89]	[−0.06, 0.00]	[2.78, 3.74]

The paired ablation reduces overall count accuracy by 1.64 percentage points, with a 95 % event-bootstrap interval of [1.39, 1.89] percentage points. Almost the entire effect occurs in true $3e$ events, for which removing TPad information reduces accuracy by

3.25 percentage points. The interval excludes zero, providing evidence that this trained network uses the TPad input for the variable-count task.

Together with the fixed-count ablation in Table VII, these results show a task-dependent role for TPad information: its measured effect is small for baseline hit assignment when multiplicity is given, but larger when TRACE must infer multiplicity. The present ablation still introduces a distribution shift because TPad inputs are removed only at inference time. A separately trained ECal-only TRACE is needed to complete the controlled architecture comparison and is treated as follow-up work rather than folded into the current estimate.

4.4 Overall Interpretation and Limitations

4.4.1 Summary of Findings

The principal result is that both set-based architectures recover substantial per-electron structure from overlapping calorimeter showers, although the assignment remains imperfect. The final Transformer reaches pooled permutation-aligned hit accuracies of 84.20 % for $2e$ events and 74.84 % for $3e$ events. Energy weighting, which is relevant because capturing the energy in the detector is the task of the detector, raises these values to 89.76 % and 83.18 %, which shows that incorrectly assigned hits carry less reconstructed energy on average. The reason for *why* that is the case is not clearly laid out from the plots but some results go hand-in-hand with this such as the fact that the accuracy per layer is highest at the first layers, and in the immediate following layers around the same region most of the hits and reconstructed energy is occupied. The reduction for $3e$ events accuracy is consistent with their smaller closest-pair separation and the additional pairwise combinations that can overlap.

The detailed diagnostics support this interpretation without reducing event difficulty to a single geometric variable. Mean accuracy rises with shower separation and falls through the later, sparsely populated part of the ECal. A broad spread remains at any fixed separation, showing that the introduced centroid distance and shower width do not fully describe the assignment problem. Confidence and entropy can rank difficult predictions and expose an accuracy–coverage trade-off, but one should be careful to establish that the softmax scores are calibrated probabilities that is not necessarily the case, though it is an interesting metric quite unique to ML-based methods, as compared to rule-based ones.

The Transformer and GravNet baselines are close in practical performance, very close. Paired comparisons resolve a Transformer a small advantage of 0.4–1.0 percentage points across the reported event-level metrics. The Transformer also trained faster in these implementations despite having more parameters. This makes it a useful representative model for the present study, though not concluded as a generally superior architecture even if this was somewhat of a expected result. Only one trained instance of each configuration was compared, and neither architecture underwent an exhaustive hyperparameter search.

The contributor-set model demonstrates that the output space can be expanded without sacrificing most of the dominant-origin performance, even though the different tasks share a common internal learned mapping analyzing either a $2e$ or $3e$ event from input. After removing truth-noise hits and aligning labels, its pooled accuracies are 84.13 % for $2e$ and 74.39 % for $3e$, close to the fixed-multiplicity Transformer values. It simultaneously derives the correct electron count for 99.983 % of the mixed test sample. It is a promising result, but it should be taken into account that it was restricted to a simpler problem to solve, namely a binary distinction between $2e$ and $3e$ and not general electron counting. Exact contributor-set accuracy is 72.41 %, mixed-hit detection reaches $F_1 = 0.551$ and the validity-masked fraction error is 0.115. These results seem to locate a remaining challenge in identifying mixed deposits and dividing their energy,

but the results might as well also illustrate the boundary of how much ambiguity one can remove, as we expect some information of the original events' electrons to be lost completely, though unknown how much that is.

The two ablation studies indicate task-dependent sensitivity to the TPad tokens, which makes sense considering which one that show sensitivity. For the final fixed-count $3e$ Transformer, intact-input accuracy exceeds the token-removed evaluation by only 0.119 percentage points in pooled hit accuracy and 0.125 points in mean event accuracy. The paired confidence intervals exclude zero, so the change is statistically resolved for that checkpoint, while its practical magnitude is small. In the earlier 100,000-event contributor development model, token removal reduces count accuracy by 1.64 points overall and 3.25 points for true $3e$ events. The latter study uses a different model and a smaller test sample than the final contributor result, so the ablation magnitudes should not be treated as one controlled cross-task comparison. More fundamentally, inference-time removal changes both the input distribution and the attention context. The studies establish sensitivity to removal, but do not necessarily isolate the causal informational value of the TPad modality.

4.4.2 Scope and Limitations

The conclusions are limited to simulated overlays of independent inclusive non-signal electron events with multiplicities $2e$ and $3e$. They do not establish performance on detector data, dark-matter signal samples, other beam conditions or a wider multiplicity range. The fixed-count baselines were also deliberately simplified. They receive the true multiplicity, use one hard dominant-origin label per hit and exclude simulated noise. The TRACE relaxes these assumptions, but remains a preliminary study within the same restricted multiplicity domain.

Canonical y -ordering supplies a stable training convention but is not a physical identity for an electron. Event-wise permutation alignment removes this arbitrary label choice from the reported baseline accuracies, while the training objective itself remains ordering dependent. Several diagnostic variables, including truth-centroid separation, contributor sets and energy fractions, also require simulation truth. They explain model behaviour but are not observables that would be available directly in deployed inference.

The reported confidence intervals resample held-out events for fixed checkpoints. They quantify finite-test-sample uncertainty, not variation from retraining, random initialisation, simulated sample production or detector systematics. Model capacity was chosen with a performance-first strategy and the study does not map the accuracy–capacity frontier. The recorded training times are implementation-specific and are not measurements of online inference latency. Finally, the models use full-resolution offline ECal hits rather than the coarser trigger-cell representation. The results should therefore be viewed as an offline reconstruction proof of concept and a reusable basis for further study, not as a production-ready reconstruction system.

5 Conclusions and Outlook

5.1 Conclusions

This thesis investigated whether modern set-based neural networks can separate overlapping electron showers in the LDMX ECal and whether heterogeneous trigger-pad information adds useful context that these models can extract. Transformer and GravNet models were trained for fixed $2e$ and $3e$ hit assignment, followed by an experimental TRACE model that predicts slot validity, hit contributors and energy fractions jointly. The common sharded dataset, deterministic event splits and held-out test samples made it possible to compare the implemented models and analyse their failure modes consistently.

Both baseline architectures learned substantial shower separation. The final Transformer reached permutation-aligned pooled hit accuracies of 84.20% for $2e$ events and 74.84% for $3e$ events. Energy-weighted accuracy was higher in both samples, which shows that errors preferentially affect lower-energy hits. The poorer $3e$ result is consistent with the smaller closest-pair separation in that sample and with the observed dependence of accuracy on shower overlap. GravNet performed similarly, while paired fixed-checkpoint comparisons gave the Transformer a small advantage of approximately one percentage point or less. The Transformer was therefore the pragmatic reference architecture in this work, although the single-seed and implementation-specific comparison does not establish a universal ranking.

TRACE retained nearly the same aligned dominant-origin performance while supporting a richer reconstruction output with both hit-level and event-level metrics. It achieved 99.983% electron-count accuracy in the restricted $2e$ -versus- $3e$ setting, while exact contributor sets, mixed-hit identification and energy fractions remained more difficult. This demonstrates the flexibility of a particle-flow-inspired set prediction, but also shows that mixed deposits are the central unresolved part of the problem. Removing TPad tokens caused a small but statistically resolved change in fixed-count hit assignment and a larger count change in an earlier TRACE study. Since both tests removed a complete modality only at inference time, they demonstrate sensitivity rather than the causal value of the detector information.

An additional contribution is the `ml_ldmx` software platform developed for the study. It connects ROOT-data extraction and tensorisation, sharded datasets, variable-size model inputs, checkpointed training and reproducible evaluation in one documented workflow. The project is intended to be reusable by later students and researchers rather than to support only the models reported here. Taken together, the results establish that set-based learning is a feasible route to per-electron reconstruction in simulated LDMX pile-up, while shower overlap, mixed hits and validation beyond the present simulation domain remain the main barriers to application.

5.2 Outlook

The first priority is a controlled test of heterogeneous detector value. ECal-only and ECal-plus-TPad models should be trained on identical event splits across several random seeds and evaluated on the same held-out events. The study should cover a wider range of electron multiplicities and explicitly separate events with complete, missing and merged TPad candidates. An alternative trigger-scintillator geometry that provides information in both transverse coordinates (introducing x-coordinates) could also be tested, since it may offer more complementary context than the present one-dimensional centroids.

TRACE should next replace or reduce its dependence on canonical y -ordering. A permutation-invariant loss with event-wise matching, such as Hungarian assignment, would align predicted slots to truth during training rather than only during evaluation. Multi-task loss balancing should then be studied together with broader electron counts and more varied event conditions. Additional detector streams or reconstructed object-level inputs can be introduced through controlled modality comparisons so that their incremental value is measurable.

Before online reconstruction is considered for production, the models should be tested on the intended hardware using realistic trigger-level inputs. These tests should measure accuracy, memory consumption and inference time for models of different sizes. If a smaller or faster model is required, methods such as pruning, quantisation and knowledge distillation can then be evaluated.

After robustness and latency have been established, the reconstruction can be evaluated with trigger-cell inputs and downstream event-level physics metrics. The documented data, training and evaluation components in `ml_ldmx` provide a handover point for these studies and for other related machine-learning reconstruction tasks within LDMX.

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Abbreviations and Glossary

Glossary

Bremsstrahlung	Photon radiation emitted by electrons traversing material. It constitutes a major background and drives electromagnetic shower development. 10
Dark Photon	A hypothetical dark-sector gauge boson A' that acts as a mediator between SM and dark-sector particles; one of the primary physics targets of LDMX. 5, 6
Electron Shower	An electromagnetic cascade initiated when electrons enter the ECal. Its topology and energy profile are central to background rejection. 10
Pile-Up	The presence of multiple overlapping electrons or interactions within the same detector readout window, leading to overlapping showers and increased ambiguity in reconstruction. 12

Acronyms

ldmx-sw	LDMX-Software 19, 21–23, 71
CMS	Compact Muon Solenoid 8, 14, 18
CNN	Convolutional Neural Network 15
CPU	Central Processing Unit 19
DM	Dark Matter 1, 3–8, 13
ECal	Electromagnetic Calorimeter , 1, 2, 8, 10, 16, 20, 23, 24, 26, 32, 35–37, 39, 42, 43, 52, 72, 74
EoT	Electrons On Target 7, 12, 13
Geant4	GEometry ANd Tracking 4 19, 21
GNN	Graph Neural Network 1, 14–18
GPU	Graphics Processing Unit 14, 19, 40
HCal	Hadronic Calorimeter 8, 9
HEP	High-Energy Physics 13, 15, 16
HGCAL	High Granularity Calorimeter 8
HPC	High-Performance Computing 19, 21
LDM	Light Dark Matter 3–5
LDMX	Light Dark Matter EXperiment , 1, 3, 4, 6, 10, 11, 20, 22, 28, 62, 63, 68
ML	Machine Learning 1, 13, 14, 19, 27, 46, 60
MLPF	Machine Learning Particle Flow , 1, 14, 18, 27, 28, 40, 53
RecHit	Reconstructed Hit Quantities 23, 25
RMS	For A Set Of Values $x_1, \dots, x_N$, The Root Mean Square Is Defined As $\text{RMS}(x) = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$. 36–38, 45

SimHit	Simulated Hit Quantities 23
SLAC	SLAC National Accelerator Laboratory 6
SM	Standard Model 1, 3–6, 21, 68
TPad	Trigger-Pad 1, 11, 26, 39, 40, 50, 51, 57–59, 61–63
TPU	Tensor Processing Unit 14
TRACE	Transformer For Reconstruction And Attribution Of Calorimeter Energy , 1, 2, 21, 25, 52, 59, 61
TS	Trigger Scintillator 1, 8, 11, 20
WIMP	Weakly Interacting Massive Particle 5

7 Appendix

This appendix collects implementation and supporting material that is useful for reproducibility but is not required to follow the scientific method in Section 3, and a statement on the use of generative AI tools for this thesis project at the end.

7.1 Simulation Production Details

The source pool was distributed across ROOT files that usually contained 10^4 events, although some files contained fewer. The actual entry count was therefore read for each input file before overlay jobs were constructed.

The `ldmx-sw` overlay workflow reads one file as the *main* event stream and another as the *overlay* stream. These names describe their software roles; both contain the same kind of inclusive single-electron event. For requested multiplicity $p \in \{2, 3\}$, each output combines one main electron with $p - 1$ overlay electrons. Poisson sampling was disabled.

Jobs used disjoint source inputs. The overlay reader has a implemented wrap-around logic if the pile up electron source runs out and to prevent it from doing so and reusing an electron (which would break a guarantee of uniquely simulated electron in every event) the number of events produced by one job was bounded by

$$N_{\text{out}} \leq \min \left(N_{\text{main}}, \left\lfloor \frac{N_{\text{overlay}}}{p - 1} \right\rfloor \right), \quad (29)$$

where N_{main} and N_{overlay} are the measured input entry counts. With 10^4 -event source files, one job consequently produced 10^4 $2e$ events or 5×10^3 $3e$ events.

7.2 ROOT Extraction and Truth Matching

The simulations are stored in ROOT files. A TTree stores each event as one entry and divides its quantities into compressed, column-oriented branches [39, 40]. The conversion pipeline uses uproot to read the selected branches without the C++ ROOT runtime and Awkward arrays to represent the nested, variable-length data [44, 45].

Table XI lists the fields used by the maintained pipeline. A complete branch path has the form "collection/collection.member_"... collection and member names are shown separately for readability.

The selected collections are read in configurable chunks so that a worker holds only part of a source file in memory. Reconstructed and simulated ECal hits may use different row orders, so a lookup keyed by detector-cell identifier attaches the simulated contribution arrays to each reconstructed hit while preserving reconstructed-hit order. A non-noise reconstructed hit without a simulated match cannot receive a supported target and causes conversion to stop.

The intermediate event record contains variable-length lists of reconstructed identifiers, positions, energies and noise flags; matched deposited-energy, origin and track contributions; and a nested trigger-pad record. Track identifiers and contribution counts

Table XI
SELECTED FIELDS FROM THE LDMX_EVENTS TREE AND THEIR ROLES IN THE SUPERVISED PIPELINE.

Collection	Members read	Role
EcalRecHits_overlay	id_, xpos_, ypos_, zpos_, energy_, is_noise_	Cell matching, reconstructed ECal position and energy, and the training-time noise policy
EcalSimHitsOverlay_overlay	id_, track_id_contribs_, edep_contribs_, origin_contribs_, n_contribs_	Cell matching and target construction from deposited-energy and origin arrays
TriggerPadTracks_overlay	centroid_, pe_	Reconstructed trigger-pad context supplied to the heterogeneous models

are retained temporarily for inspection but are not model inputs. Figure 37 summarises the full IO/Root reading and tensorization conversion going from the source collections that store the simulation to the stored tensor event that stores only the relevant parts from the original simulation.

7.3 Tensor Dataset, Scaling and Provenance

The conversion writes one PyTorch .pt shard for each selected .root source file [46]. A shard is not a complete copy of the source ROOT file: it contains the inputs, targets, masks, raw values and metadata required for the tasks in this thesis. Once conversion is complete, training and evaluation do not reopen the source file, thus saving time and resources.

Each shard records its schema version, source information and preprocessing configuration. Every retained event contains the combined detector input, detector masks, raw detector values, hit-level targets and source metadata. The simulation-derived dominant-origin target is stored separately from the class index used by the training loss. Original origin identifiers and source indices are retained when applicable so that canonical classes and tensor rows can be traced back to the simulated event. Event multiplicity is retained as metadata and as a target for the slot model, but is never an input feature.

The dataset is described by two JSON files. manifest.json records the source files, accepted labels, noise treatment, input transformations, target definition and feature layout. index.json records shard locations, event ranges, the total event count and skipped source files (if there are any). During training, shards are loaded on demand and a bounded cache retains recently used shards. This avoids requiring the full dataset in memory or repeating conversion for each model run.

Standardisation is applied after ECal and trigger-pad rows have been placed in the common tensor. Means and standard deviations are calculated from the continuous columns (3–6 of Eq. 12) of the heterogeneous training representation, including zero-filled fields, before a model-specific view is selected. The binary detector-type mask columns retain the row identity after scaling. Then the model view is selected to be restricted to only the ECal then it uses the same scaling fitted on the common representation.

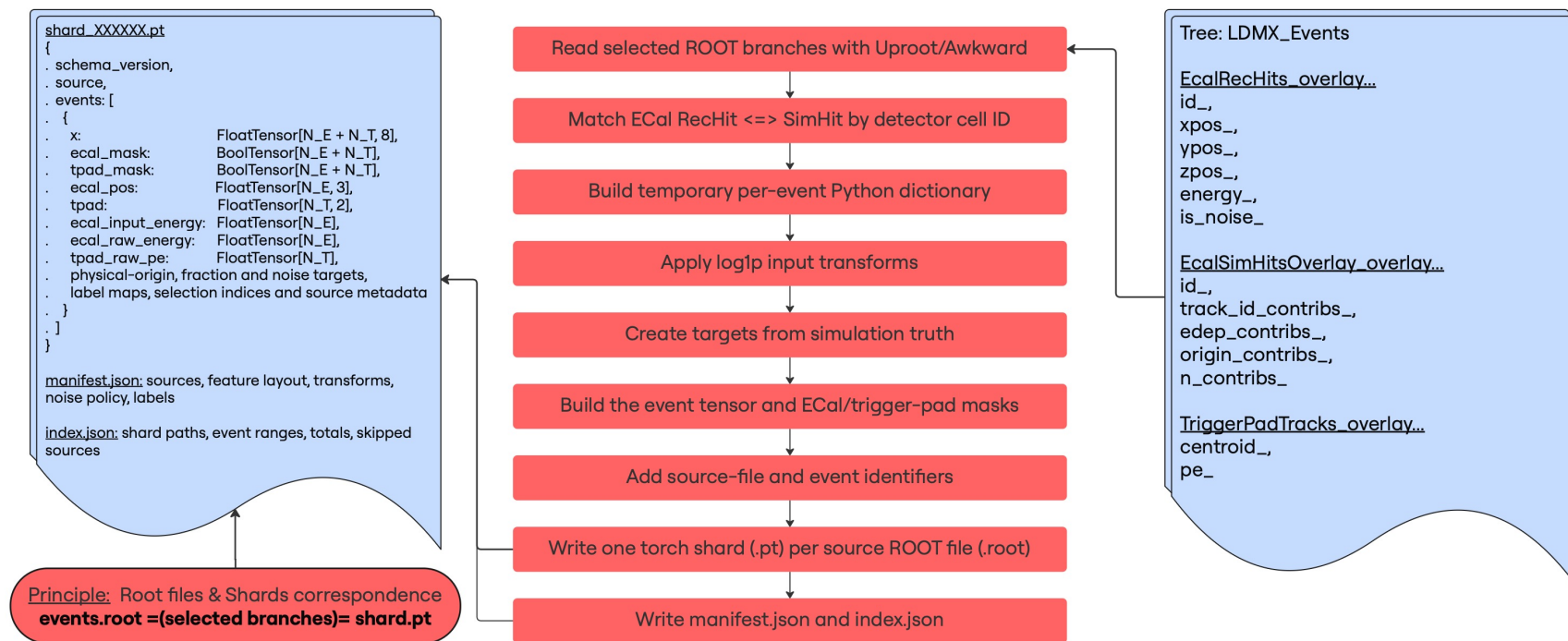


Figure 37. Detailed data flow from selected reconstructed and simulated ROOT collections through hit matching and target construction to the model-ready tensor event.

Figure 38 shows the separate logarithmic transformation of the two non-negative intensity inputs defined in Equation (13).

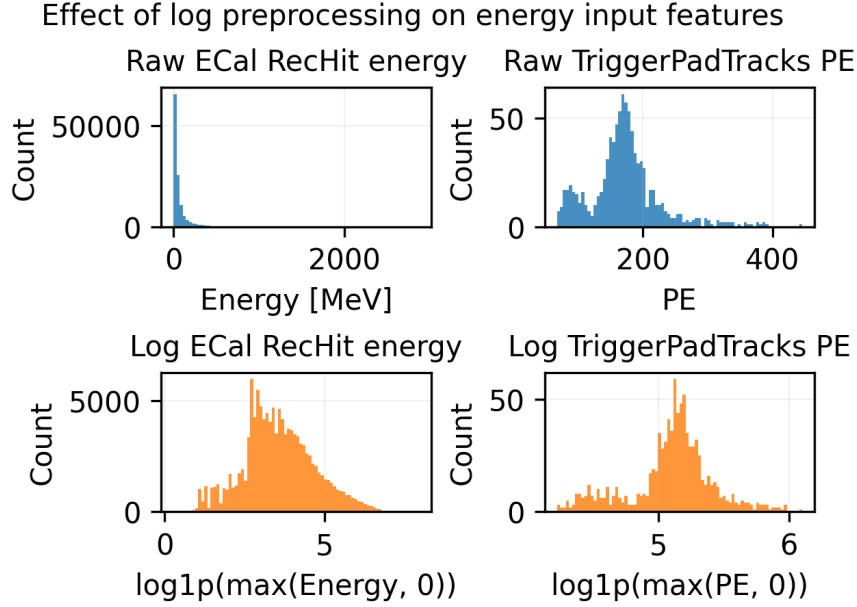


Figure 38. Reconstructed ECal-energy and trigger-pad photoelectron distributions before and after the input $\log(1+x)$ transformation.

7.4 Mini-Batch Implementation of Variable-Sized Events

Figure 39 gives the complete logic of creating mini-batches of the variable-sized event hit collections. Transformer batches are padded to the longest event in the batch, whereas GravNet batches does not introduce padded entries but it simply concatenate nodes and carry an event-assignment vector instead. Both paths retain an ECal mask so that trigger-pad objects can contribute context without entering the hit-origin loss.

7.5 Supporting Evaluation Diagnostics

Figure 40 shows the interactive event display used as a qualitative hit-level diagnostic, allowing truth assignments, incorrect predictions and additional per-hit information to be inspected in the ECal, allowing the user to form hypotheses.

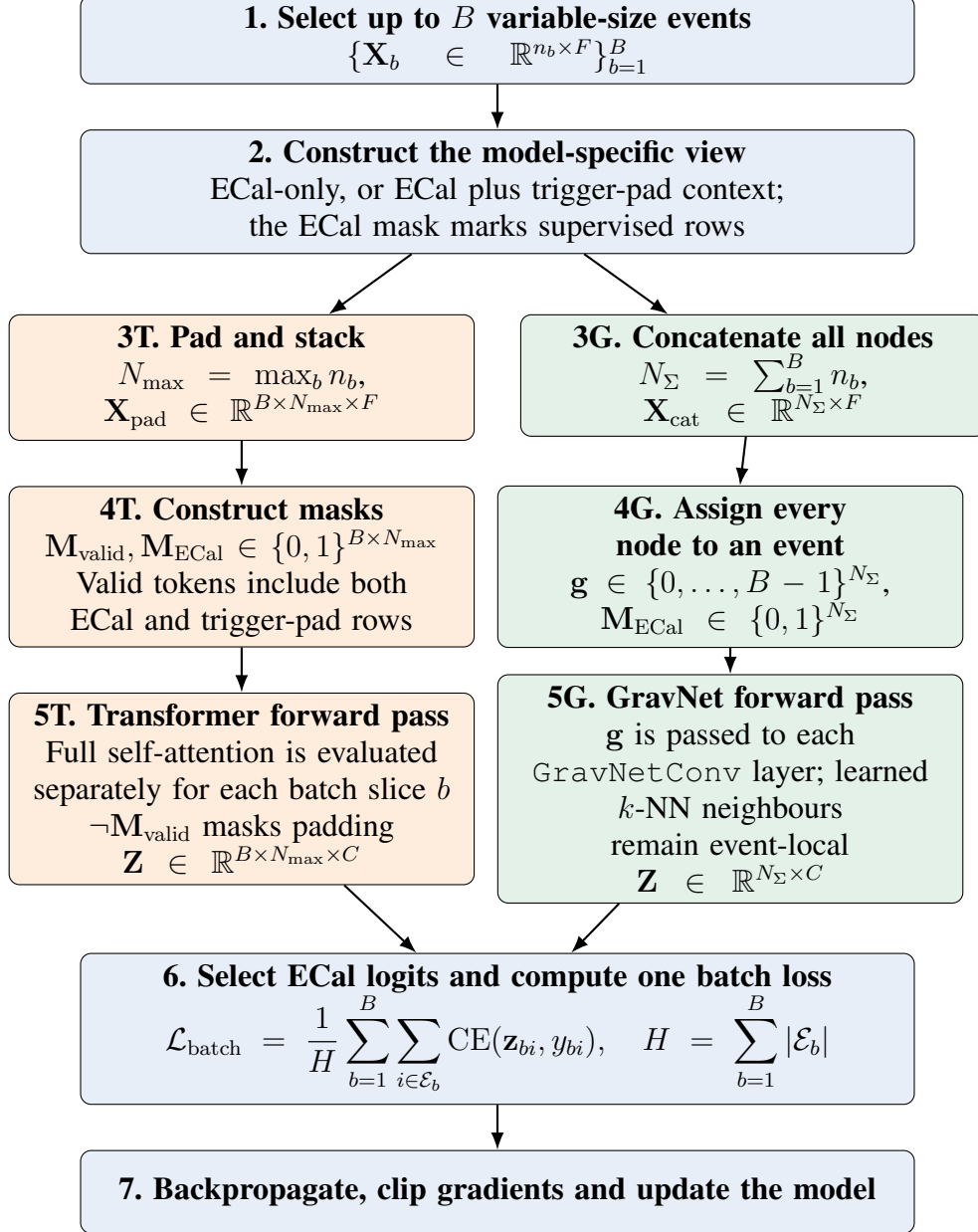


Figure 39. Mini-batch construction for the Transformer and GravNet baselines. Here F is the input-feature dimension, C is the number of output classes and $\mathcal{E}_b$ is the set of ECal rows in event b . Trigger-pad rows provide context but do not enter the hit-origin loss.

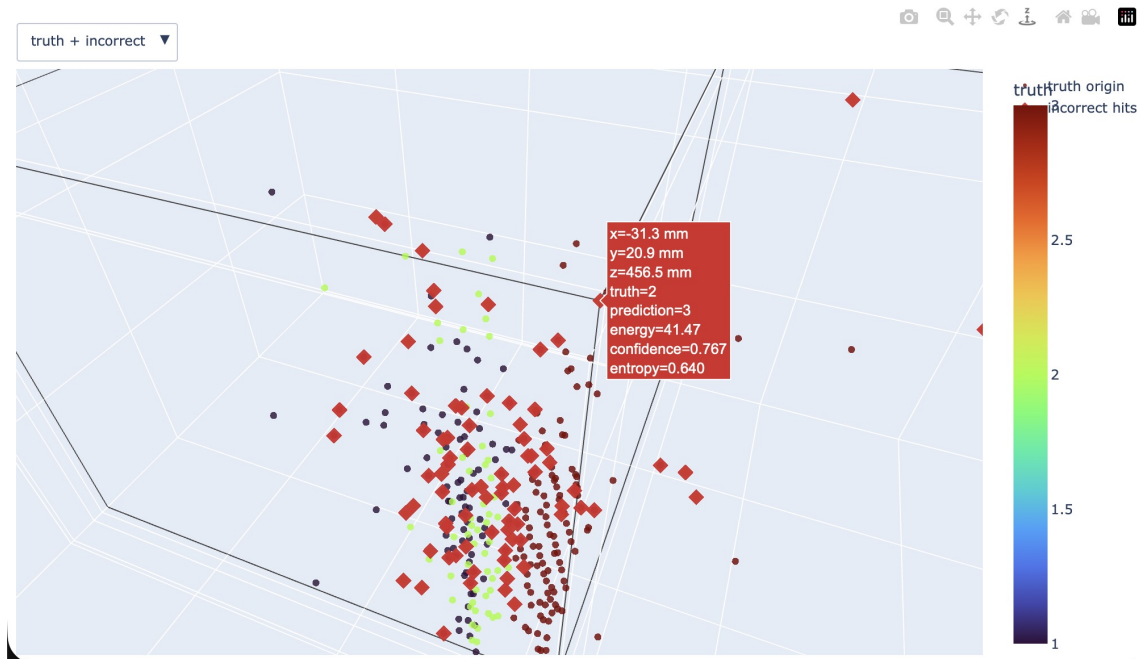


Figure 40. Screenshot of the interactive three-dimensional ECal event display for a $3e$ event. In the currently selected view "truth + incorrect" from the drop-down menu, the correctly assigned hits use their truth-group colours and incorrectly assigned hits are marked with red squares, and hover text exposes more hit-level information.

7.6 Statement on the Use of Generative AI Tools

Generative AI tools, primarily ChatGPT 5.6 Sol, were used in writing this thesis as an auxiliary tool for language editing, enhancing clarity and structure, assistance with LaTeX, and Codex as a development tool for the `ml_ldmx` framework [7]. The tools were also used to discuss technical concepts and possible approaches during development. All generated content has been reviewed and if necessary modified by the author. The scientific methodology, implementation choices, analysis of results and conclusions were made by the author with the support of the supervisors and the final content of the thesis remains the responsibility of the author.